

SECTION-WISE QUESTION BANK

Most-expected questions with detailed explanations
and topic-by-topic weightage analysis

Built for DSSSB TGT - KVS TGT/PGT - NVS TGT - EMRS - State TGT/PGT Computer Science

HOW THIS BANK IS ORGANISED

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HOW TO READ THE QUESTION LABELS

[PYQ]	Appeared verbatim in a DSSSB TGT Computer Science paper
[EXPECTED]	Written by me on a syllabus topic that recurs across DSSSB / KVS / NVS / EMRS / state TGT papers - not a copied question
[NUMERICAL]	Needs a calculation; full working is shown
[TRAP]	A commonly mis-answered question - read the explanation twice

Weightage figures are counted from the three DSSSB TGT Computer Science papers supplied by the reader and cross-checked against the published syllabi of KVS, NVS and EMRS. Always confirm the current pattern on the official notification.

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PART 1

EXAM PATTERNS AND TOPIC WEIGHTAGE

Where the marks actually are. Read this before you attempt a single question.

1.1 The Four Exams This Bank Covers

1.1.1 Why one bank works for all of them

DSSSB, KVS, NVS, EMRS and most state TGT/PGT Computer Science papers draw from **almost the same syllabus** : computer fundamentals, digital logic, architecture, operating systems, programming, data structures, DBMS, networks, web technologies, software engineering, plus teaching methodology.

What differs is the **packaging** - number of questions, time, whether there is a Tier-2, and how much weight the subject section carries. The subject content overlaps heavily, which is why a single well-organised bank serves all of them.

1.1.2 Pattern comparison

Exam	Reported pattern	Negative marking
DSSSB TGT	One-tier CBT, 200 MCQs, 200 marks, 120 minutes . Section A = 100 marks (General Awareness, Reasoning, Arithmetic, English, Hindi). Section B = 100 marks (Subject Concerned including Teaching Methodology)	0.25 per wrong answer
KVS TGT	Written test reported at 180 marks ; the "Subject Concerned" part carries the largest share , with additional parts for language proficiency, general awareness, reasoning and pedagogy. Final merit combines the written test with an interview / professional competency test	0.25 per wrong answer
NVS TGT	MCQ-based, 1 mark per question with 0.25 negative marking . Recent notifications have described both a single 150-question paper and a two-tier structure, so the number of questions has varied between cycles	0.25 per wrong answer
EMRS TGT	Multi-part paper including General Knowledge and Current Affairs, Reasoning Ability, and a dedicated Knowledge of ICT part (computer fundamentals, operating system basics, MS Office, shortcuts, abbreviations, networks, cyber security, internet), plus the subject and pedagogy	0.25 per wrong answer
State TGT /PGT	Varies by state. Usually 100-200 MCQs with a subject section of 50-70% weight	Usually 0.25, some states have none

Verify the pattern yourself - this matters

Exam patterns for these bodies **change between recruitment cycles**, and the figures above come from secondary coaching and news sources rather than a single official document I could read end to end. Two of them openly disagree on the NVS question count.

Before you build a time strategy around any of these numbers, download the **official notification PDF** for the specific cycle you are sitting and confirm: total questions, total marks, duration, sectional split, negative marking, and whether sectional qualifying marks apply.

Sources consulted for the table above: [DSSSB TGT exam pattern - pw.live](<https://www.pw.live/teaching/exams/dsssb-tgt-exam-pattern>) , [DSSSB TGT syllabus - pw.live](<https://www.pw.live/teaching/exams/dsssb-tgt-syllabus-2024>) , [DSSSB TGT pattern - testbook](<https://testbook.com/dsssb-tgt/natural-science-syllabus>) , [KVS TGT syllabus - pw.live](<https://www.pw.live/teaching/exams/kvs-tgt-syllabus>) , [KVS TGT syllabus - careerpower](<https://www.careerpower.in/blog/kvs-tgt-syllabus>) , [NVS exam pattern - pw.live](<https://www.pw.live/teaching/exams/nvs-exam-pattern>) , [NVS pattern - drishti IAS](<https://teaching.drishtiias.com/nvs>) , [EMRS TGT Computer Science syllabus](<https://www.scribd.com/document/923572136/EMRS-TGT-Computer-Science-2025>). Content from these sources was paraphrased and condensed for compliance with licensing restrictions.

1.1.3 What the marking scheme means for strategy

With **+1 for correct and -0.25 for wrong**, the arithmetic of guessing is fixed:

Situation	Expected value per question	Verdict
You know the answer	+1.00	Attempt
You eliminated 2 of 4 options (50% chance)	$0.5(+1) + 0.5(-0.25) = +0.375$	ATTEMPT - clearly worth it
You eliminated 1 of 4 options (33% chance)	$0.33(+1) + 0.67(-0.25) = +0.16$	Attempt, marginally worth it
Pure blind guess (25% chance)	$0.25(+1) + 0.75(-0.25) = +0.0625$	Barely positive. Skip if a distractor looks deliberately attractive
You have no idea and cannot eliminate anything	Effectively 0 or negative	SKIP

The single most profitable habit

Do not aim to answer everything. Aim to **eliminate two options** on every question you are unsure about. That converts a coin-flip into a clearly positive-expectation attempt, and it is a trainable skill - it comes from knowing the *neighbouring* concepts, which is exactly what the explanations in this bank drill.

1.2 Topic Weightage - The Master Table

1.2.1 How these numbers were produced

Two independent inputs:

- 1. Observed count.** I counted every subject question in the three DSSSB TGT Computer Science papers supplied, classified it by topic, and averaged per paper. These are hard counts from real papers.
- 2. Syllabus cross-check.** I compared that distribution against the published topic lists for KVS, NVS and EMRS Computer Science to catch topics that DSSSB happened not to ask in those three papers but which sit squarely on the common syllabus.

Where the two disagree, the "Expected range" column widens. Treat the range as a planning tool, not a prophecy.

1.2.2 The table

Rank	Topic	Observed in DSSSB papers (avg per paper)	Expected range	Priority
1	Operating Systems and Linux	10 - 12	8 - 14	CRITICAL
2	Computer Networks and Security	9 - 12	8 - 14	CRITICAL
3	Programming - C, C++, Java, OOP	9 - 15	8 - 16	CRITICAL
4	Data Structures and Algorithms	6 - 9	6 - 12	CRITICAL
5	DBMS and SQL	5 - 8	5 - 10	HIGH
6	Digital Logic and Number Systems	4 - 7	4 - 8	HIGH
7	Computer Fundamentals and Organisation	4 - 7	4 - 9	HIGH
8	Web Technologies (HTML, CSS, JS, XML, PHP)	4 - 8	3 - 8	HIGH
9	Software Engineering	3 - 5	3 - 6	MEDIUM
10	MIS, Knowledge Management, E-Commerce	3 - 6	2 - 6	MEDIUM
11	Computer Architecture (addressing, cache, pipelining)	3 - 5	3 - 6	MEDIUM
12	Computer Graphics and Multimedia	2 - 4	1 - 4	MEDIUM
13	Emerging technology - AI/ML, Cloud, IoT, Cyber security	0 - 2	2 - 6 and RISING	MEDIUM-HIGH
14	Mathematics and Statistics for CS	4 - 8	2 - 8	MEDIUM
15	Physics / Accounting / Economics (DSSSB-specific)	6 - 10	0 - 10	VARIABLE
16	Teaching Methodology and Pedagogy	10 - 20	10 - 25	CRITICAL

Read the last three rows carefully - they decide your strategy

Row 13 (Emerging technology). The three DSSSB papers you shared barely touched AI, cloud, IoT or cyber security. But newer papers across all these bodies are adding them, and EMRS already has an explicit ICT and cyber-security component. I have therefore included a full section on it (Chapter 9.4) even though its historical DSSSB count is near zero. This is the one place where I am deliberately preparing you for a paper that may not look like the last one.

Row 15 (Physics / Accounting / Economics). This is a DSSSB peculiarity - its Discipline sections pull from the wider graduate stream. KVS and NVS Computer Science papers generally do **not** do this. If you are sitting only KVS or NVS, you may de-prioritise this row. If you are sitting DSSSB, you cannot - it was worth 6 to 10 marks per paper.

Row 16 (Teaching Methodology). The highest marks-per-hour-of-study in the entire syllabus, across every one of these exams. Most candidates under-prepare it because it feels less serious than Computer Science. That is

exactly why it is the cheapest place to gain 15 marks.

1.2.3 Where to spend your study hours

If you have 100 hours left, this is roughly how to allocate them:

Hours	Spend on	Reason
20	Teaching Methodology	Highest return per hour; almost pure recall
18	Operating Systems + Linux	Biggest CS block, very predictable question types
16	Computer Networks + Security	Second biggest, formula-driven and learnable
16	Programming output-tracing	Cannot be crammed - needs daily practice
10	Data Structures and Algorithms complexity tables	Pure recall, high frequency
8	DBMS and SQL	Compact syllabus, reliable marks
6	Digital Logic and Number Systems	Short, mechanical, quick wins
6	Everything else (SE, graphics, MIS, emerging tech)	Breadth cover, one pass each

The mistake almost everyone makes

Candidates spend 60% of their time on the topic they already enjoy - usually programming - and arrive at the exam having never once revised Teaching Methodology or MIS. Your weakest *high-weightage* topic is always worth more than your strongest one. Rank your topics by (weightage x weakness) and study the top of that list, not the top of your comfort list.

1.3 How to Use This Question Bank

1.3.1 The method

1. Pick one chapter. **Cover the answers.** Attempt every question, writing your choice on paper.
2. Score yourself with the real scheme: +1, -0.25, 0.
3. Read **every** explanation, including for questions you answered correctly. The explanations teach the *neighbouring* concepts, which is where the next paper's questions come from.
4. Keep a "leak list" - one line per question you got wrong, naming the concept, not the question.
5. After finishing all chapters, your leak list is your revision syllabus. Nothing else.
6. Re-attempt each chapter after two weeks. Anything wrong twice needs the underlying theory re-read, not just the answer memorised.

1.3.2 What the labels mean

Label	Meaning
[PYQ]	This question appeared verbatim in one of the three DSSSB TGT Computer Science papers supplied. Verified.
[EXPECTED]	I wrote this question on a syllabus topic that recurs across DSSSB, KVS, NVS, EMRS and state TGT papers. It is not copied from any paper - it is a practice question built to the same pattern.
[NUMERICAL]	Requires calculation. Full working is always shown.
[TRAP]	A question most candidates get wrong, usually because two options are both plausible. Read the explanation twice.

An honest statement about sources

I have the three DSSSB papers you shared, and I have read the **published syllabi** of KVS, NVS and EMRS. I do **not** have the actual question papers of KVS, NVS or any state TGT exam.

So every question here is either marked **[PYQ]** - genuinely from your DSSSB papers - or **[EXPECTED]**, meaning I constructed it from the common syllabus and the question patterns these bodies use. I have not labelled anything as being "from the KVS 2023 paper", because I cannot verify such a claim and it would be dishonest to imply it.

This matters practically too: an expected question that teaches the right concept is far more useful than a misattributed one. The concepts are what repeat across these exams, not the exact wordings.

PART 2

**FUNDAMENTALS, NUMBER SYSTEMS,
DIGITAL LOGIC AND ARCHITECTURE**

Weightage: 11 to 19 marks combined. Mechanical, learnable, and the fastest marks in the paper.

2.1 Computer Fundamentals and Organisation

WEIGHTAGE: 4-7 marks in DSSSB (observed) | 4-9 expected across KVS / NVS / EMRS | Priority HIGH

2.1.1 Topic checklist

- Block diagram, ALU / CU / registers, characteristics of computers
- Generations of computers and of programming languages
- Hardware vs software, system vs application software, firmware, open source
- Language translators: compiler, interpreter, assembler, linker, loader
- Input devices, output devices, both-way devices, printer classification
- Memory hierarchy, RAM types, ROM types, cache, secondary storage
- Units of memory and conversions

2.1.2 Questions

Q. [PYQ] Which type of computer program acts as a bridge between a computer user and the computer hardware?

- (a) Random Access Memory
- (b) User thread
- (c) **Operating system**
- (d) Kernel

Why: The OS is the complete interface layer between user and hardware. The **kernel** is only its innermost core - if a question asks for the "core" or "always-resident part", choose kernel; for the user-to-hardware bridge, choose operating system.

Q. [PYQ] Which of the following is a translator that translates source code into object target code?

- (a) Language processor
- (b) Interpreter
- (c) **Compiler**
- (d) Assembler

Why: A compiler translates the entire program at once into object code. An interpreter executes line-by-line and produces no stored object file. An assembler translates *assembly*, not high-level source. "Language processor" is the parent category, too general to be the answer.

Q. [EXPECTED] Which of the following converts assembly language into machine code?

- (a) Compiler
- (b) Interpreter
- (c) **Assembler**
- (d) Loader

Why: Assembler is specifically the assembly-to-machine-code translator. Remember the chain: **Preprocessor to Compiler to Assembler to Linker to Loader**.

Q. [EXPECTED] [TRAP] Which one joins several object files and library routines into a single executable?

- (a) Loader
- (b) **Linker**
- (c) Assembler
- (d) Debugger

Why: **Linker** combines object modules and resolves external references. **Loader** then brings the finished executable from disk into memory for execution. Candidates routinely swap these two - linker builds, loader launches.

Q. [PYQ] Which of the following is an example of the fifth-generation language?

- (a) Java
- (b) Cobol
- (c) **Prolog**
- (d) Python

Why: Prolog is a logic/constraint programming language used in AI, which defines 5GL. Java, Cobol and Python are all third-generation. Remember: **1GL machine, 2GL assembly, 3GL high-level procedural, 4GL non-procedural like SQL, 5GL logic/AI like Prolog and LISP.**

Q. [PYQ] 'FORTRAN' stands for ____.

- (a) Format Translation
- (b) Format Transfer
- (c) **Formula Translation**
- (d) Formula Transfer

Why: FORMula TRANSlation - the first successful high-level language, built for scientific computing. Related: COBOL = Common Business Oriented Language, BASIC = Beginner's All-purpose Symbolic Instruction Code.

Q. [PYQ] ____ are a set of computer programs with necessary documentation.

- (a) Relative programs
- (b) **Software packages**
- (c) File packages
- (d) Application programs

Why: The phrase "**with necessary documentation**" is the signature of a *package*. A bare program has no documentation attached; a package (like MS Office) ships with it.

Q. [PYQ] [TRAP] Which of the following is NOT true about software?

- (a) **System software is written in a high-level language.**
- (b) Application software is written in a high-level language.
- (c) System software plays a vital role for the effective functioning of a system.
- (d) System software is written in a low-level language.

Why: System software must talk directly to hardware, so it is normally written in a **low-level** language (assembly or C). Note that the option list deliberately contains both "high-level" and "low-level" versions of the same claim - spot that pattern and you know which one is being tested.

Q. [PYQ] Which of the following is an example of open-source application software?

- (a) **GIMP**
- (b) CorelDRAW
- (c) Microsoft Excel
- (d) Adobe Illustrator

Why: GIMP (GNU Image Manipulation Program) is the free open-source image editor. Other open-source names worth knowing: **Linux, LibreOffice, MySQL, Apache, Firefox, Blender, VLC, Audacity.**

Q. [EXPECTED] Software that is free to use but whose source code is NOT released is called:

- (a) Open source
- (b) **Freeware**
- (c) Shareware
- (d) Proprietary

Why: **Freeware** = free, closed source (Adobe Reader). **Open source** = source code public. **Shareware** = free trial then paid (WinRAR). Free-of-cost and open-source are not the same thing.

Q. [EXPECTED] Software permanently embedded into a hardware chip is called:

- (a) System software
- (b) Application software
- (c) **Firmware**
- (d) Utility software

Why: Firmware sits between hardware and software - BIOS, a router's operating code, a washing-machine controller. It is stored in ROM or flash.

Q. [PYQ] Which of the following is frequently used to ensure that data is correctly entered into a computer?

- (a) **Keyboards**
- (b) Speakers
- (c) Input controls
- (d) Digitisers

Why: The keyboard is the standard device for entering and visually verifying data. Speakers are output only; a digitiser converts hand-drawn images.

Q. [PYQ] Which of the following is an example of an output device?

- (a) Mouse
- (b) Light pen
- (c) Keyboard
- (d) **Visual Display Unit**

Why: A **light pen is an INPUT device** despite operating on the screen - information flows *into* the computer. Apply the one rule: which way is information flowing?

Q. [EXPECTED] Which of the following is used to read the magnetic-ink numbers printed on a bank cheque?

- (a) OCR
- (b) OMR
- (c) **MICR**
- (d) Barcode reader

Why: **MICR** = Magnetic Ink Character Recognition, used exclusively for cheque processing. **OMR** reads pencil bubbles on answer sheets; **OCR** converts printed text images into editable text.

Q. [EXPECTED] Which of the following is a page printer?

- (a) Dot matrix printer
- (b) Daisy wheel printer
- (c) **Laser printer**
- (d) Drum printer

Why: Laser printers compose and print an entire **page** at once - the fastest class. Dot-matrix and daisy-wheel are character printers; drum and chain printers are line printers. Resolution is measured in **DPI**.

Q. [PYQ] Memory that does not change its contents due to external causes is referred to as:

- (a) dynamic memory
- (b) fixed memory
- (c) **static memory**
- (d) temporary memory

Why: SRAM ("static") retains its value as long as power is applied and needs **no refreshing**. DRAM stores charge in capacitors that leak, so it must be refreshed thousands of times per second.

Q. [EXPECTED] [TRAP] Which statement about SRAM and DRAM is correct?

- (a) DRAM is faster and used as cache
- (b) **SRAM is faster and used as cache; DRAM is denser and used as main memory**
- (c) Both need periodic refreshing
- (d) SRAM is cheaper per bit than DRAM

Why: SRAM uses a 6-transistor flip-flop per bit: fast, expensive, low density, no refresh - ideal for **cache**. DRAM uses one transistor plus a capacitor: slow, cheap, high density, needs refresh - ideal for **main memory**.

Q. [PYQ] Select an appropriate ROM type where data can be erased or destroyed using ultraviolet light.

- (a) EEPROM
- (b) Flash ROM
- (c) **EPROM**
- (d) PROM

Why: EPROM has a quartz window for **UV** erasure. EEPROM and Flash erase **electrically**; PROM can be written once and never erased; masked ROM is fixed at manufacture.

Q. [PYQ] Which storage media supports only sequential access?

- (a) Hard disk
- (b) CD
- (c) **Magnetic tape**
- (d) DVD

Why: Magnetic tape must be wound through from the start, exactly like an audio cassette. Hard disks, CDs, DVDs and SSDs are all **direct (random) access**. Tape survives because it is the cheapest medium per GB for archival

backup.

Q. [EXPECTED] Arrange in order of increasing access time (fastest first).

- (a) Cache, Register, RAM, Hard disk
- (b) **Register, Cache, RAM, Hard disk**
- (c) RAM, Register, Cache, Hard disk
- (d) Register, RAM, Cache, Hard disk

Why: The hierarchy from fastest/smallest/costliest to slowest/largest/cheapest: **Register, Cache (L1-L2-L3), Main memory (RAM), Secondary storage (SSD/HDD), Tertiary (tape/optical).**

Q. [PYQ] [NUMERICAL] Which of the following statements is NOT true?

- (a) 5 kilobytes is equal to 10,240 half bytes
- (b) 3 kilobytes is equal to 24,576 bits
- (c) **4 kilobytes is equal to 4069 bytes**
- (d) 2 kilobytes is equal to 4096 nibbles

Why: 4 KB = 4 x 1024 = **4096** bytes, not 4069 - a digit-swap trap. Verify the others: 5 KB = 5120 bytes = 10,240 nibbles (a nibble is a half byte); 3 KB = 3072 bytes x 8 = 24,576 bits; 2 KB = 2048 bytes x 2 = 4096 nibbles. **Method:** convert everything to bytes first, then look for the number that is slightly off.

Q. [EXPECTED] [NUMERICAL] How many bits are there in 2 MB?

- (a) 2,097,152
- (b) 8,388,608
- (c) **16,777,216**
- (d) 1,048,576

Why: 2 MB = 2 x 1024 x 1024 = 2,097,152 bytes. Multiply by 8: **16,777,216 bits**. Note the distractors are the byte count and the 1 MB byte count - always finish the conversion the question asks for.

Q. [EXPECTED] 1 nibble equals:

- (a) 2 bits
- (b) **4 bits**
- (c) 8 bits
- (d) 16 bits

Why: Nibble = 4 bits = half a byte. One nibble represents exactly one hexadecimal digit, which is why hex is grouped in fours.

Q. [PYQ] Which of the following computer programs provides instructions for computer operations?

- (a) **System software**
- (b) System utility
- (c) Magnetic card
- (d) System application

Why: System software controls and coordinates the machine itself. A *utility* is a narrower helper program (antivirus, defragmenter) that falls under system software.

Q. [EXPECTED] Which of the following is NOT a characteristic of a computer?

- (a) Speed
- (b) Accuracy
- (c) Diligence
- (d) **Intelligence**

Why: A computer has **no intelligence of its own** - it only follows instructions. Errors come from bad data or bad programs, summarised as **GIGO: Garbage In, Garbage Out**.

Q. [EXPECTED] The first generation of computers used which technology?

- (a) Transistors
- (b) Integrated circuits
- (c) **Vacuum tubes**
- (d) Microprocessors

Why: 1st = vacuum tubes, 2nd = transistors, 3rd = integrated circuits (IC), 4th = microprocessors (VLSI), 5th = AI / ULSI. ENIAC and UNIVAC are first-generation machines.

Q. [EXPECTED] A hybrid computer is best described as one that:

- (a) Uses two processors
- (b) **Combines analog and digital operation**
- (c) Runs two operating systems
- (d) Has both RAM and ROM

Why: A hybrid computer handles continuous analog inputs and converts them for digital processing. The standard example is an **ICU patient monitor**, which measures analog vital signs and displays them digitally.

2.2 Number Systems and Codes

WEIGHTAGE: part of the 4-7 mark Digital Logic block | Almost always 1-3 direct questions |
Priority HIGH

2.2.1 Topic checklist

- Binary, octal, decimal, hexadecimal and all conversions
- BCD, Gray code, ASCII, EBCDIC, Unicode
- 1's and 2's complement, r's and (r-1)'s complement, signed representations
- Binary arithmetic, shift-as-multiply/divide

2.2.2 Questions

Q. [PYQ] [NUMERICAL] The decimal equivalent of the binary number 111111 is:

- (a) 46
- (b) **63**
- (c) 52
- (d) 36

Why: Positional weights $32+16+8+4+2+1 = 63$. **Shortcut:** n consecutive 1s always equal $2^n - 1$. Here $2^6 - 1 = 63$.

Q. [EXPECTED] [NUMERICAL] Convert (1101011) binary to octal.

- (a) 151
- (b) **153**
- (c) 155
- (d) 143

Why: Group in threes from the right: 1 101 011, pad the left to 001 101 011, giving 1, 5, 3 = **153**. Groups of three because $8 = 2^3$.

Q. [EXPECTED] [NUMERICAL] Convert (2AF) hexadecimal to decimal.

- (a) 683
- (b) **687**
- (c) 675
- (d) 691

Why: $2 \times 256 + 10 \times 16 + 15 \times 1 = 512 + 160 + 15 = 687$. Remember A=10, B=11, C=12, D=13, E=14, F=15.

Q. [EXPECTED] [NUMERICAL] Convert decimal 45 to binary.

- (a) 101110
- (b) **101101**
- (c) 110101
- (d) 101011

Why: Divide repeatedly by 2 and read remainders bottom-up: 45 to 101101. Check: $32+8+4+1 = 45$.

Q. [PYQ] [NUMERICAL] Represent (743) in octal in Gray code.

- (a) **100010010**
- (b) 101011101
- (c) 110100000
- (d) 100010000

Why: Step 1: octal to binary, 3 bits each - $7=111$, $4=100$, $3=011$, giving 111100011. Step 2: Gray - copy the first bit, then XOR each adjacent pair: 1, $1^1=0$, $1^1=0$, $1^1=0$, $1^0=1$, $0^0=0$, $0^0=0$, $0^1=1$, $1^1=0$ = **100010010**.

Q. [EXPECTED] [NUMERICAL] The Gray code equivalent of binary 1011 is:

- (a) 1101
- (b) **1110**
- (c) 1010
- (d) 1001

Why: $G1 = B1 = 1$; $G2 = 1^0 = 1$; $G3 = 0^1 = 1$; $G4 = 1^1 = 0$. Result **1110**.

Q. [EXPECTED] Why is Gray code used in rotary position encoders?

- (a) It is faster to compute
- (b) **Only one bit changes between consecutive values, avoiding transient errors**
- (c) It uses fewer bits than binary
- (d) It supports negative numbers

Why: In binary, 0111 to 1000 flips four bits at once; if the sensor reads mid-transition it can output a wildly wrong value. Gray code changes exactly **one bit** per step, so the worst-case misread is off by one. This is also why **K-map rows and columns are labelled in Gray order**.

Q. [PYQ] [NUMERICAL] How many INVALID input combinations occur at the input of a BCD adder?

- (a) 58
- (b) 56
- (c) **156**
- (d) 256

Why: A BCD adder takes two 4-bit digits = 8 input lines = $2^8 = 256$ total combinations. Valid BCD digits are 0000-1001, so valid pairs = $10 \times 10 = 100$. Invalid = $256 - 100 = 156$. Memorise the triple: 256 total, 100 valid, 156 invalid.

Q. [EXPECTED] Which of the following is an invalid BCD code?

- (a) 1001
- (b) 0111
- (c) **1100**
- (d) 0101

Why: BCD encodes single decimal digits 0-9, so only 0000-1001 are legal. **1100 = 12**, which exceeds 9. The six invalid patterns are 1010 through 1111.

Q. [PYQ] [NUMERICAL] What is the 2's complement of the binary number 1100?

- (a) **0100**
- (b) 1111
- (c) 11
- (d) 1001

Why: Invert to 0011, then add 1 to get **0100**. Fast trick: copy bits from the right up to and including the first 1, then flip everything to the left - 1100 gives keep "100", flip "1" to 0, giving 0100.

Q. [EXPECTED] [NUMERICAL] The 1's complement of 10110 is:

- (a) 10111
- (b) **01001**
- (c) 01010
- (d) 10101

Why: 1's complement simply inverts every bit: 10110 to **01001**. Adding 1 to this would give the 2's complement.

Q. [PYQ] [NUMERICAL] [TRAP] A register contains a 3's complement number 10100. If it is divided by 2, find the value of the register.

- (a) 01100
- (b) 00110
- (c) **01010**
- (d) 11010

Why: The complement wording is a distraction. In a register, **dividing by 2 = shifting all bits one place RIGHT**. 10100 shifted right gives **01010**. Corollary: left shift multiplies by 2, and shifting by k multiplies or divides by 2^k .

Q. [EXPECTED] In an 8-bit signed 2's complement representation, the range of values is:

- (a) -127 to +127
- (b) **-128 to +127**
- (c) -128 to +128
- (d) 0 to 255

Why: 2's complement is asymmetric: one extra negative value exists because there is only one zero. General formula for n bits: **$-2^{(n-1)}$ to $+2^{(n-1)} - 1$** .

Q. [EXPECTED] Which character code supports every script in the world including Devanagari?

- (a) ASCII
- (b) EBCDIC
- (c) BCD
- (d) **Unicode**

Why: Unicode (UTF-8 / UTF-16) covers all writing systems. ASCII is 7-bit English-only (128 characters); EBCDIC is an 8-bit IBM mainframe code. **UTF-8 is backward compatible with ASCII.**

Q. [EXPECTED] The ASCII value of 'a' is:

- (a) 65
- (b) **97**
- (c) 48
- (d) 90

Why: 'A' = 65, 'a' = 97 (exactly 32 more), '0' = 48. From these three anchors you can derive any letter or digit value.

2.3 Boolean Algebra and Logic Gates

WEIGHTAGE: 2-4 marks | Very high certainty - gates or K-map appear in almost every paper | Priority HIGH

2.3.1 Topic checklist

- All seven gates and truth tables; universal gates
- Boolean laws, De Morgan's theorems, distributive law
- SOP / POS, minterms / maxterms, canonical vs standard form
- K-map rules, prime implicants, don't-care conditions
- XOR properties and parity

2.3.2 Questions

Q. [PYQ] What does the XNOR gate output when A=1 and B=0?

- (a) B
- (b) 0
- (c) A
- (d) 1

Why: XNOR outputs 1 only when inputs are **identical**. Here they differ, so output is 0. Hook: **XOR = "X marks the difference"; XNOR = "No difference"**.

Q. [PYQ] Which of the following statements is false about universal logic gates?

- (a) All basic logic gates can be implemented through universal logic gates.
- (b) NOR gate is known as universal logic gate.
- (c) NAND gate is known as universal logic gate.
- (d) **All basic logic gates cannot be implemented using universal logic gates.**

Why: This contradicts the definition. **NAND and NOR are the two universal gates** - either one alone can build NOT, AND, OR, XOR and XNOR, and therefore any digital circuit.

Q. [EXPECTED] [NUMERICAL] How many NAND gates are required to implement a 2-input AND gate?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Why: A NAND gives the inverted AND, so you need a second NAND wired as an inverter: $A \text{ AND } B = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$. Hence **2**. For comparison, OR from NAND needs 3.

Q. [EXPECTED] The output of a 3-input XOR gate is 1 when:

- (a) All inputs are 1
- (b) All inputs are 0
- (c) **An odd number of inputs are 1**
- (d) An even number of inputs are 1

Why: XOR is the **parity** function: output 1 for an odd count of 1s. This is why XOR chains are used for parity generation and error detection.

Q. [PYQ] [TRAP] A three-variable function $(A \text{ XOR } B \text{ XOR } C)'$ = ?

- (a) $(A \text{ XNOR } B \text{ XNOR } C)'$
- (b) $A \text{ XOR } B \text{ XNOR } C'$
- (c) $A \text{ XNOR } B \text{ XOR } C$
- (d) **$A \text{ XOR } B \text{ XNOR } C$**

Why: Complementing an XOR chain equals complementing **any one** operand, and XOR against a complemented operand is exactly XNOR. Useful identities: $A \text{ XOR } 0 = A$, $A \text{ XOR } 1 = A'$, $A \text{ XOR } A = 0$, $A \text{ XOR } A' = 1$.

Q. [PYQ] Find the value of $F = xy + x'z$ using distributive law.

- (a) $(x + z)(y + z)$
- (b) $(x + y')(x + z)$

- (c) $(x' + y)(y + z)$
 (d) $(x' + y)(x + z)(y + z)$

Why: Apply $A + BC = (A+B)(A+C)$ twice: $xy + x'z = (xy + x')(xy + z) = (x + x')(y + x')(x + z)(y + z)$. Since $(x + x') = 1$ by the complement law, that factor vanishes, leaving $(x' + y)(x + z)(y + z)$.

Q. [EXPECTED] According to De Morgan's theorem, $(A + B)'$ equals:

- (a) $A' + B'$
 (b) $A' \cdot B'$
 (c) $A \cdot B$
 (d) $(A \cdot B)'$

Why: "Break the bar, change the sign": the OR becomes AND and each variable is individually complemented. The companion theorem is $(A \cdot B)' = A' + B'$.

Q. [EXPECTED] The Boolean expression $A + AB$ simplifies to:

- (a) AB
 (b) A
 (c) B
 (d) $A + B$

Why: The **absorption law**: $A + AB = A(1 + B) = A(1) = A$. Its dual is $A(A + B) = A$.

Q. [EXPECTED] $A + A'B$ simplifies to:

- (a) A
 (b) B
 (c) $A + B$
 (d) AB

Why: The **redundancy law**. Verify with a truth table, or expand: $A + A'B = (A + A')(A + B) = 1(A + B) = A + B$.

Q. [PYQ] [TRAP] Which statement is false about simplification of Boolean functions using the Karnaugh Map technique?

- (a) K-map is used to simplify POS form.
 (b) In Karnaugh Map, the rows and columns are represented in binary code sequence.
 (c) You group adjacent squares in powers of two, covering as many minterms or maxterms as you can.
 (d) K-map is used to simplify SOP form.

Why: K-map rows and columns use **GRAY code** (00, 01, 11, 10), never plain binary order. That is precisely what makes physically adjacent cells differ in exactly one variable, which is the whole basis of the method. This is one of the most repeated "false statement" questions in the subject.

Q. [EXPECTED] In a K-map, a group of 4 adjacent 1s eliminates how many variables?

- (a) 1
 (b) 2
 (c) 3
 (d) 4

Why: A group of 2^k cells eliminates k variables. Group of 2 removes 1, group of 4 removes 2, group of 8 removes 3. Bigger groups always give simpler expressions, so make each group as large as possible.

Q. [EXPECTED] To simplify a function into POS form using a K-map, you group:

- (a) the 1s
 (b) the 0s
 (c) the don't-cares
 (d) alternate cells

Why: Group the **0s** for POS (product of sums) and the **1s** for SOP (sum of products). Don't-care cells (X) may be included in either if doing so enlarges a group.

Q. [EXPECTED] A minterm of a 3-variable function contains:

- (a) Any number of literals
 (b) All three variables, each exactly once
 (c) Only the complemented variables
 (d) At most two variables

Why: A minterm is a full AND term containing every variable once, either plain or complemented. For n variables there

are 2^n minterms. A **canonical** expression is made only of full minterms or maxterms.

Q. [EXPECTED] Which method is preferred over the K-map when the number of variables exceeds six?

- (a) Boolean algebra
- (b) Truth table
- (c) **Quine-McCluskey method**
- (d) De Morgan's theorem

Why: The Quine-McCluskey tabular method scales beyond the visual limit of K-maps and, crucially, **can be programmed**, whereas K-map simplification relies on human pattern recognition.

2.4 Combinational and Sequential Circuits

WEIGHTAGE: 1-3 marks | Counters and adders are the recurring favourites | Priority MEDIUM-HIGH

2.4.1 Questions

Q. [EXPECTED] In a half adder, the SUM output is produced by which gate?

- (a) AND
- (b) OR
- (c) **XOR**
- (d) NAND

Why: Sum = A XOR B, Carry = A AND B. A **full adder** adds three bits: Sum = A XOR B XOR Cin, and it can be built from two half adders plus one OR gate.

Q. [EXPECTED] A BCD adder adds which value to correct a result that exceeds 9?

- (a) 0100
- (b) **0110**
- (c) 1010
- (d) 1001

Why: It adds **6 (0110)** to skip the six invalid codes 1010-1111 and return to valid BCD.

Q. [EXPECTED] A multiplexer with 8 data inputs requires how many select lines?

- (a) 2
- (b) **3**
- (c) 4
- (d) 8

Why: A 2^n -to-1 MUX needs **n** select lines, and $2^3 = 8$, so 3 lines. A MUX is "many inputs, one output" - a data selector - and a single 2^n -to-1 MUX can implement any Boolean function of n variables.

Q. [EXPECTED] [TRAP] Which of the following converts an n-bit code into 2^n output lines, activating exactly one?

- (a) Encoder
- (b) **Decoder**
- (c) Multiplexer
- (d) Comparator

Why: Decoder: few inputs, many outputs, one active - used for memory address selection and seven-segment displays. **Encoder** is the reverse: 2^n inputs to n outputs. Students swap these constantly; remember a decoder *expands*.

Q. [EXPECTED] Which flip-flop has a forbidden input combination?

- (a) JK flip-flop
- (b) D flip-flop
- (c) **SR flip-flop**
- (d) T flip-flop

Why: In an SR flip-flop, **S = R = 1** is invalid (it tries to set and reset simultaneously). The JK flip-flop resolves this by making J = K = 1 **toggle** the output.

Q. [EXPECTED] Which flip-flop simply passes its input to the output at each clock edge?

- (a) SR
- (b) JK
- (c) **D**
- (d) T

Why: The **D (data/delay)** flip-flop sets Q = D at the active clock edge. It is the simplest and the most used in registers and shift registers.

Q. [PYQ] [NUMERICAL] A Mod-6 counter and a Mod-8 counter in cascade give a ___ counter.

- (a) Mod-14
- (b) Mod-68
- (c) Mod-2
- (d) **Mod-48**

Why: Cascading counters **multiplies** their moduli: $6 \times 8 = 48$. The second counter advances once per complete cycle of the first. Never add them.

Q. [EXPECTED] [NUMERICAL] How many flip-flops are needed to build a MOD-10 (decade) counter?

- (a) 3
- (b) **4**
- (c) 5
- (d) 10

Why: You need n flip-flops such that 2^n is at least 10. $2^3 = 8$ (too few), $2^4 = 16$ (sufficient), so **4** flip-flops. Six of the sixteen states are unused.

Q. [PYQ] [NUMERICAL] For what minimum value of propagation delay in each flip-flop will a 10-bit ripple counter skip a count when clocked at 10 MHz?

- (a) **10 ns**
- (b) 25 ns
- (c) 100 ns
- (d) 50 ns

Why: Clock period $T = 1/(10 \times 10^6) = 100$ ns. In a **ripple** counter delays accumulate, so total = $10t$. Setting $10t = 100$ ns gives $t = 10$ ns. General formula: **$t = \text{clock period} / \text{number of flip-flops}$** .

Q. [EXPECTED] [TRAP] Which is the key advantage of a synchronous counter over an asynchronous (ripple) counter?

- (a) It needs fewer flip-flops
- (b) **All flip-flops are clocked together, so delays do not accumulate**
- (c) It can count higher
- (d) It needs no clock

Why: In a synchronous counter every flip-flop shares one clock, so total delay equals **one** propagation delay rather than n of them. This is exactly why the ripple counter in the previous question miscounts at high frequency.

Q. [EXPECTED] Output depends only on present inputs in which type of circuit?

- (a) Sequential
- (b) **Combinational**
- (c) Both
- (d) Neither

Why: **Combinational** circuits (adders, MUX, decoders) have no memory. **Sequential** circuits (flip-flops, registers, counters) depend on present inputs *and* stored state.

2.5 Computer Architecture

WEIGHTAGE: 3-5 marks in DSSSB | 3-6 expected | Priority MEDIUM

2.5.1 Topic checklist

- Von Neumann vs Harvard, ISA, micro-architecture
- Boot process, BIOS, POST, CMOS, bootstrap loader
- Registers (PC, IR, MAR, MDR, accumulator, flags)
- Buses, instruction cycle, pipelining and hazards
- RISC vs CISC, addressing modes
- Cache mapping, TAG-bit calculation, write policies, replacement
- I/O: programmed, interrupt-driven, DMA

2.5.2 Questions

Q. [PYQ] Which architecture is the most energy efficient?

- (a) CISC
- (b) IANA
- (c) ISA
- (d) **RISC**

Why: RISC uses simple fixed-length instructions and a **hardwired** control unit, so far fewer transistors switch per instruction. That is why every phone and tablet uses an ARM (RISC) processor. Note the distractor **IANA** is a networking body, not an architecture.

Q. [PYQ] Which of the following is NOT a type of computer architecture?

- (a) Micro Architecture
- (b) Harvard Architecture
- (c) **Software Architecture**
- (d) Instruction Set Architecture

Why: "Software architecture" belongs to **Software Engineering** - it describes module organisation, not hardware design.

Q. [EXPECTED] [TRAP] The main advantage of Harvard architecture over von Neumann is:

- (a) It uses less memory
- (b) **Instructions and data can be fetched simultaneously through separate buses**
- (c) It supports more instructions
- (d) It requires no control unit

Why: Von Neumann shares one memory and bus for both code and data, creating the "**von Neumann bottleneck**". Harvard separates them so both fetches can happen at once. Modern CPUs use **modified Harvard**: separate L1 instruction and data caches over one unified main memory.

Q. [PYQ] Which of the following statements is true?

- (a) POST is performed by BIOS immediately after the system is shutdown.
- (b) The bootstrap program is stored in RAM.
- (c) **CMOS chip is powered by the battery located on the mother board.**
- (d) BIOS was developed by Charles Babbage.

Why: POST runs at **start-up**, not shutdown. The bootstrap loader must be in **ROM** because RAM is empty at power-on. Babbage died in 1871, long before BIOS. The CMOS chip holding BIOS settings and the clock is kept alive by the motherboard's **CR2032 button cell**.

Q. [EXPECTED] Which register holds the address of the next instruction to be executed?

- (a) Instruction Register
- (b) Accumulator
- (c) **Program Counter**
- (d) Memory Data Register

Why: The **PC** (also called Instruction Pointer) holds the *address*; the **IR** holds the *instruction itself* once fetched.

MAR holds the address being accessed in memory, MDR the data.

Q. [PYQ] What is the size of the computer accumulator register?

- (a) 8 bytes
- (b) 16 bytes
- (c) 8 KB
- (d) **4 bytes**

Why: The accumulator matches the machine **word size**; on a standard 32-bit machine that is 32 bits = **4 bytes**.

Q. [PYQ] A computer bus line is made up of which of the following components?

- (a) Registers
- (b) **Set of parallel lines**
- (c) Accumulators
- (d) RAM and ROM

Why: A bus is physically nothing more than a bundle of **parallel conductors**. Three kinds: address bus (one-way), data bus (bidirectional), control bus.

Q. [EXPECTED] [NUMERICAL] If a CPU has 20 address lines, how much memory can it directly address?

- (a) 512 KB
- (b) **1 MB**
- (c) 2 MB
- (d) 4 MB

Why: $2^{20} = 1,048,576$ locations = **1 MB**. Learn the ladder: 16 lines = 64 KB, 20 = 1 MB, 24 = 16 MB, 32 = 4 GB, 36 = 64 GB.

Q. [PYQ] Which cycle is primarily managed by the Control Unit?

- (a) I/O cycle
- (b) Memory refresh cycle
- (c) **Instruction cycle (Fetch-Decode-Execute cycle)**
- (d) Arithmetic cycle

Why: The **CU** fetches, decodes and sequences; the **ALU** only computes. If a question asks who *performs the operation*, the answer is ALU; who *directs* it, the CU.

Q. [PYQ] Select the INCORRECT option among the phases of the CPU Instruction Cycle.

- (a) Execute
- (b) **Encode**
- (c) Store
- (d) Fetch

Why: The phases are **Fetch, Decode, Execute, Store**. An instruction is *decoded*, never encoded, by the CPU.

Q. [PYQ] Which of the following is an addressing mode used in an instruction of the form ADD R1, [R2]?

- (a) **Register Indirect**
- (b) Register
- (c) Indexed Register
- (d) Assembly Register

Why: The **square brackets** are decisive: R2 alone would be register mode (value in R2), while [R2] means "the value at the memory address held in R2" - the register acts as a pointer. "Assembly Register" is not a real addressing mode.

Q. [EXPECTED] In which addressing mode is the operand value contained within the instruction itself?

- (a) Direct
- (b) Register
- (c) **Immediate**
- (d) Indirect

Why: **Immediate** mode, e.g. MOV R1, #25. Fastest, since no memory access is needed, but the value is fixed at assembly time.

Q. [EXPECTED] Which addressing mode is used for relocatable code and branch instructions?

- (a) Immediate
- (b) **PC-relative**

- (c) Register
- (d) Implied

Why: PC-relative computes the effective address as **PC + offset**, so the code works regardless of where it is loaded in memory.

Q. [PYQ] Which of the following is a key design goal of an Instruction Set Architecture?

- (a) Minimizing the transistor count
- (b) Ensuring only complex instructions are supported
- (c) Maximizing the instruction decode time
- (d) **Balancing simplicity and performance for ease of programming and efficiency**

Why: You want to **minimise** decode time, not maximise it, and transistor count is a chip-fabrication concern rather than an ISA goal. Good ISA design aims for completeness, orthogonality, regularity and efficiency.

Q. [PYQ] [NUMERICAL] A four-way set-associative cache of 32 KB uses a block size of 1 K words. Word length is 8 bits and the physical address space is 32 GB. The number of TAG bits is:

- (a) **22**
- (b) 24
- (c) 25
- (d) 23

Why: Address space 32 GB = 2^{35} bytes, and words are 1 byte, so **35 address bits**. Block = 1 K words = 2^{10} , so **10 offset bits**. Cache = 32 KB = 2^{15} bytes, lines = $2^{15}/2^{10} = 32$; four-way means $32/4 = 8$ sets = 2^3 , so **3 set bits**. TAG = $35 - 3 - 10 = 22$.

Q. [EXPECTED] [TRAP] Which cache mapping technique suffers most from conflict misses?

- (a) Fully associative
- (b) **Direct mapping**
- (c) Two-way set associative
- (d) Four-way set associative

Why: In **direct mapping** each memory block has exactly one permitted cache line, so two frequently used blocks competing for that line evict each other repeatedly. Fully associative has no conflict misses at all (a block may go anywhere) but needs expensive parallel tag comparison.

Q. [EXPECTED] In the write-back cache policy:

- (a) Memory is updated on every write
- (b) **Memory is updated only when the block is evicted, tracked by a dirty bit**
- (c) Writes bypass the cache
- (d) Writes are not allowed

Why: **Write-back** is faster but leaves memory temporarily stale. **Write-through** updates both cache and memory immediately - simpler and always consistent, but slower.

Q. [EXPECTED] [NUMERICAL] If the cache hit ratio is 0.9, cache access time is 10 ns and memory access time is 100 ns, the average access time is:

- (a) 10 ns
- (b) **19 ns**
- (c) 55 ns
- (d) 100 ns

Why: Average = $H \times T_c + (1 - H) \times T_m = 0.9(10) + 0.1(100) = 9 + 10 = 19$ ns. This formula appears in some form in most architecture papers.

Q. [EXPECTED] Which I/O technique transfers a whole block between device and memory with minimal CPU involvement?

- (a) Programmed I/O
- (b) Interrupt-driven I/O
- (c) **DMA**
- (d) Polling

Why: **DMA** (Direct Memory Access) uses a separate controller and interrupts the CPU only once, at completion. Programmed I/O wastes the CPU in a polling loop; interrupt-driven I/O still involves the CPU per byte.

Q. [EXPECTED] In pipelining, a branch instruction causes which type of hazard?

- (a) Structural
- (b) Data
- (c) **Control**
- (d) Resource

Why: A **control (branch) hazard** occurs because instructions already fetched may be from the wrong path. Mitigated by **branch prediction**. Data hazards come from unavailable results (fixed by forwarding); structural hazards from two instructions needing the same unit.

Q. [PYQ] Which logical operation does the ALU perform that results in a 1 output only when both inputs are different?

- (a) AND
- (b) **XOR**
- (c) OR
- (d) NOR

Why: XOR is the difference detector. AND needs both 1; OR needs at least one 1; NOR needs both 0.

PART 3

OPERATING SYSTEMS AND LINUX

Weightage: 10 to 12 marks in DSSSB, 8 to 14 expected. The single highest-return Computer Science topic in every one of these exams.

3.1 OS Concepts, Processes and Threads

WEIGHTAGE: 4-6 marks | Process states and definitions are near-certain | Priority CRITICAL

3.1.1 Topic checklist

- OS definition, services, types (batch, multiprogramming, time-sharing, RTOS, distributed)
- Kernel types, system calls, user vs kernel mode
- Process vs program, PCB, process states and legal transitions
- Threads, user vs kernel threads, multithreading models
- fork, exec, wait, join, zombie and orphan processes
- IPC: pipe, message queue, shared memory, socket, signal
- Synchronisation: critical section, mutex, semaphore, monitor
- Deadlock: four conditions, prevention, avoidance, Banker's algorithm

3.1.2 Questions

Q. [PYQ] Which of the following services is NOT provided by the operating system?

- (a) Resource management
- (b) Inter process for communication
- (c) **Translate source into machine code**
- (d) Program execution

Why: Translation is the **compiler's** job. The OS *runs* the compiler but performs no translation itself. Genuine OS services: process management, memory management, file management, I/O, IPC, protection, error detection, accounting.

Q. [EXPECTED] Which of the following is NOT a function of an operating system?

- (a) Memory management
- (b) Process scheduling
- (c) File management
- (d) **Designing the database schema**

Why: Schema design is a human/DBA activity in DBMS. Keep the OS's remit clear: it manages the machine's resources.

Q. [EXPECTED] The core part of the operating system that is always resident in memory is called:

- (a) Shell
- (b) **Kernel**
- (c) Compiler
- (d) BIOS

Why: The **kernel** handles scheduling, memory, drivers, system calls and interrupts, and never leaves memory. The **shell** is the outer command interpreter.

Q. [EXPECTED] [TRAP] Which kernel type runs file systems and device drivers in user space?

- (a) Monolithic kernel
- (b) **Microkernel**
- (c) Hybrid kernel
- (d) Exokernel

Why: A **microkernel** keeps only scheduling, memory and IPC in kernel space; everything else runs as user processes. Result: very reliable and modular, but slower because of message-passing overhead. **Monolithic** (Linux, UNIX) puts everything in kernel space - fast but a single bug can crash the system.

Q. [EXPECTED] A system call causes a transition from:

- (a) kernel mode to user mode
- (b) **user mode to kernel mode**
- (c) user mode to user mode
- (d) kernel mode to kernel mode

Why: A system call is the controlled doorway into privileged execution. This **dual-mode** design is what stops a user

program from touching hardware directly.

Q. [EXPECTED] Which of the following is a process-control system call?

- (a) open()
- (b) read()
- (c) **fork()**
- (d) ioctl()

Why: The five system-call categories: **process control** (fork, exec, exit, wait), **file management** (open, close, read, write), **device management** (ioctl), **information maintenance** (getpid, time), **communication** (pipe, socket, shmget).

Q. [PYQ] What is typically returned by the getpid system call in Unix-based operating systems?

- (a) The process ID of the kernel
- (b) **Process ID of the current process**
- (c) Process ID of the first process
- (d) Process ID of the parent process

Why: getpid() returns the caller's own PID; **getppid()** - note the extra p - returns the **parent's** PID. A guaranteed pair question.

Q. [EXPECTED] [TRAP] What is the difference between a program and a process?

- (a) They are identical
- (b) **A program is passive code on disk; a process is a program in execution**
- (c) A process is stored on disk and a program in memory
- (d) A program can have only one process

Why: A program is a **passive** file (a recipe); a process is an **active** entity with a program counter, registers, stack and memory (actually cooking). One program can spawn many processes.

Q. [EXPECTED] Which of the following is NOT stored in the Process Control Block?

- (a) Process state
- (b) Program counter
- (c) CPU register values
- (d) **The complete source code of the program**

Why: The PCB holds process **metadata**: PID, state, PC, registers, scheduling info, memory-management info, accounting data, I/O status. Source code is never in the PCB.

Q. [PYQ] An operating system completely releases the memory allocated to a process after the process transitions from which of the following states?

- (a) Sleeping to ready
- (b) Ready to running
- (c) **Running to terminated**
- (d) Running to sleeping

Why: Memory is reclaimed only when the process **ends**. A sleeping or ready process will resume later and must find its data intact.

Q. [EXPECTED] [TRAP] Which process state transition is NOT possible?

- (a) Running to Ready
- (b) Waiting to Ready
- (c) Running to Waiting
- (d) **Waiting to Running**

Why: A blocked process whose event completes always returns to the **Ready queue** and must be re-dispatched by the scheduler - it can never jump straight onto the CPU. Also impossible: Ready to Waiting, and anything out of Terminated.

Q. [PYQ] If a process arrives at time 5, what is the latest possible time for it to start execution if there are no other processes in the system?

- (a) Time 6
- (b) Time 0
- (c) **Time 5**
- (d) Time 4

Why: With no competition the scheduler dispatches it immediately on arrival, and it cannot begin before it exists. So $\text{earliest} = \text{latest} = 5$.

Q. [EXPECTED] A thread shares which of the following with other threads of the same process?

- (a) Program counter
- (b) Stack
- (c) Register set
- (d) **Code, data and open files**

Why: Threads share the **address space** (code, data, heap) and open files. Each thread keeps its own **program counter, registers and stack**. Hence a thread is "lightweight".

Q. [EXPECTED] [TRAP] What is the main disadvantage of user-level threads?

- (a) They are slow to create
- (b) **If one thread blocks, the entire process blocks**
- (c) They cannot share memory
- (d) They need more memory than kernel threads

Why: The kernel is unaware of user-level threads, so a blocking system call in one blocks the whole process, and the threads cannot be spread across multiple cores. Kernel-level threads avoid both problems at the cost of slower management.

Q. [PYQ] ___ and ___ are used in conjunction for making a complete process subroutine.

- (a) **fork(), join()**
- (b) fork(), sleep()
- (c) sleep(), join()
- (d) join(), fork()

Why: `fork()` splits execution into parent and child; `join()/wait()` merges by waiting for completion. Order matters - you cannot join before you fork. Failing to wait leaves **zombie** processes.

Q. [EXPECTED] What does fork() return in the child process?

- (a) The child's PID
- (b) The parent's PID
- (c) **0**
- (d) -1

Why: `fork()` returns **0 to the child**, the **child's PID to the parent**, and **-1 on failure**. This asymmetry is how a single piece of code knows which process it is running in.

Q. [EXPECTED] [NUMERICAL] How many total processes exist after three successive fork() calls?

- (a) 3
- (b) 4
- (c) 6
- (d) **8**

Why: Each fork doubles the process count, so n forks give 2^n processes: $2^3 = 8$ total, meaning 7 new children.

Q. [EXPECTED] A process that has terminated but whose parent has not yet collected its exit status is called:

- (a) Orphan process
- (b) **Zombie process**
- (c) Daemon process
- (d) Suspended process

Why: A **zombie** is dead but still holds a PCB entry awaiting `wait()`. An **orphan** is a live process whose parent died; it is adopted by `init/systemd` (PID 1).

Q. [PYQ] What is an endpoint of an inter-process communication?

- (a) Tunnel
- (b) Port
- (c) **Socket**
- (d) Pipe

Why: A **socket** is one endpoint of a two-way communication link, identified by **IP address + port + protocol**. A **port** is merely one component of a socket address; a **pipe** is a channel, not an endpoint, and is not network-capable.

Q. [EXPECTED] Which IPC mechanism is the fastest for transferring large amounts of data?

- (a) Pipe
- (b) Message queue
- (c) **Shared memory**
- (d) Signal

Why: **Shared memory** maps one region into both processes, so after setup no kernel copying occurs. Pipes and message queues copy through the kernel; signals carry almost no data.

Q. [EXPECTED] A binary semaphore whose value can only be 0 or 1 is functionally similar to a:

- (a) Monitor
- (b) **Mutex lock**
- (c) Counting semaphore
- (d) Condition variable

Why: A **mutex** provides mutual exclusion for one resource. A **counting semaphore** permits up to N concurrent holders. Semaphore operations: `wait ( )/P()/down()` decrements, `signal ( )/V()/up()` increments.

Q. [EXPECTED] Which of the following is NOT one of the four necessary conditions for deadlock?

- (a) Mutual exclusion
- (b) Hold and wait
- (c) No preemption
- (d) **Starvation**

Why: The four are **mutual exclusion, hold and wait, no preemption, circular wait**. **Starvation** is a different problem - the system still progresses, just not for one unlucky process. Its cure is **aging**.

Q. [EXPECTED] [TRAP] The Banker's algorithm is used for deadlock:

- (a) prevention
- (b) **avoidance**
- (c) detection
- (d) recovery

Why: Avoidance - it permits the four conditions but tests every request to keep the system in a **safe state**.

Prevention structurally forbids one condition; *detection* lets deadlock happen and finds it with a wait-for graph. Dijkstra devised the Banker's algorithm.

Q. [EXPECTED] Real operating systems such as UNIX and Windows handle deadlock mainly by:

- (a) the Banker's algorithm
- (b) resource ordering
- (c) **largely ignoring it and relying on restart**
- (d) full detection and rollback

Why: Deadlock is rare enough in practice that the overhead of avoidance is not worth it - this is sometimes called the "ostrich algorithm". Good to know because it counters the textbook assumption.

3.2 CPU Scheduling

WEIGHTAGE: 2-4 marks | Definitions and one numerical are typical | Priority CRITICAL

3.2.1 Questions

Q. [PYQ] [TRAP] Which of the following statements is FALSE?

- (a) **Burst Time: Time required by a process to start its execution**
- (b) Turn Around Time: difference between completion time and arrival time
- (c) Completion Time: time at which a process completes its execution
- (d) Arrival Time: time at which a process arrives in the ready queue

Why: Burst time is the time a process needs **ON THE CPU to execute**, not the time to start. Examiners change "to execute" into "to start its execution" - a one-word swap that flips the statement.

Q. [EXPECTED] Turn Around Time is calculated as:

- (a) Burst Time - Arrival Time
- (b) **Completion Time - Arrival Time**
- (c) Completion Time - Burst Time
- (d) Waiting Time + Arrival Time

Why: $TAT = CT - AT$, and $WT = TAT - BT$. These two formulas solve every scheduling numerical. Build a table with columns AT, BT, CT, TAT, WT.

Q. [EXPECTED] [NUMERICAL] Three processes arrive at time 0 with burst times 5, 3 and 8, scheduled by FCFS in that order. What is the average waiting time?

- (a) 2.67
- (b) **4.33**
- (c) 5.33
- (d) 6.00

Why: Waiting times are 0, 5 and 8 (each waits for all previous bursts). Average = $(0 + 5 + 8)/3 = 13/3 = 4.33$.

Q. [EXPECTED] Which scheduling algorithm gives the minimum average waiting time among non-preemptive algorithms?

- (a) FCFS
- (b) **SJF**
- (c) Round Robin
- (d) Priority

Why: **Shortest Job First** is provably optimal for average waiting time, but it needs burst times known in advance and can **starve** long jobs. Its preemptive form is **SRTF**.

Q. [EXPECTED] The convoy effect is a problem associated with:

- (a) SJF
- (b) Round Robin
- (c) **FCFS**
- (d) Priority scheduling

Why: In **FCFS**, one long CPU-bound process at the head makes all short processes wait behind it, like a convoy stuck behind a slow truck. This is why FCFS has poor average waiting time.

Q. [EXPECTED] [TRAP] In Round Robin scheduling, what happens if the time quantum is made very large?

- (a) It becomes SJF
- (b) **It behaves like FCFS**
- (c) It causes starvation
- (d) Context switching increases

Why: With a quantum longer than every burst, no preemption ever occurs, so RR degenerates to **FCFS**. A very *small* quantum has the opposite problem: excessive context-switch overhead.

Q. [EXPECTED] Which scheduling algorithm is most suitable for a time-sharing system?

- (a) FCFS
- (b) SJF
- (c) **Round Robin**
- (d) Priority without aging

Why: Round Robin guarantees each process the CPU within a bounded time, which is exactly what interactive response requires.

Q. [EXPECTED] Starvation in priority scheduling is solved by:

- (a) Preemption
- (b) **Aging**
- (c) Increasing the time quantum
- (d) Using FCFS

Why: **Aging** gradually raises the priority of long-waiting processes so they eventually run.

Q. [EXPECTED] Which scheduler controls the degree of multiprogramming?

- (a) Short-term scheduler
- (b) **Long-term scheduler**
- (c) Medium-term scheduler
- (d) Dispatcher

Why: The **long-term (job) scheduler** decides which jobs enter the ready queue, thereby fixing how many processes are in memory. The **short-term** scheduler picks who runs next (milliseconds); the **dispatcher** performs the actual context switch.

Q. [EXPECTED] A context switch is best described as:

- (a) useful computation
- (b) **pure overhead with no useful work done**
- (c) a form of scheduling algorithm
- (d) an I/O operation

Why: During a context switch the CPU only saves and restores state. Minimising switches is therefore a real performance goal.

3.3 Memory Management and Virtual Memory

WEIGHTAGE: 3-4 marks | Paging and page replacement recur constantly | Priority CRITICAL

3.3.1 Questions

Q. [PYQ] [TRAP] How does the Best Fit algorithm impact system performance?

- (a) It improves system performance by reducing memory wastage.
- (b) **It may reduce system performance due to increased search time for free blocks.**
- (c) It always improves system performance.
- (d) It improves system performance by reducing external fragmentation.

Why: Best Fit must scan the **entire** free list on every request, and the tiny leftover holes it creates actually **worsen** external fragmentation. Note that three of the four options praise it - the word "always" alone disqualifies one of them.

Q. [EXPECTED] Which allocation algorithm scans from the start and takes the first sufficiently large hole?

- (a) Best Fit
- (b) **First Fit**
- (c) Worst Fit
- (d) Next Fit

Why: **First Fit** is the fastest to search and performs well in practice. **Worst Fit** takes the largest hole and performs poorly. **Next Fit** resumes scanning from the last position.

Q. [EXPECTED] [TRAP] Wasted space inside an allocated block is called:

- (a) External fragmentation
- (b) **Internal fragmentation**
- (c) Compaction
- (d) Thrashing

Why: **Internal** fragmentation is unused space **within** an allocated block (typical of fixed partitions and **paging**). **External** fragmentation is free memory scattered in pieces too small to use (typical of variable partitions and **segmentation**), cured by **compaction**.

Q. [EXPECTED] Paging eliminates which problem?

- (a) Internal fragmentation
- (b) **External fragmentation**
- (c) Thrashing
- (d) Page faults

Why: Because any page can occupy any free frame, contiguity is no longer required, so external fragmentation disappears. A little internal fragmentation remains in each process's last page.

Q. [EXPECTED] [TRAP] Which statement correctly distinguishes paging from segmentation?

- (a) Paging uses variable-size blocks; segmentation uses fixed-size
- (b) **Paging uses fixed-size blocks decided by the OS; segmentation uses variable-size logical divisions visible to the programmer**
- (c) Both use fixed-size blocks
- (d) Segmentation causes internal fragmentation

Why: **Paging:** fixed size, OS-decided, invisible to the programmer, internal fragmentation. **Segmentation:** variable size, logical (code/data/stack segments), visible to the programmer, external fragmentation.

Q. [EXPECTED] The hardware cache that speeds up page-table lookups is called:

- (a) Page table base register
- (b) **TLB**
- (c) Frame buffer
- (d) Cache line

Why: The **Translation Lookaside Buffer** stores recent page-to-frame mappings. Without it every memory reference would need two memory accesses - one for the page table, one for the data.

Q. [PYQ] What is the main purpose of virtual memory?

- (a) **To extend the physical memory using secondary storage**
- (b) To provide cache storage
- (c) To store BIOS settings
- (d) To increase CPU speed

Why: Virtual memory keeps only the currently needed pages in RAM and the remainder in **swap space on disk**, so a program larger than physical RAM can run.

Q. [PYQ] Page fault means:

- (a) **accessing the page that is not currently in the memory**
- (b) removing the page whenever it is needed
- (c) fetching the page whenever it is needed
- (d) an error that always occurs in the page

Why: A page fault is a **normal, expected event** in demand paging, not a program error. The OS fetches the page from disk, updates the page table and restarts the instruction.

Q. [EXPECTED] Thrashing occurs when:

- (a) The CPU is idle
- (b) **The system spends more time swapping pages than executing processes**
- (c) The page table is full
- (d) A deadlock occurs

Why: Caused by too high a degree of multiprogramming, so no process holds enough frames for its **working set**. Cure: reduce multiprogramming or apply the working-set model.

Q. [EXPECTED] [TRAP] Belady's anomaly is observed in which page replacement algorithm?

- (a) LRU
- (b) Optimal
- (c) **FIFO**
- (d) LFU

Why: Belady's anomaly - more frames producing *more* page faults - occurs in **FIFO** (and Second-Chance). **LRU and Optimal** are "stack algorithms" and never suffer it. A classic one-mark question.

Q. [EXPECTED] Which page replacement algorithm gives the lowest possible fault rate but cannot be implemented?

- (a) LRU
- (b) FIFO
- (c) **Optimal (OPT)**
- (d) LFU

Why: Optimal evicts the page not needed for the longest time in the **future**, which requires knowledge no OS has. It is used only as a benchmark against which real algorithms are measured.

Q. [EXPECTED] [NUMERICAL] With 3 frames and the reference string 1, 2, 3, 4, 1, 2, 5 using FIFO, how many page faults occur?

- (a) 6
- (b) **7**
- (c) 8
- (d) 9

Why: Every one of the seven references faults: 1,2,3 fill the frames; 4 evicts 1; 1 evicts 2; 2 evicts 3; 5 evicts 4. Total **7 faults**. Work these row by row, writing the frame contents after each reference.

3.4 File Systems and Disk Scheduling

WEIGHTAGE: 2-3 marks | Disk scheduling trigger phrases are near-certain | Priority HIGH

3.4.1 Questions

Q. [PYQ] In an acyclic graph directory structure, what is a common method used to implement shared directories or files?

- (a) File duplication
- (b) **Symbolic (soft) links**
- (c) Hard links
- (d) Directory pointers

Why: Hard links to **directories** are forbidden on UNIX because they could create cycles, destroying the acyclic property. A **symbolic link** stores a path, can point to a directory and can cross file systems - so it is the standard sharing mechanism here.

Q. [EXPECTED] [TRAP] Which statement about hard and soft links is correct?

- (a) Both have their own inode
- (b) **A hard link shares the target's inode; a soft link has its own inode containing a path**
- (c) A soft link cannot cross file systems
- (d) Deleting the original breaks a hard link

Why: **Hard link:** same inode, cannot cross file systems, cannot link directories, and the file survives deletion of the original name. **Soft link:** own inode holding a path, crosses file systems, links directories, becomes a dangling link if the target is deleted.

Q. [EXPECTED] Which file allocation method suffers from external fragmentation?

- (a) Linked allocation
- (b) Indexed allocation
- (c) **Contiguous allocation**
- (d) FAT

Why: Contiguous allocation gives excellent sequential and direct access but the file cannot grow easily and free space becomes fragmented. **Linked** allocation avoids fragmentation but makes direct access slow; **indexed** allocation supports direct access via an index block.

Q. [PYQ] The default file system type in Linux is:

- (a) Ext3
- (b) Ext1
- (c) **Ext4**
- (d) Ext2

Why: **Ext4** adds journaling, extents, delayed allocation and support for very large files and volumes. Ext2 had no journaling.

Q. [PYQ] [TRAP] Which of the following is NOT true about NTFS?

- (a) It utilises disk space efficiently using smaller clusters.
- (b) NTFS attributes include read-only, hidden, system, archive, not content indexed, off-line, temporary and compressed.
- (c) It introduced improved performance and scalability compared with its precursor.
- (d) **It does not support the compression of files and directories to optimise storage space.**

Why: NTFS **does** support built-in file and folder compression, along with journaling, ACL-based file-level security, EFS encryption and disk quotas.

Q. [EXPECTED] Journaling in a file system primarily provides:

- (a) Faster read speed
- (b) **Crash consistency by logging intended changes before applying them**
- (c) File compression
- (d) Encryption

Why: The journal is written first, so after a crash it can be replayed and the file system is never left in an inconsistent

state.

Q. [PYQ] In which scheduling algorithm does the disk head move in one direction, satisfying all requests until it reaches the end, and then reverses direction?

- (a) **SCAN**
- (b) SSTF
- (c) LOOK
- (d) FCFS

Why: "Until it reaches the end" is the decisive phrase. Learn the three triggers: **SCAN** goes to the physical end then reverses; **LOOK** reverses at the last *request*; **C-SCAN** jumps back to the start without serving on the return.

Q. [PYQ] [TRAP] The advantage of SSTF (Shortest Seek Time First) is:

- (a) high variance of response time as SSTF favours only some requests
- (b) overhead to calculate seek time in advance
- (c) **average response time decreases**
- (d) throughput decreases

Why: Three options are **disadvantages** dressed up as advantages. SSTF actually *increases* throughput, and its real drawbacks are starvation of distant requests and high response-time variance.

Q. [EXPECTED] Which component of disk access time is usually the largest?

- (a) Rotational latency
- (b) **Seek time**
- (c) Transfer time
- (d) Controller overhead

Why: Moving the arm mechanically dominates. **Access time = seek time + rotational latency + transfer time**, and rotational latency averages half a revolution.

3.5 Linux and UNIX

WEIGHTAGE: 2-4 marks in DSSSB | EMRS also tests basic OS and ICT commands | Priority HIGH

3.5.1 Questions

Q. [PYQ] In which of the following directories are the configuration files found?

- (a) /dev
- (b) /root
- (c) **/etc/**
- (d) /bin

Why: **/etc** = configuration files and administration commands. **/bin** = essential command binaries. **/dev** = device files. **/root** = the superuser's home directory (not the same as /).

Q. [PYQ] [TRAP] Which of the following statements is NOT true?

- (a) etc: contains additional commands related to system maintenance and administration.
- (b) **usr: It contains temporary files created by Linux or other users.**
- (c) bin: It contains executable files for the UNIX commands.
- (d) dev: It contains device drivers that control various peripherals.

Why: Temporary files live in **/tmp**, not **/usr**. **/usr** holds user programs and read-only shareable data (**/usr/bin**, **/usr/lib**, **/usr/share**). **/var** holds variable data such as logs.

Q. [PYQ] Identify the command that is used to delete files in Linux.

- (a) **rm**
- (b) Rmdir
- (c) Delete
- (d) Rdir

Why: **rm** removes files; **rm -r** removes directories recursively; **rm -f** forces. **rmdir** deletes only **empty** directories. "Delete" and "Rdir" are not Linux commands at all.

Q. [PYQ] Grep command is used to:

- (a) change the case of alphabets
- (b) **search for a specific pattern from a specified file and display those lines containing the pattern**
- (c) cut the columns/fields of a specified file
- (d) redirect the output of one command to another command

Why: **GREP = Globally search for a Regular Expression and Print**. Knowing the full form makes the definition unforgettable. The distractors describe **tr**, **cut** and the pipe respectively.

Q. [EXPECTED] Which command displays the first 10 lines of a file by default?

- (a) tail
- (b) cat
- (c) **head**
- (d) less

Why: **head** shows the beginning, **tail** the end. **tail -f** follows a growing log file in real time - the standard way to watch server logs.

Q. [EXPECTED] Which command is used to change file permissions in Linux?

- (a) chown
- (b) **chmod**
- (c) chgrp
- (d) umask

Why: **chmod** changes **mode** (permissions); **chown** changes **owner**; **chgrp** changes **group**; **umask** sets the *default* permissions for newly created files.

Q. [PYQ] [NUMERICAL] Which of the following is a default permission for the text file in Linux?

- (a) **rwX r-- r--**
- (b) **rw- r-- r--**

- (c) `rwX rwX rwX`
- (d) `rwX --- ---`

Why: Files start from base **666** and the standard umask **022** removes group and other write permission, giving **644** = **rw-r--r--**. A plain data file is never executable by default. Directories start from 777 and become **755**.

Q. [EXPECTED] [NUMERICAL] What does `chmod 750 file.sh` set?

- (a) `rwX rwX r-X`
- (b) **`rwX r-X ---`**
- (c) `rw- r-X ---`
- (d) `rwX r-- r--`

Why: Read as three octal digits: 7 = `rwX` (owner), 5 = `r-X` (group), 0 = `---` (others). Weights are **r=4, w=2, x=1**.

Q. [PYQ] Which of the following is NOT a shell in Linux?

- (a) C shell
- (b) Korn shell
- (c) Bourne shell
- (d) **Alpha cell**

Why: No such shell exists - and note the deliberate misspelling "cell" for "shell". Real shells: **sh** (Bourne), **bash** (Bourne Again, the Linux default), **csh** (C shell), **tcsh**, **ksh** (Korn), **zsh**.

Q. [EXPECTED] Which symbol appends output to an existing file without overwriting it?

- (a) `>`
- (b) **`>>`**
- (c) `<`
- (d) pipe

Why: `>` overwrites, `>>` appends, `<` takes input from a file, and the pipe sends one command's output into another's input.

Q. [EXPECTED] Which command shows the currently running processes?

- (a) `ls`
- (b) **`ps`**
- (c) `pwd`
- (d) `du`

Why: `ps` reports processes (`ps -ef` shows all); `top/htop` give a live view; `kill -9 PID` force-terminates.

Q. [PYQ] Which of the following is NOT a benefit of distributed systems?

- (a) Scalability
- (b) **High communication latency**
- (c) Resource sharing
- (d) Fault tolerance

Why: Latency is the classic **drawback**, because messages must cross a network instead of staying inside one machine. Genuine benefits: resource sharing, scalability, fault tolerance, parallel speed-up, geographic distribution.

PART 4

PROGRAMMING: C, C++, OOP AND JAVA

Weightage: 9 to 15 marks in DSSSB, 8 to 16 expected. Output-tracing questions dominate. This is the one topic you cannot cram - practise daily.

4.1 C Language - Basics, Operators and Control Flow

WEIGHTAGE: 4-7 marks | "What is the output" questions are guaranteed | Priority CRITICAL

4.1.1 Topic checklist

- Data types and sizes, format specifiers, header files
- Operators and full precedence table, associativity
- if/else, switch rules, all three loops, break/continue/goto
- Functions, call by value vs reference, actual vs formal parameters
- Storage classes and their default values
- Arrays, strings, pointers, structures, unions, file handling

4.1.2 Questions

Q. [PYQ] scanf() is a predefined function in which of the following header files?

- (a) ctype.h
- (b) **stdio.h**
- (c) unistd.h
- (d) string.h

Why: Both printf and scanf are declared in **stdio.h**. Other essentials: **stdlib.h** (malloc, free, exit, atoi), **string.h** (strlen, strcpy, strcmp), **math.h** (sqrt, pow), **stdarg.h** (va_list for variable arguments).

Q. [PYQ] Which arithmetic operator has the highest precedence in C?

- (a) -
- (b) =
- (c) +
- (d) **%**

Why: The group `* / %` sits above `+ -`. All three of `*`, `/`, `%` share the same level, so if `*` were offered it would be equally valid. `=` has almost the lowest precedence of all operators.

Q. [EXPECTED] Which operators in C are right-to-left associative?

- (a) Arithmetic operators
- (b) Relational operators
- (c) **Unary, ternary and assignment operators**
- (d) Bitwise operators

Why: Only three groups associate right-to-left: **unary** (`!`, `~`, `++`, `--`, `sizeof`, type casts), the **ternary** `?:`, and **assignment** (`=`, `+=`). Everything else is left-to-right - which is exactly what makes `a > b > c` evaluate as `((a>b)>c)`.

Q. [PYQ] [TRAP] Output of: int a = 5, b = 3, c = 0; if (a > b > c) printf("True"); else printf("False");

- (a) compile time error
- (b) undefined behavior
- (c) **TRUE**
- (d) FALSE

Why: Relational operators are **left-to-right**, so this is `((a>b)>c)`. Step 1: `5 > 3` gives **1**. Step 2: `1 > 0` gives **1**, which is true. Prints **True**. Never read chained comparisons as mathematics.

Q. [PYQ] [TRAP] Output of: int x = -1, y = 0, z = 1; int result = (x > 0) ? x : (y == 0) ? z : y;

- (a) compile time error
- (b) -1
- (c) **1**
- (d) 0

Why: `?:` is **right-associative**, so it reads `(x>0) ? x : ((y==0) ? z : y)`. `x > 0` is false, so evaluate the inner ternary: `y == 0` is true, giving **z = 1**.

- Q. [EXPECTED] Output of:** `int a = 10, b = 20; int max = (a > b) ? a : b; printf("%d", max);`
- (a) 30
 - (b) compile time error
 - (c) 10
 - (d) 20

Why: `10 > 20` is false, so `b` is selected. The ternary is shorthand for a complete if-else expression.

- Q. [PYQ] [TRAP] Output of:** `int i = 5; do { printf("%d ", i); i--; } while (i > 5);`
- (a) No output printed
 - (b) Infinite loop
 - (c) 5 4 3 2 1
 - (d) 5

Why: A **do-while** executes its body at least once before testing. It prints 5, decrements to 4, then tests `4 > 5` which is false and exits. This is the single most-asked loop trap.

- Q. [PYQ] Which of the following causes an infinite loop?**

- (a) `for (int i = 10; i > 5; i--) { }`
- (b) `do { } while (0);`
- (c) **`while (1) { }`**
- (d) `for (int i = 0; i < 10; i++) { }`

Why: `while(1)` never terminates. The others run 5 times, exactly once, and 10 times respectively. Other infinite forms: `for(;;){}` and `do{}while(1);`.

- Q. [PYQ] Which of the following is true about the for loop in C?**

- (a) A for loop cannot be nested inside another for loop.
- (b) **The condition in a for loop is optional, and it can result in an infinite loop if omitted.**
- (c) A for loop must always have all three components.
- (d) The increment/decrement part of the for loop is mandatory.

Why: All three parts of a for loop are optional - `for(;;)` is perfectly legal. Nesting is also allowed, so the first option is false too.

- Q. [PYQ] [NUMERICAL] How many times will the innermost loop execute?** `for(i=0;i<3;i++)
for(j=0;j<2;j++) printf(...)`
- (a) **6 times**
 - (b) 3 times
 - (c) 5 times
 - (d) 2 times

Why: Multiply the iteration counts: $3 \times 2 = 6$. Printed output would be 0 0 1 1 2 2.

- Q. [PYQ] Which of the following will terminate the loop immediately in C?**

- (a) **When a break statement is encountered**
- (b) When the loop variable reaches a certain value
- (c) When a continue statement is encountered
- (d) When the loop condition is not given

Why: `break` exits the innermost loop or switch at once. `continue` only skips the remainder of the current iteration and jumps to the next test.

- Q. [PYQ] Which of the following is true about nested switch statements in C?**

- (a) Both inner and outer switch statements must have a default case.
- (b) A switch statement cannot contain another switch.
- (c) Inner switch cases should not contain break statement
- (d) **A switch statement can contain another switch statement inside any of its cases.**

Why: Nesting is allowed and each switch is independent - an inner `break` exits only the inner switch. `default` is always optional.

- Q. [EXPECTED] [TRAP] Which data type is NOT allowed in a switch expression?**

- (a) `int`
- (b) `char`

- (c) short
- (d) float

Why: A switch expression must be an **integral** type (int, char, short, long, enum). **Floating-point values and strings are illegal** because case labels must be exact constant integers.

Q. [EXPECTED] What happens if break is omitted from a case in a switch?

- (a) Compilation error
- (b) The switch exits
- (c) Execution falls through into the next case
- (d) The default case runs

Why: Fall-through is deliberate C behaviour, sometimes used intentionally to let several cases share code.

Q. [PYQ] [NUMERICAL] Output of: `int a = 4; int b = 3; result = a ^ b; printf("%d", result);`

- (a) 0
- (b) 3
- (c) 4
- (d) 7

Why: ^ is **bitwise XOR**, not exponentiation. 4 = 100, 3 = 011; XOR gives 111 = 7. (For powers you need `pow()` from `math.h`.)

Q. [EXPECTED] [NUMERICAL] What is the value of x after `int x = 5; x = x << 2;`?

- (a) 10
- (b) 20
- (c) 25
- (d) 7

Why: Left shift by n multiplies by 2^n : $5 \times 4 = 20$. Right shift divides. Compilers use shifts because they are far cheaper than multiplication.

Q. [PYQ] [NUMERICAL] Given `int a, b, c; a = b = c = 15; c = c + 10;` what is the value of c?

- (a) Undefined
- (b) 10
- (c) 25
- (d) 15

Why: Assignment is **right-to-left**, so c, b and a all become 15; then $c = 15 + 10 = 25$.

Q. [PYQ] [TRAP] Output of nested blocks: `outer int x=10, y=20; inner block declares int y = 40; then does x++; y++;` and after the inner block ends we print x and y.

- (a) x = 11, y = 41
- (b) x = 10, y = 41
- (c) x = 10, y = 20
- (d) x = 11, y = 20

Why: The inner y is a **new variable that shadows** the outer one and dies when the block ends, so the outer y is untouched at 20. x has no inner declaration, so `x++` modifies the outer x, giving 11.

Q. [PYQ] [NUMERICAL] Output of: `int a[5]; for(i=0;i<5;i++) a[i] = i*i - 2*i + 1; then printing the array.`

- (a) 1 0 2 4 9
- (b) 1 0 1 4 9
- (c) 1 1 2 4 9
- (d) 1 2 3 4 9

Why: Recognise the identity: $i*i - 2i + 1 = (i-1)^2$. For $i = 0..4$ that gives 1, 0, 1, 4, 9. Spotting the algebraic form is faster and less error-prone than substituting each value.

Q. [PYQ] [NUMERICAL] Output of: `char buffer[50]; int n = sprintf(buffer, "Hello, World!"); printf("%d %s", n, buffer);`

- (a) Prints '13 Hello'
- (b) Prints '12 Hello'
- (c) Prints '12 Hello, World!'

(d) Prints '13 Hello, World!'

Why: `printf` writes formatted text **into the buffer** (it prints nothing itself) and **returns the character count excluding the terminating '\0'**. "Hello, World!" has 13 characters including the comma and space.

Q. [PYQ] Which of the following techniques can be used to modify the actual variable passed to a function?

- (a) Use a local variable instead of passing any parameters.
- (b) Pass a copy of the actual parameter to function.
- (c) Standard C has no options for modifying actual variables passed to a function.
- (d) Pass the address of the actual parameters to function.**

Why: C is call-by-value by default, so a copy cannot affect the original. Passing the **address** (a pointer) lets the function write through to the caller's variable.

Q. [PYQ] Which of the following options is correct about the functions in C language?

- (a) Actual parameters are declared in the function definition.
- (b) Actual parameters are always variables.
- (c) Actual parameters are used to define the function signature.
- (d) Actual parameters are the values that are passed to a function when it is called.**

Why: **Actual parameters** are the values supplied at the call site (and may be literals or expressions, e.g. `add(3, x+2)`). **Formal parameters** are the variables declared in the definition that receive them.

Q. [PYQ] [TRAP] Which of the following statements is INCORRECT about the storage classes in C?

- (a) The scope of a variable in automatic storage class is local to the block in which it is defined.
- (b) In Register storage class, the default initial value of the variable is zero.**
- (c) In static storage class, the default initial value of the variable is zero.
- (d) In Register storage class, the storage of a variable is CPU register.

Why: register variables, like `auto`, start with **garbage**. Only **static** and **extern** are guaranteed to be zero-initialised. Also note you cannot take the address of a register variable with `&`.

Q. [EXPECTED] A static local variable inside a function:

- (a) is destroyed when the function returns
- (b) retains its value between function calls**
- (c) is visible to other files
- (d) is stored on the stack

Why: `static` gives the variable **program lifetime** but keeps its **block scope**. It lives in the data segment, initialised to 0, and is why returning a pointer to a static array is safe.

Q. [PYQ] [TRAP] [NUMERICAL] A function returns a pointer to a static `int arr[5] = {1,2,3,4,5};` and main loops five times printing `*ptr` without ever incrementing `ptr`. Output?

- (a) 1 1 3 4 1
- (b) 2 3 4 5 0
- (c) 1 2 3 4 5
- (d) 1 1 1 1 1**

Why: `ptr` is **never advanced**, so `*ptr` dereferences the first element every time; the loop variable only counts. To print all five you would need `*(ptr+i)` or `ptr++`. Note `static` is what makes returning the array's address legal.

Q. [PYQ] Which of the following is NOT a correct way of declaring and initializing array in C?

- (a) `char arr[6] = {2};`
- (b) `char arr[ ] = "consultancy";`
- (c) `char arr[6]; arr = "consultancy";`**
- (d) `char arr[6] = { };`

Why: An **array name is not a modifiable lvalue**, so it cannot be assigned after declaration - you must use `strcpy`. Independently, "consultancy" needs 12 bytes and `arr` has only 6.

Q. [PYQ] What do the first and second dimensions represent in a two-dimensional character array?

- (a) The first stores the length of each name, the second the number of names.
- (b) The first represents the number of names, and the second the maximum length of each name.**
- (c) The first is the maximum length, the second the number of names.

(d) The first holds the total characters, the second the number of names.

Why: `char names[5][20] = 5 names, each up to 20 characters`. Think "how many, then how long".

Q. [EXPECTED] [NUMERICAL] How many bytes does the string "HELLO" occupy in C?

- (a) 5
- (b) 6
- (c) 7
- (d) 10

Why: A C string is terminated by the null character `'\0'`, so 5 characters need **6 bytes**. `strlen("HELLO")` returns 5, but `sizeof("HELLO")` returns 6 - a favourite distinction.

Q. [EXPECTED] [TRAP] Which is the correct way to compare two strings in C?

- (a) `if (s1 == s2)`
- (b) `if (strcmp(s1, s2) == 0)`
- (c) `if (s1.equals(s2))`
- (d) `if (s1 = s2)`

Why: `==` on char arrays compares **addresses**, not contents. `strcmp` returns **0 when equal**, negative if `s1 < s2` and positive if `s1 > s2`. (`.equals()` is Java, not C.)

Q. [EXPECTED] `p+1` on a pointer `int *p` advances the address by:

- (a) 1 byte
- (b) 2 bytes
- (c) `sizeof(int)` bytes
- (d) 8 bytes

Why: Pointer arithmetic is scaled by the **pointed-to type's size**, which is why `a[i]` is exactly equivalent to `*(a + i)`.

Q. [EXPECTED] [TRAP] Which function allocates memory and initialises it to zero?

- (a) `malloc()`
- (b) `calloc()`
- (c) `realloc()`
- (d) `free()`

Why: `calloc(count, size)` zeroes the memory; `malloc(n)` leaves it with garbage. `realloc` resizes; `free` releases. Forgetting `free` causes a **memory leak**.

Q. [EXPECTED] [TRAP] What is the difference between a structure and a union?

- (a) They are identical
- (b) **In a structure each member has its own memory; in a union all members share the same memory**
- (c) A union can hold more members
- (d) Structures cannot be nested

Why: Structure size = sum of members (plus padding); **union size = size of the largest member**, and only one member is valid at a time. Unions save memory when only one field is needed.

Q. [EXPECTED] Which file mode erases the existing contents of a file?

- (a) `"r"`
- (b) `"w"`
- (c) `"a"`
- (d) `"r+"`

Why: `"w"` truncates or creates. `"a"` appends without destroying. `"r"` requires the file to exist and cannot write.

4.2 C++ and Object Oriented Programming

WEIGHTAGE: 2-4 marks | OOP pillar definitions are near-certain | Priority CRITICAL

4.2.1 Questions

Q. [PYQ] What is the objective of encapsulation in object-oriented programming?

- (a) To create multiple instances of a class
- (b) To enable one class to inherit from another class
- (c) To allow objects to take multiple forms
- (d) **To combine data and methods that operate on that data within a single unit**

Why: The distractors each define a *different* concept - instantiation, inheritance and polymorphism. Learn one crisp phrase per pillar so you can tell them apart instantly.

Q. [EXPECTED] [TRAP] Which OOP concept hides complexity and shows only essential features?

- (a) Encapsulation
- (b) **Abstraction**
- (c) Inheritance
- (d) Polymorphism

Why: **Abstraction** hides **COMPLEXITY** (design level - "what it does"); **encapsulation** hides **DATA** (implementation level - "how it is stored"). A car's steering wheel is abstraction; the sealed engine block is encapsulation.

Q. [PYQ] [TRAP] What feature of OOP allows an instance of a class to take on many forms?

- (a) Operator overriding
- (b) Multiple inheritance
- (c) Nested class
- (d) **Operator overloading**

Why: The concept is **polymorphism**; since that word was not offered, the mechanism implementing it is **operator overloading**. Note that "operator overriding" is not a real term - operators are *overloaded*, methods are *overridden*.

Q. [EXPECTED] [TRAP] Function overloading is resolved at:

- (a) run time
- (b) **compile time**
- (c) link time
- (d) load time

Why: **Overloading** = same name, different parameter lists = **compile-time (static / early) binding**. **Overriding** = same signature in a subclass = **run-time (dynamic / late) binding**, achieved through virtual functions.

Q. [EXPECTED] A class that cannot be instantiated is called:

- (a) Final class
- (b) Static class
- (c) **Abstract class**
- (d) Nested class

Why: In C++ a class containing a **pure virtual function** (`virtual void f() = 0;`) becomes abstract. It exists only to be inherited from and may still contain concrete methods.

Q. [EXPECTED] Which of the following is NOT supported for classes in Java but is supported in C++?

- (a) Single inheritance
- (b) Encapsulation
- (c) **Multiple inheritance**
- (d) Polymorphism

Why: Java forbids multiple inheritance of **classes** to avoid the **Diamond Problem** ambiguity, providing **interfaces** instead. C++ allows it directly. Java also has no operator overloading.

Q. [EXPECTED] A constructor differs from an ordinary method in that it:

- (a) can be called explicitly any number of times
- (b) **has the same name as the class and no return type**
- (c) must return void
- (d) cannot be overloaded

Why: Constructors run automatically at object creation, share the class name, and have **no return type at all** (not even void). They **can** be overloaded. Destructors, by contrast, cannot be overloaded.

Q. [PYQ] What does the code snippet `myObject.doActivity()`; represent?

- (a) **Calling a method named doActivity on the object myObject**
- (b) Sending a message to the doActivity class
- (c) Creating an object myObject
- (d) Sending a message to the child class of myObject

Why: doActivity is a **method**, not a class, and object creation requires new. This is the standard OOP notion of **message passing** to an object.

Q. [EXPECTED] [TRAP] Which relationship describes "a Car HAS-A Engine"?

- (a) Inheritance
- (b) **Composition / Aggregation**
- (c) Polymorphism
- (d) Abstraction

Why: HAS-A = **composition/aggregation**; IS-A = **inheritance**. Composition is a strong part-of relation (destroy the house and the rooms go too); aggregation is weaker (a department has professors who exist independently).

Q. [PYQ] In C++ language, the instruction `ignore(x, y);` ____.

- (a) skips the last x characters and stops if character y is encountered
- (b) skips the first y characters and stops if character x is encountered
- (c) **skips the first x characters and stops if character y is encountered**
- (d) skips the last y characters and stops if character x is encountered

Why: `cin.ignore(count, delimiter)` discards characters until it has removed **count** of them or has consumed the **delimiter**, whichever comes first. Its everyday use is `cin.ignore(1000, '\n')` to clear a leftover newline before `getline()`.

Q. [EXPECTED] Which C++ feature allows a non-member function to access private members of a class?

- (a) Inline function
- (b) Virtual function
- (c) **Friend function**
- (d) Static function

Why: A friend declaration grants specific external functions or classes access to private and protected members. It deliberately breaks encapsulation, so it should be used sparingly.

4.3 Java

WEIGHTAGE: 3-5 marks | Exception handling and access modifiers recur | Priority CRITICAL

4.3.1 Questions

Q. [EXPECTED] [TRAP] Java achieves platform independence through:

- (a) Machine code
- (b) **Bytecode executed by the JVM**
- (c) Assembly language
- (d) Direct hardware compilation

Why: javac produces **bytecode** (a .class file), and any platform's **JVM** runs that same bytecode - "Write Once, Run Anywhere". **JDK** = development kit (includes compiler), **JRE** = runtime only, **JVM** = the executing engine.

Q. [PYQ] _____ does NOT have any child ClassLoaders.

- (a) Bootstrap ClassLoader
- (b) Extension ClassLoader
- (c) Primordial ClassLoader
- (d) **System ClassLoader**

Why: The delegation hierarchy is **Bootstrap to Extension to System**, so the **System (Application) ClassLoader** sits at the bottom with nothing beneath it. Note the trap: "Primordial" is just another name for Bootstrap, which **does** have a child.

Q. [PYQ] What does the super keyword in Java do?

- (a) Refers to high performing objects
- (b) Refers to the current object
- (c) Used to define static variables
- (d) **Refers to the parent class**

Why: super accesses the **parent** class (super.method(), super() for the parent constructor); this refers to the **current object**. Do not swap them.

Q. [PYQ] [TRAP] Which of the following statements is FALSE?

- (a) **Final methods can be overridden.**
- (b) "static public void main(String...abcd){}" is a valid statement.
- (c) Abstract classes cannot be instantiated.
- (d) Interface allows default method definitions.

Why: A **final method** is by definition one that cannot be overridden. The others are all true: modifier order is free and varargs String... is legal for main; abstract classes cannot be instantiated; and since **Java 8** interfaces may contain default methods with bodies.

Q. [EXPECTED] Applying final to a class means:

- (a) its methods cannot be called
- (b) **it cannot be inherited from**
- (c) its variables become static
- (d) it cannot be instantiated

Why: **final class** = no subclass (e.g. String). **final method** = cannot be overridden. **final variable** = a constant. A class that cannot be **instantiated** is abstract, not final.

Q. [PYQ] Which access modifier can be used to access members in a deriving class?

- (a) Void
- (b) **Protected**
- (c) Private
- (d) Public

Why: protected is designed precisely for inheritance - visible in the same package **and** in subclasses in other packages. private blocks subclasses entirely, and void is a return type, not an access modifier.

Q. [PYQ] Identify the descending order of strict access protection.

- (a) Public, protected, private, package
- (b) **Private, protected, package, public**
- (c) Public, private, protected, package
- (d) Private, package, protected, public

Why: Look for the option beginning with **Private** and ending with **public**. Strictly, Java's order is private > default(package) > protected > public, because protected additionally admits subclasses in other packages - but this paper's key placed protected before package, so use the start/end test.

Q. [PYQ] A Parent class has a protected void protectedMethod(); Child extends Parent and calls it from a public method; main creates a Child and calls that method. What happens?

- (a) **The code will compile and print the message.**
- (b) Compilation Error: protectedMethod is not visible in the Child class.
- (c) Compilation Error: childMethod is not visible due to access restrictions.
- (d) Runtime Error: Cannot access protectedMethod from the Child class.

Why: A protected member is **fully accessible inside a subclass**, and childMethod is public so main may call it. Nothing here violates any access rule.

Q. [PYQ] Select the true statement about try-catch blocks.

- (a) A try block must have at least 1 catch to have a finally block.
- (b) A try block must have only 1 finally block for each catch block.
- (c) **A try block can have many catches but only 1 finally block.**
- (d) A try block can have many catches and many finally blocks.

Why: Many catch blocks, exactly **one** finally. A try may also have finally with **no** catch at all. Catch blocks must be ordered most-specific first, or the code will not compile.

Q. [EXPECTED] [TRAP] When does a finally block NOT execute?

- (a) When an exception is thrown
- (b) When the try block returns a value
- (c) When no exception occurs
- (d) **Practically never - only if the JVM exits via System.exit() or crashes**

Why: finally runs even when the try block executes return or throws. That reliability is why it is used for closing files and connections.

Q. [PYQ] An ExceptionTest calls calculate(10, 0). The inner catch catches ArithmeticException and throws new Exception("Division by zero not allowed"); both levels have finally blocks.

Output?

- (a) Inner finally block executed. General exception caught: / by zero In finally block.
- (b) **Inner finally executed / General exception caught: Division by zero not allowed / In finally block.**
- (c) Inner finally executed. Result: 0 In finally block.
- (d) Inner finally executed. ArithmeticException caught: / by zero In finally block.

Why: Two rules combine. (1) The **inner finally runs before control leaves** the method, even though an exception is propagating. (2) The re-thrown object's type is now plain **Exception**, so the catch (ArithmeticException) block does **not** match and the general catch (Exception) does - carrying the **new** message, not "/ by zero".

Q. [PYQ] A CustomException extends Exception; validateAge(15) throws it because 15 < 18; main catches it and has a finally block. Output?

- (a) Caught Exception: null / Execution finished.
- (b) Compilation error because custom exceptions cannot extend Exception.
- (c) Age is valid. / Execution finished.
- (d) **Caught Exception: Age must be 18 or above. / Execution finished.**

Why: The exception is thrown, so "Age is valid." never prints. super (message) passes the text up to Throwable, so getMessage() returns it rather than null. finally always runs.

Q. [EXPECTED] [TRAP] Which of the following is a checked exception?

- (a) NullPointerException
- (b) ArrayIndexOutOfBoundsException
- (c) **IOException**
- (d) ArithmeticException

Why: **Checked** exceptions are detected at compile time and must be caught or declared: IOException, SQLException, ClassNotFoundException. The other three are **unchecked** (subclasses of RuntimeException).

- Q. [PYQ] [TRAP] Output of:** `int Integer = 24; char String = 'I';
System.out.print(Integer); System.out.print(String);`
- (a) I
 - (b) Compiler Error
 - (c) Throws Exception
 - (d) **24I**

Why: Integer and String are **class names, not reserved keywords**, so they are legal identifiers. `print()` adds no newline, so the outputs run together as **24I**.

- Q. [PYQ] Which of the following is FALSE about packages?**

- (a) **Packages cannot provide controlled access.**
- (b) Packages are used to prevent naming conflicts.
- (c) Packages are considered data encapsulation.
- (d) Packages prevent naming conflicts and are considered data encapsulation.

Why: Packages **do** provide controlled access, because default (package-private) and protected members are visible only within the package.

- Q. [PYQ] What is the name of the .NET collection class that enables an element to be accessed using a unique key?**

- (a) Linked list
- (b) **Hashtable**
- (c) Double linked list
- (d) Array list

Why: A **Hashtable** stores key-value pairs and retrieves values directly from the key by hashing, giving average **O(1)** access. Lists and arrays are accessed by position or traversal.

- Q. [EXPECTED] [TRAP] Which Java collection allows duplicate elements and maintains insertion order?**

- (a) HashSet
- (b) TreeSet
- (c) **ArrayList**
- (d) HashMap

Why: **List** implementations (ArrayList, LinkedList, Vector) are ordered and allow duplicates. **Set** forbids duplicates. **Map** stores key-value pairs and is not a Collection subinterface.

- Q. [EXPECTED] [TRAP] Which of the following is thread-safe (synchronised)?**

- (a) ArrayList
- (b) HashMap
- (c) **Vector**
- (d) LinkedList

Why: **Vector** and **Hashtable** are the legacy synchronised classes. ArrayList and HashMap are not synchronised and are therefore faster. Also: **HashMap allows one null key**, whereas **Hashtable allows none**.

4.4 .NET Framework

WEIGHTAGE: 1-3 marks in DSSSB | Appeared in all three papers supplied | Priority MEDIUM

4.4.1 Questions

Q. [PYQ] Which of the following is the execution engine for .NET applications and servers, acting as the interface between .NET applications and the operating system?

- (a) Framework Class Library (FCL)
- (b) Common Type System (CTS)
- (c) Common Language Specifications (CLS)
- (d) **Common Language Runtime (CLR)**

Why: The **CLR** executes managed code and provides garbage collection, memory management, security, exception handling and thread management. FCL is a class library; CTS defines types; CLS defines cross-language rules. None of those *execute* anything.

Q. [PYQ] ___ is a code that consists of CPU and platform-independent set of instructions, which can be easily converted to the native code.

- (a) DLL
- (b) FCL
- (c) JIT
- (d) **MSIL**

Why: **MSIL** (Microsoft Intermediate Language, also CIL/IL) is the intermediate output of a .NET compiler; the **JIT** compiler is what *converts* it to native code at run time. A **DLL** is a compiled file; **FCL** is a library.

Q. [PYQ] Which of the following is NOT a feature of ADO.net?

- (a) Maintainability
- (b) Interoperability
- (c) **Reachability**
- (d) Scalability

Why: ADO.NET's stated features are **interoperability, maintainability, programmability, performance and scalability**. "Reachability" belongs to graph theory and garbage collection.

Q. [PYQ] How to write the dot net server control?

- (a) `<asp:controlType ID="ControlID" runin="server" ...>`
- (b) `<asp:controlType ID="ControlID" runas="server" ...>`
- (c) `<asp:controlType ID="ControlID" run="server" ...>`
- (d) **`<asp:controlType ID="ControlID" runat="server" ...>`**

Why: Every ASP.NET server control requires `runat="server"` - that attribute is what makes the framework process the tag on the server. Memorise the exact spelling: r-u-n-a-t. The distractors are runin, runas and run.

Q. [EXPECTED] Which ADO.NET object provides fast, forward-only, read-only access to query results?

- (a) DataSet
- (b) DataAdapter
- (c) **DataReader**
- (d) Connection

Why: **DataReader** streams rows while connected - fastest for read-only traversal. **DataSet** is a disconnected in-memory copy that can be modified and reconciled later via a **DataAdapter**.

PART 5

DATA STRUCTURES AND ALGORITHMS

Weightage: 6 to 9 marks in DSSSB, 6 to 12 expected. Almost pure recall once the complexity tables are memorised.

5.1 Complexity Analysis

WEIGHTAGE: 2-3 marks | Complexity of a named algorithm is asked in every paper | Priority CRITICAL

5.1.1 Questions

Q. [PYQ] Which of the following asymptotic notation gives asymptotically tight upper bound as well as asymptotically tight lower bound?

- (a) Omega
- (b) Big Oh
- (c) **Theta**
- (d) Small Omega

Why: The words "as well as" demand both bounds simultaneously - that is **Theta**. Big Oh gives only the upper bound, Omega only the lower, and small omega is not even tight.

Q. [PYQ] Which of the following cases DOES NOT exist when one computes the time complexity of an algorithm?

- (a) Average case
- (b) Worst case
- (c) **Test Case**
- (d) Best Case

Why: Only **best, average and worst** exist. "Test case" belongs to software testing.

Q. [PYQ] Arrange the following in the order of growth: $O(n)$, $O(n^2)$, $O(2^n)$, $O(\log n)$, $O(n \log n)$, $O(n^2 \log n)$

- (a) $O(n)$, $O(n^2)$, $O(\log n)$, $O(2^n)$, $O(n \log n)$, $O(n^2 \log n)$
- (b) $O(n)$, $O(2^n)$, $O(\log n)$, $O(n^2 \log n)$, $O(n \log n)$
- (c) **$O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, $O(n^2 \log n)$, $O(2^n)$**
- (d) $O(n^2)$, $O(n)$, $O(2^n)$, $O(\log n)$, $O(n \log n)$, $O(n^2 \log n)$

Why: Substitute $n = 1024$ (so $\log n = 10$) and compare: $10 < 1024 < 10,240 < 1,048,576 < 10,485,760 < 2^{1024}$. **Rule:** polynomials always beat exponentials, and a log factor is weaker than one extra power of n .

Q. [EXPECTED] Which of the following grows fastest for large n ?

- (a) n^3
- (b) $n^2 \log n$
- (c) **2^n**
- (d) $n \log n$

Why: Exponential growth eventually dominates every polynomial, no matter how large the exponent. Full chain: **$O(1) < O(\log n) < O(\sqrt{n}) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!) < O(n^n)$** .

Q. [PYQ] [TRAP] Which statement is correct about the divide and conquer approach of binary search?

- (a) The recurrence relation of binary search is $T(n) = 2T(n/2) + O(1)$.
- (b) **The worst-case time complexity of binary search is $O(\log n)$.**
- (c) The best-case time complexity of binary search is $O(\log n)$.
- (d) The average-case time complexity of binary search is $O(n)$.

Why: Binary search discards half and searches **only one** half, so the recurrence is **$T(n) = T(n/2) + O(1)$** - the coefficient is 1, not 2. Its **best case is $O(1)$** (target is the middle element immediately), average and worst are $O(\log n)$.

Q. [EXPECTED] The recurrence $T(n) = 2T(n/2) + O(n)$ solves to:

- (a) $O(n)$
- (b) $O(\log n)$
- (c) **$O(n \log n)$**
- (d) $O(n^2)$

Why: This is the **merge sort** recurrence. Learn the standard set: $T(n)=T(n/2)+O(1)$ gives $O(\log n)$ (binary search); $T(n)=2T(n/2)+O(1)$ gives $O(n)$; $T(n)=T(n-1)+O(n)$ gives $O(n^2)$; $T(n)=2T(n-1)+O(1)$ gives $O(2^n)$ (Tower of Hanoi).

Q. [EXPECTED] A precondition for applying binary search is that the array must be:

- (a) of even length
- (b) **sorted**
- (c) of numeric type
- (d) stored in a linked list

Why: Binary search relies on discarding half based on an ordering comparison, so the data **must already be sorted**. This is also why binary search is impractical on linked lists - no random access to the middle.

5.2 Linear Data Structures

WEIGHTAGE: 2-4 marks | Linked-list properties are a repeat favourite | Priority CRITICAL

5.2.1 Questions

Q. [PYQ] What are the data structures categorised as a linear list?

- (a) Binary tree
- (b) Graphs
- (c) Circular queue
- (d) **Array**

Why: A **linear list** means a strict sequence where each element has one predecessor and one successor. Trees and graphs are non-linear; a circular queue links its last element back to the first, breaking the plain sequence.

Q. [PYQ] In a singly linked list, each node of the list consists of ____.

- (a) **data value and address of next node**
- (b) NULL and address of the next node
- (c) NULL and address of the previous node
- (d) data value and address of previous node

Why: Only the **last** node holds NULL. An address of the *previous* node indicates a **doubly** linked list.

Q. [PYQ] [TRAP] Which of the following is a disadvantage of singly linked list?

- (a) **Accessing an element requires traversal from the head, making it slower than arrays.**
- (b) Deletion at the beginning is performed in $O(1)$ time.
- (c) Insertion at the beginning is performed in $O(1)$ time.
- (d) Linked lists have a variable size.

Why: Three options are genuine **advantages** disguised as answers. The real weakness is **no random access**: reaching the i -th element costs $O(n)$, whereas an array does it in $O(1)$.

Q. [PYQ] [NUMERICAL] Calculate the time complexity of deleting an element from the beginning of a dynamic array.

- (a) $O(\log)$
- (b) $O(1)$
- (c) $O(\log n)$
- (d) **$O(n)$**

Why: After removing $a[0]$, all remaining $n-1$ elements must **shift left** to preserve contiguity. Only deletion at the **end** is $O(1)$.

Q. [EXPECTED] [NUMERICAL] What is the time complexity of inserting at the beginning of a singly linked list?

- (a) $O(n)$
- (b) **$O(1)$**
- (c) $O(\log n)$
- (d) $O(n^2)$

Why: Just create the node and re-point the head - no shifting. This is exactly where linked lists beat arrays.

Q. [EXPECTED] A stack follows which order?

- (a) FIFO
- (b) **LIFO**
- (c) Priority order
- (d) Random

Why: **LIFO** - Last In First Out, like a pile of plates. push, pop and peek are all **$O(1)$** .

Q. [EXPECTED] [TRAP] Which of the following is NOT an application of a stack?

- (a) Recursion / function call management
- (b) Infix to postfix conversion
- (c) Balanced parentheses checking

(d) CPU scheduling ready queue

Why: The ready queue is FIFO, so it uses a **queue**. Stack applications: function call stack, recursion, expression conversion and evaluation, parenthesis matching, undo/redo, browser back button, **DFS**, backtracking, string reversal.

Q. [EXPECTED] Which traversal technique uses a queue?

- (a) DFS
- (b) BFS**
- (c) Inorder traversal
- (d) Postorder traversal

Why: **BFS uses a QUEUE** and explores level by level; **DFS uses a STACK** (or recursion) and dives deep first. Both are $O(V + E)$.

Q. [EXPECTED] [TRAP] What problem does a circular queue solve?

- (a) Overflow
- (b) Wastage of space at the front after repeated deletions**
- (c) Underflow
- (d) Slow enqueue

Why: In a simple linear queue, once front advances, the vacated cells at the start cannot be reused and the queue appears full while space remains. A **circular queue** wraps rear around to index 0.

Q. [EXPECTED] In a priority queue, the element dequeued first is:

- (a) the one inserted first
- (b) the one inserted last
- (c) the one with the highest priority**
- (d) a random element

Why: Arrival order is irrelevant. Priority queues are normally implemented with a **heap**, giving $O(\log n)$ insert and $O(\log n)$ extract.

Q. [EXPECTED] Which notation requires no parentheses and no precedence rules for evaluation?

- (a) Infix
- (b) Postfix**
- (c) Both infix and prefix
- (d) None

Why: **Postfix (Reverse Polish)** can be evaluated in one left-to-right pass with a single stack, which is why compilers convert infix expressions to postfix.

5.3 Trees and Graphs

WEIGHTAGE: 2-3 marks | Traversal order and BST properties recur | Priority HIGH

5.3.1 Questions

Q. [EXPECTED] [TRAP] Which traversal of a Binary Search Tree yields elements in sorted ascending order?

- (a) Preorder
- (b) **Inorder**
- (c) Postorder
- (d) Level order

Why: Inorder (Left, Root, Right) on a BST always produces sorted output - a very frequently asked property. Memory hook: the word tells you where the ROOT goes - **Pre**order root first, **In**order root in between, **Post**order root last. Left always precedes right.

Q. [EXPECTED] [NUMERICAL] For the tree with root A, left child B, right child C, and B having children D and E, the preorder traversal is:

- (a) D B E A C
- (b) **A B D E C**
- (c) D E B C A
- (d) A B C D E

Why: Preorder = Root, Left subtree, Right subtree: A, then B's subtree (B, D, E), then C = **A B D E C**. (Inorder would be D B E A C; postorder D E B C A; level order A B C D E.)

Q. [EXPECTED] [NUMERICAL] The maximum number of nodes in a binary tree of height 3 (root at level 0) is:

- (a) 8
- (b) **15**
- (c) 16
- (d) 7

Why: Maximum nodes = $2^{(h+1)} - 1 = 2^4 - 1 = 15$. Maximum nodes at a single level l is 2^l . A tree with n nodes always has exactly $n - 1$ edges.

Q. [EXPECTED] [TRAP] The worst-case time complexity of searching in a Binary Search Tree is:

- (a) $O(\log n)$ always
- (b) **$O(n)$ when the tree is skewed**
- (c) $O(1)$
- (d) $O(n \log n)$

Why: Inserting keys in **sorted order** degenerates a BST into a linked list, making search $O(n)$. **Self-balancing trees** fix this: AVL and Red-Black guarantee $O(\log n)$.

Q. [EXPECTED] In an AVL tree, the balance factor of every node must be:

- (a) 0 only
- (b) **-1, 0 or +1**
- (c) between -2 and +2
- (d) equal for all nodes

Why: Balance factor = height(left) - height(right). Violations are repaired by **rotations** (LL, RR, LR, RL). AVL is more strictly balanced than Red-Black, so lookups are faster but insertions cost more rotations.

Q. [EXPECTED] [TRAP] Which tree structure is standard for database indexing?

- (a) AVL tree
- (b) Binary search tree
- (c) **B+ tree**
- (d) Heap

Why: B+ trees keep all data in linked leaf nodes, which makes **range queries** efficient, and their high branching factor minimises **disk reads** - the dominant cost in databases. B-trees and B+ trees are designed around disk block

size.

Q. [EXPECTED] In a max heap, the largest element is located:

- (a) at the last leaf
- (b) **at the root**
- (c) at the middle
- (d) anywhere

Why: Max heap: every parent is at least as large as its children, so the maximum sits at the root and can be read in **O(1)**. Insert and delete-root are $O(\log n)$; building a heap from n elements is **$O(n)$** .

Q. [EXPECTED] [NUMERICAL] For a node at index i in a 0-based array heap, the left child is at:

- (a) $2i$
- (b) **$2i + 1$**
- (c) $2i + 2$
- (d) $i/2$

Why: Left = $2i+1$, right = $2i+2$, parent = $(i-1)/2$. (In 1-based indexing it is $2i$ and $2i+1$ with parent $i/2$.)

Q. [PYQ] [NUMERICAL] What are the articulation points in the given graph (vertices 1-10, with 9 and 10 hanging off 3, and 6 hanging off 5)?

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 3, 4 and 5
- (d) **2, 3 and 5**

Why: Remove each vertex and check whether the graph splits. Removing **3** isolates 9 and 10; removing **2** separates the left cluster from the right; removing **5** isolates 6. Vertices 1 and 4 leave the graph connected. **A leaf can never be an articulation point**, and any vertex that is the sole gateway to a region always is.

Q. [PYQ] [NUMERICAL] In a path-graph having n vertices where $n > 2$, ___ is the number of vertices with degree 2.

- (a) 3
- (b) **$n - 2$**
- (c) 2
- (d) 0

Why: A path is a straight line $v_1-v_2-\dots-v_n$. The **two end vertices have degree 1**; every one of the remaining $n - 2$ middle vertices has degree 2. Check with $n=5$: degrees are 1,2,2,2,1, so three vertices of degree 2 = $5-2$.

Q. [EXPECTED] [NUMERICAL] How many edges does a complete graph with 6 vertices have?

- (a) 30
- (b) **15**
- (c) 12
- (d) 36

Why: $n(n-1)/2 = 6 \times 5 / 2 = 15$ for an undirected complete graph. The same formula gives the number of links in a full mesh network topology.

Q. [EXPECTED] [TRAP] Which algorithm finds the shortest path in a graph with **NEGATIVE** edge weights?

- (a) Dijkstra
- (b) Prim
- (c) **Bellman-Ford**
- (d) Kruskal

Why: **Dijkstra fails with negative weights** because its greedy choice becomes invalid. **Bellman-Ford** handles them and additionally detects negative cycles, at $O(VE)$. Prim and Kruskal build minimum spanning trees, not shortest paths.

Q. [EXPECTED] A minimum spanning tree of a graph with V vertices has exactly:

- (a) V edges
- (b) **$V - 1$ edges**
- (c) $V + 1$ edges
- (d) $E - V$ edges

Why: Any spanning tree on V vertices is a tree, so it has **$V - 1$** edges and no cycles. Prim and Kruskal are the two

standard greedy algorithms.

Q. [EXPECTED] Which representation is preferable for a sparse graph?

- (a) Adjacency matrix
- (b) **Adjacency list**
- (c) Incidence matrix
- (d) Edge weight matrix

Why: Adjacency **list** costs $O(V + E)$ space and lists a vertex's neighbours quickly. Adjacency **matrix** costs $O(V^2)$ regardless of edge count but answers "is there an edge $u-v$?" in $O(1)$ - better for dense graphs.

5.4 Searching, Sorting and Design Techniques

WEIGHTAGE: 2-4 marks | Sorting complexity is the most repeated fact in this topic | Priority CRITICAL

5.4.1 Questions

Q. [PYQ] What is the best case and worst-case complexity of selection sort?

- (a) $O(n \log n)$
- (b) $O(n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

Why: Selection sort always scans the whole unsorted portion to find the minimum, so the comparison count $n(n-1)/2$ is **independent of the input order**. Contrast with bubble and insertion sort, which do have an $O(n)$ best case on already-sorted data.

Q. [PYQ] What is the average case time complexity of merge sort?

- (a) $O(n \log n)$
- (b) $O(n)$
- (c) $O(1)$
- (d) $O(n^2 \log n)$

Why: Merge sort always splits exactly in half, so **best = average = worst = $O(n \log n)$** . Its one weakness is needing $O(n)$ **extra space**, so it is not in-place.

Q. [EXPECTED] [TRAP] Which sorting algorithm has the worst case $O(n^2)$ but is usually fastest in practice for arrays?

- (a) Merge sort
- (b) Heap sort
- (c) **Quick sort**
- (d) Bubble sort

Why: **Quick sort** degrades to $O(n^2)$ when the pivot is always the smallest or largest element (e.g. sorted input with a first-element pivot), but its low constant factors and cache friendliness make it the practical winner. Cure the worst case with a random or median-of-three pivot.

Q. [PYQ] [TRAP] Which sorting technique is best for sorting a linked list with n elements?

- (a) Insertion sort
- (b) Bubble sort
- (c) **Merge sort**
- (d) Selection sort

Why: Quick sort needs **random access** to reach the pivot, which a linked list cannot provide. Merge sort only ever traverses sequentially and relinks pointers - and on a linked list it needs **no extra array**, so its auxiliary space drops to $O(1)$. It is also stable.

Q. [EXPECTED] Which of the following sorting algorithms is NOT stable?

- (a) Insertion sort
- (b) Merge sort
- (c) Bubble sort
- (d) **Quick sort**

Why: **Stable** (equal elements keep relative order): bubble, insertion, merge, counting, radix. **Unstable:** selection, quick, heap.

Q. [EXPECTED] Which comparison-based sorting algorithm has $O(n \log n)$ worst case AND $O(1)$ auxiliary space?

- (a) Merge sort
- (b) Quick sort
- (c) **Heap sort**
- (d) Insertion sort

Why: **Heap sort** is the only common algorithm achieving both. Merge sort needs $O(n)$ space; quick sort's worst case is $O(n^2)$.

Q. [EXPECTED] The theoretical lower bound for any comparison-based sorting algorithm is:

- (a) $O(n)$
- (b) **$O(n \log n)$**
- (c) $O(\log n)$
- (d) $O(n^2)$

Why: No comparison sort can beat $O(n \log n)$ in the worst case. **Counting, radix and bucket sort** appear to beat it only because they do not compare elements - they use the key values directly.

Q. [PYQ] Which of the following algorithm design techniques is used in the merge sort algorithm?

- (a) Greedy algorithm
- (b) Backtracking
- (c) **Divide and conquer**
- (d) Divide and conquer, backtracking and greedy algorithm

Why: Merge sort divides into halves, conquers recursively, then combines - the textbook three steps.

Q. [PYQ] [TRAP] Which of the following is NOT the application of Divide and Conquer?

- (a) Binary Search
- (b) **All Pairs Shortest Path Problem**
- (c) Quick Sort
- (d) Strassen's Matrix Multiplication

Why: All-pairs shortest path is solved by **Floyd-Warshall**, which is **dynamic programming** - its subproblems **overlap** and are stored and reused. Divide and conquer requires **independent** subproblems. That overlap is the exact dividing line between the two techniques.

Q. [EXPECTED] [TRAP] Which technique solves problems with overlapping subproblems by storing results?

- (a) Divide and conquer
- (b) Greedy
- (c) **Dynamic programming**
- (d) Backtracking

Why: DP uses **memoisation or tabulation** to compute each subproblem once. Classic DP: Floyd-Warshall, Bellman-Ford, **0/1 knapsack**, longest common subsequence, matrix chain multiplication.

Q. [EXPECTED] [TRAP] The fractional knapsack problem is solved optimally by which technique?

- (a) Dynamic programming
- (b) **Greedy**
- (c) Backtracking
- (d) Branch and bound

Why: **Fractional** knapsack yields to a greedy choice (highest value-to-weight ratio first). The **0/1** knapsack does **not** - it needs dynamic programming. This pair is a favourite discriminator.

Q. [EXPECTED] Which technique abandons a partial solution as soon as it cannot lead to a valid answer?

- (a) Greedy
- (b) Dynamic programming
- (c) **Backtracking**
- (d) Divide and conquer

Why: **Backtracking** prunes the search tree. Classic problems: N-Queens, Sudoku, **sum of subsets**, graph colouring, Hamiltonian cycle, maze solving.

Q. [PYQ] [NUMERICAL] What are the solutions for the sum of subset problem $w[1:5] = \{2, 7, 8, 9, 15\}$, sum required 17?

- (a) 11100; 10101; 00110
- (b) **11100; 10001; 00110**
- (c) 11000; 10001; 00110
- (d) 11100; 10101; 00111

Why: Map each bit onto the weights and add. **11100** = $2+7+8 = 17$; **10001** = $2+15 = 17$; **00110** = $8+9 = 17$. Check a

rejected option: $10101 = 2+8+15 = 25$, not 17. **Do not run the backtracking tree - just add up the selected weights.**

Q. [EXPECTED] Which of the following uses a greedy strategy?

- (a) Floyd-Warshall
- (b) 0/1 Knapsack
- (c) **Huffman coding**
- (d) Longest common subsequence

Why: Greedy algorithms: **Dijkstra, Prim, Kruskal, Huffman coding, fractional knapsack, activity selection, job sequencing.** Greedy makes one locally best choice and never reconsiders - fast, but only optimal for problems with the greedy-choice property.

Q. [EXPECTED] The average-case time complexity of searching in a hash table is:

- (a) $O(n)$
- (b) $O(\log n)$
- (c) **$O(1)$**
- (d) $O(n \log n)$

Why: Hashing computes the index directly. The **worst case is $O(n)$** if every key collides into one slot. Collision resolution: **separate chaining** (linked list per slot) or **open addressing** (linear probing, quadratic probing, double hashing).

Q. [EXPECTED] [TRAP] Which collision resolution technique suffers from primary clustering?

- (a) Separate chaining
- (b) **Linear probing**
- (c) Double hashing
- (d) Quadratic probing

Why: **Linear probing** checks $\text{index}+1$, $\text{index}+2$, ... so occupied slots coalesce into long runs, lengthening every subsequent probe. **Quadratic probing** reduces primary clustering; **double hashing** distributes best.

PART 6

DATABASE MANAGEMENT SYSTEMS AND SQL

Weightage: 5 to 8 marks in DSSSB, 5 to 10 expected. A compact, highly memorisable syllabus - among the most reliable marks available.

6.1 DBMS Concepts, Keys and the ER Model

WEIGHTAGE: 2-4 marks | Schema/instance and key definitions are near-certain | Priority HIGH

6.1.1 Topic checklist

- DBMS vs file system, advantages, data independence
- Three-schema architecture, schema vs instance, metadata
- Database models, relational model terminology, degree and cardinality
- All key types and integrity constraints
- ER model symbols, cardinality, descriptive attributes, generalisation

6.1.2 Questions

Q. [PYQ] What is the overall design of a database called?

- (a) **Database Schema**
- (b) Database Instance
- (c) Database Table
- (d) Database Model

Why: **Schema** = structure (defined once, rarely changes). **Instance** = the actual data at a moment (changes constantly). The analogy to hold: **schema is to instance as a class is to an object**, or as a variable's type is to its value.

Q. [PYQ] What makes a database relational?

- (a) **Tables**
- (b) Field
- (c) Records
- (d) Tuple

Why: The defining feature of the relational model (E. F. Codd, 1970) is organising all data into **tables (relations)**. Fields, records and tuples are merely the *parts* of a table - and "record" and "tuple" mean the same thing, so neither could be the distinguishing answer.

Q. [EXPECTED] [TRAP] In relational terminology, degree and cardinality refer to:

- (a) rows and columns respectively
- (b) **columns and rows respectively**
- (c) keys and attributes
- (d) tables and views

Why: **Degree** = number of **COLUMNS (attributes)**. **Cardinality** = number of **ROWS (tuples)**. Memory hook: "Degree for Descriptions - columns describe; Cardinality for Count of records."

Q. [EXPECTED] Which key can accept NULL values?

- (a) Primary key
- (b) Candidate key
- (c) **Foreign key**
- (d) Super key

Why: A **primary key can never be NULL** (entity integrity). A **foreign key may be NULL**, meaning "no related row yet". A **candidate key** is a minimal super key, one of which is chosen as primary.

Q. [EXPECTED] A minimal super key is called:

- (a) Primary key
- (b) **Candidate key**
- (c) Foreign key
- (d) Composite key

Why: A **super key** is any uniquely identifying attribute set, possibly with redundant attributes. Remove the redundancy and you have a **candidate key**. The designer promotes one candidate key to **primary**; the rest become **alternate** keys.

Q. [EXPECTED] Referential integrity is enforced by which constraint?

- (a) Primary key
- (b) **Foreign key**
- (c) NOT NULL
- (d) CHECK

Why: A foreign key value must either match an existing primary key in the referenced table or be NULL. Related actions: **ON DELETE CASCADE / SET NULL / RESTRICT**.

Q. [PYQ] Descriptive attributes are used to:

- (a) **record the information about relationships**
- (b) record the information about attributes
- (c) record the information about participating entities
- (d) record the information about data

Why: A **descriptive attribute belongs to a RELATIONSHIP**, not to either entity. In "Student ENROLS IN Course", the *grade* and *date of enrolment* describe the enrolment itself, not the student or the course.

Q. [EXPECTED] In an ER diagram, a weak entity is represented by:

- (a) A single rectangle
- (b) **A double rectangle**
- (c) A diamond
- (d) A double ellipse

Why: **Double rectangle** = weak entity (no key of its own, only a partial key; identified through an owner entity via a **double diamond** identifying relationship). Single rectangle = entity, ellipse = attribute, **double ellipse** = **multivalued attribute**, dashed ellipse = derived attribute, diamond = relationship.

Q. [EXPECTED] [TRAP] Generalisation in the ER model is:

- (a) a top-down process splitting an entity into sub-entities
- (b) **a bottom-up process combining similar entities into a higher-level entity**
- (c) the same as aggregation
- (d) used only for weak entities

Why: **Generalisation is BOTTOM-UP** (Car and Truck become Vehicle). **Specialisation is TOP-DOWN** (Employee becomes Engineer, Manager, Clerk). **Aggregation** treats a whole relationship as one higher-level entity.

Q. [EXPECTED] A many-to-many relationship, when converted to tables, requires:

- (a) two tables only
- (b) **a separate junction/bridge table**
- (c) a single merged table
- (d) no change

Why: The junction table holds the primary keys of both participants as a composite key, plus any descriptive attributes of the relationship.

Q. [EXPECTED] Logical data independence means:

- (a) storage structure can change without affecting the logical schema
- (b) **the logical schema can change without affecting application programs**
- (c) data can be stored in any format
- (d) indexes can be added freely

Why: **Physical** independence: change *how* data is stored without touching the logical schema (easier to achieve).

Logical independence: change the logical schema without rewriting applications - achieved largely through **views**.

6.2 Functional Dependencies and Normalisation

WEIGHTAGE: 1-3 marks | Normal-form definitions and Armstrong's axioms recur | Priority HIGH

6.2.1 Questions

Q. [PYQ] [TRAP] Which of the following DBMS scenarios will NOT really follow Armstrong's Axiom?

- (a) Reflexivity rule
- (b) **Pseudo transitivity rule**
- (c) Transitivity rule
- (d) Armstrong's axioms

Why: Armstrong's axioms **proper** are exactly three: **Reflexivity, Augmentation, Transitivity**. Union, decomposition, **pseudo-transitivity** and composition are *derived* rules - they follow from the axioms but are not themselves axioms. Memorise the three-word list.

Q. [PYQ] [NUMERICAL] Consider $R(PQRSTU)$ with $F = \{P \rightarrow RT, Q \rightarrow S, R \rightarrow PS, QS \rightarrow TU\}$. Find the closure of PQ .

- (a) PQRST
- (b) PQRTU
- (c) PQRSU
- (d) **PQRSTU**

Why: Start with $\{P, Q\}$. $P \rightarrow RT$ adds R and T. $Q \rightarrow S$ adds S. Now we hold Q and S, so $QS \rightarrow TU$ adds U. Result is all six attributes = **PQRSTU**, which means PQ is a superkey. **Method: loop through the FD list repeatedly until one full pass adds nothing new.**

Q. [EXPECTED] A relation is in 1NF if:

- (a) it has no transitive dependency
- (b) it has no partial dependency
- (c) **every attribute holds a single atomic value**
- (d) every determinant is a super key

Why: **1NF = atomic values** (no repeating groups or multi-valued cells). **2NF** removes **partial** dependencies; **3NF** removes **transitive** dependencies; **BCNF** requires every determinant to be a **super key**.

Q. [EXPECTED] [TRAP] Removing partial dependency achieves which normal form?

- (a) 1NF
- (b) **2NF**
- (c) 3NF
- (d) BCNF

Why: A partial dependency exists when a non-key attribute depends on only **part of a composite** primary key - so 2NF is only an issue when the key is composite. Hook: "**One is Atomic, Two is Partial, Three is Transitive, BC is Super.**"

Q. [EXPECTED] A relation in 3NF where every determinant is a super key is in:

- (a) 2NF
- (b) **BCNF**
- (c) 4NF
- (d) 5NF

Why: BCNF is **stricter than 3NF** and handles anomalies arising from overlapping candidate keys. Important caveat: **BCNF decomposition may fail to preserve dependencies**, whereas a lossless *and* dependency-preserving 3NF decomposition always exists.

Q. [EXPECTED] The purpose of normalisation is primarily to:

- (a) speed up queries
- (b) **reduce redundancy and eliminate update, insert and delete anomalies**
- (c) reduce the number of tables
- (d) improve security

Why: Normalisation usually **increases** the number of tables and can slow reads by requiring joins. That is why data

warehouses deliberately **denormalise** (star schema) for query speed.

6.3 SQL

WEIGHTAGE: 2-4 marks | Command classification and NULL logic are recurring | Priority HIGH

6.3.1 Questions

Q. [PYQ] Which of the following is used to define the structure of a relation as well as delete relations and relate schemas?

- (a) Integrity constraint
- (b) View
- (c) DML (Data Manipulation Language)
- (d) **DDL (Data Definition Language)**

Why: Every verb in the question - define the structure, delete relations, relate schemas - is **structural**, which is DDL's domain (CREATE, ALTER, DROP, TRUNCATE). DML only touches the rows inside an existing structure.

Q. [PYQ] Which SQL statement is used to update data in a table?

- (a) CHANGE
- (b) MODIFY
- (c) **UPDATE**
- (d) SET

Why: CHANGE and MODIFY are clauses used **inside ALTER TABLE** to rename or retype a column; SET is a clause **inside UPDATE**. Only **UPDATE** is a standalone data-modification statement.

Q. [EXPECTED] [TRAP] Which of the following is a DDL command that cannot normally be rolled back?

- (a) DELETE
- (b) **TRUNCATE**
- (c) UPDATE
- (d) INSERT

Why: **TRUNCATE** is DDL, removes all rows, takes no WHERE clause, is very fast and is normally not rollback-able. **DELETE** is DML, accepts WHERE, logs each row and **can** be rolled back. **DROP** removes the table structure entirely.

Q. [PYQ] Select the correct SQL statement to insert a new row into Customer specifying only CustomerID, Name and Product (other columns allow NULL).

- (a) INSERT Customer SET CustomerID=101, Name='Alice', Product='Pen';
- (b) INSERT INTO Customer VALUES (101, 'Alice', 'Pen');
- (c) INSERT INTO Customer (101, 'Alice', 'Pen');
- (d) **INSERT INTO Customer (CustomerID, Name, Product) VALUES (101, 'Alice', 'Pen');**

Why: To insert into a subset of columns you must name the columns **AND** use **VALUES**. The bare VALUES form supplies values for ***all*** columns in order and would fail here. The SET form is MySQL-specific, and the third option omits VALUES entirely.

Q. [PYQ] [TRAP] Which of the following SQL expressions evaluates to TRUE, according to SQL-99?

- (a) NULL <= NULL returns UNKNOWN
- (b) NULL >= NULL returns UNKNOWN
- (c) NULL IS NULL returns FALSE
- (d) **NULL IS NULL returns TRUE**

Why: Read the question precisely - it asks which **expression evaluates to TRUE**. The first two statements are **factually accurate** but the expressions themselves evaluate to UNKNOWN, not TRUE. **Golden rule: never test NULL with = or <>; use IS NULL / IS NOT NULL.**

Q. [EXPECTED] [TRAP] What does COUNT(*) do differently from COUNT(column)?

- (a) They are identical
- (b) **COUNT(*) counts all rows including NULLs; COUNT(column) ignores NULLs**
- (c) COUNT(*) is slower always
- (d) COUNT(column) counts distinct values

Why: All aggregate functions except COUNT(*) **ignore NULLs**. This is why AVG(col) is not the same as

SUM(col)/COUNT(*) when NULLs are present.

Q. [EXPECTED] [TRAP] Which clause filters groups after aggregation?

- (a) WHERE
- (b) **HAVING**
- (c) GROUP BY
- (d) ORDER BY

Why: WHERE filters individual rows BEFORE grouping and cannot contain aggregate functions. HAVING filters groups AFTER aggregation and can. Execution order: FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY.

Q. [EXPECTED] Which join returns all rows from the left table plus matching rows from the right?

- (a) INNER JOIN
- (b) **LEFT OUTER JOIN**
- (c) RIGHT OUTER JOIN
- (d) CROSS JOIN

Why: Unmatched right-side columns come back as NULL. INNER JOIN returns only matches; FULL OUTER returns everything from both; CROSS JOIN is the Cartesian product.

Q. [PYQ] [TRAP] ___ is NOT allowed in a SQL:1999 view definition.

- (a) Use of subqueries
- (b) Use of nested queries
- (c) Use of aggregate formulas
- (d) **Use of ORDER BY clause**

Why: A view is defined to be a **relation**, and a relation is an **unordered set of tuples** - so ordering has no meaning in the definition. Apply ORDER BY when you SELECT *from* the view. Subqueries and aggregates are legal (they merely make the view read-only).

Q. [EXPECTED] A view is best described as:

- (a) a physical copy of a table
- (b) **a virtual table produced by a stored query**
- (c) an index
- (d) a backup

Why: A standard view stores **no data** - rows are generated when queried. A **materialised view** does store results physically and must be refreshed, trading freshness for speed.

Q. [EXPECTED] Which of the following is NOT a benefit of using views?

- (a) Security by hiding columns
- (b) Simplifying complex joins
- (c) Logical data independence
- (d) **Faster write performance on the base tables**

Why: Views do nothing for write speed and can complicate updates. Their genuine benefits are security, simplification and logical independence.

Q. [EXPECTED] [TRAP] How many clustered indexes can a table have?

- (a) Unlimited
- (b) **One**
- (c) Two
- (d) None

Why: A **clustered index determines the physical row order**, so there can be only **one** per table (usually on the primary key). **Non-clustered** indexes are separate structures and many are permitted.

6.4 Transactions and Concurrency Control

WEIGHTAGE: 1-3 marks | ACID and serialisability appear regularly | Priority HIGH

6.4.1 Questions

Q. [EXPECTED] Which ACID property guarantees that a transaction is all-or-nothing?

- (a) Consistency
- (b) **Atomicity**
- (c) Isolation
- (d) Durability

Why: **Atomicity** = all operations complete or none do (partial work is rolled back). **Consistency** = valid state to valid state. **Isolation** = concurrent transactions do not interfere. **Durability** = committed changes survive a crash.

Q. [EXPECTED] Which ACID property is ensured by write-ahead logging?

- (a) Atomicity
- (b) Consistency
- (c) Isolation
- (d) **Durability**

Why: The log record is forced to stable storage **before** the data change, so committed work can always be reconstructed after a crash. WAL supports both durability and atomicity (via UNDO).

Q. [EXPECTED] [TRAP] Reading a value written by an uncommitted transaction is called:

- (a) Lost update
- (b) **Dirty read**
- (c) Unrepeatable read
- (d) Phantom read

Why: **Dirty read** - if the writer later rolls back, the reader has used data that never officially existed. **Unrepeatable read** : same row read twice gives different values. **Phantom read**: a range query returns extra rows the second time. **Lost update**: one transaction's write overwrites another's.

Q. [EXPECTED] Two operations conflict if they belong to different transactions, access the same data item, and:

- (a) both are reads
- (b) **at least one is a write**
- (c) both are writes
- (d) they occur simultaneously

Why: **Read-read never conflicts.** Only read-write, write-read and write-write pairs do. This definition is the foundation of conflict serialisability.

Q. [PYQ] [NUMERICAL] Which of the following is a conflict serializable schedule? S1: R1(A); R1(B); R2(A); R2(B); W2(B); W1(A) and S2: R1(A); R2(A); R2(B); W2(B); R1(B); W1(A)

- (a) S1 is conflict serializable but S2 is not.
- (b) **S1 is not conflict serializable but S2 is conflict serializable.**
- (c) Both are not conflict serializable.
- (d) Both are conflict serializable.

Why: Build the **precedence graph**. In **S1**: on item A, R2(A) precedes W1(A) giving edge T2 to T1; on item B, R1(B) precedes W2(B) giving edge T1 to T2 - both directions exist, so there is a **cycle** and S1 fails. In **S2**: R2(A) before W1(A) gives T2 to T1, and W2(B) before R1(B) also gives T2 to T1 - all edges point one way, **no cycle**, so S2 is conflict serialisable (equivalent to running T2 then T1). **A schedule is conflict serialisable if and only if its precedence graph is acyclic.**

Q. [EXPECTED] In two-phase locking, once a transaction releases its first lock it:

- (a) may acquire more locks
- (b) **may not acquire any new locks**
- (c) must commit immediately
- (d) must abort

Why: 2PL has a **growing phase** (acquire only) and a **shrinking phase** (release only). This guarantees serialisability but can still deadlock. **Conservative (static) 2PL** acquires all locks upfront and is **deadlock free**.

Q. [EXPECTED] [TRAP] Which lock type allows several transactions to read the same item concurrently?

- (a) Exclusive lock
- (b) **Shared lock**
- (c) Update lock
- (d) Binary lock

Why: Multiple **shared (S)** locks can coexist for reading. An **exclusive (X)** lock permits no other lock of any kind - required for writing.

Q. [EXPECTED] Which concurrency control method never causes deadlock?

- (a) Two-phase locking
- (b) Strict 2PL
- (c) **Timestamp ordering**
- (d) Rigorous 2PL

Why: **Timestamp ordering** resolves conflicts by comparing timestamps and aborting offenders rather than making them wait, so no waiting cycle can form. Conservative 2PL is also deadlock free (it avoids hold-and-wait).

Q. [EXPECTED] [TRAP] What is the difference between OLTP and OLAP?

- (a) OLTP is for analysis, OLAP for transactions
- (b) **OLTP handles many short read/write transactions; OLAP handles few complex read-only analytical queries**
- (c) Both are identical
- (d) OLAP is highly normalised

Why: **OLTP** = day-to-day operations, highly normalised, current detailed data (booking a ticket). **OLAP** = decision support, **denormalised** star/snowflake schema, historical summarised data ("compare quarterly sales by region over five years").

Q. [EXPECTED] A data warehouse is characterised as subject-oriented, integrated, time-variant and:

- (a) volatile
- (b) **non-volatile**
- (c) normalised
- (d) transactional

Why: Bill Inmon's four adjectives: **subject-oriented, integrated, time-variant, non-volatile**. Non-volatile means data is loaded and read but not updated in place.

Q. [EXPECTED] The 5 Vs of Big Data are Volume, Velocity, Variety, Veracity and:

- (a) Validity
- (b) Visibility
- (c) **Value**
- (d) Version

Why: **Value** - data is worthless unless insight can be extracted from it. Related: **ETL** (Extract, Transform, Load) moves data into the warehouse; **fact tables** hold measures and **dimension tables** hold context.

PART 7

COMPUTER NETWORKS, SECURITY AND MOBILE COMPUTING

Weightage: 9 to 12 marks in DSSSB, 8 to 14 expected. Layer functions, IP addressing and cryptography are the reliable scorers.

7.1 Fundamentals, Topologies and Devices

WEIGHTAGE: 2-3 marks | Device-to-layer mapping is a repeat favourite | Priority CRITICAL

7.1.1 Questions

Q. [PYQ] Which of the following are the most essential components a computer system uses for communication over internet?

- (a) Applications and mouse
- (b) Monitor and keyboard
- (c) **NIC and TCP/IP**
- (d) Camera and microphone

Why: You need **hardware** to connect physically (Network Interface Card) and a **protocol suite** so both ends agree on the rules (TCP/IP). Monitors, keyboards and cameras are peripherals for the human, not for networking.

Q. [EXPECTED] [TRAP] Which topology needs the least cable but fails completely if the backbone breaks?

- (a) Star
- (b) Ring
- (c) **Bus**
- (d) Mesh

Why: Bus: cheapest, least cable, but one cable fault kills the whole network and collisions are frequent. **Star:** easy fault isolation, but the central hub/switch is a single point of failure. **Mesh:** most reliable, needs $n(n-1)/2$ links.

Q. [EXPECTED] [NUMERICAL] How many cables are needed for a full mesh network with 8 nodes?

- (a) 64
- (b) 56
- (c) **28**
- (d) 16

Why: $n(n-1)/2 = 8 \times 7 / 2 = 28$ links, and each node needs $n-1 = 7$ ports. The quadratic growth is why full mesh is rare in practice.

Q. [EXPECTED] [TRAP] A hub differs from a switch in that a hub:

- (a) works at the network layer
- (b) **broadcasts incoming data to all ports, creating one collision domain**
- (c) filters traffic by IP address
- (d) connects different networks

Why: A **hub** is a multiport repeater (Physical layer, layer 1) that blindly floods every port. A **switch** (Data Link, layer 2) learns MAC addresses and forwards a frame only to the correct port, giving each port its own collision domain.

Q. [EXPECTED] Which device connects two different networks and forwards packets using IP addresses?

- (a) Hub
- (b) Switch
- (c) Bridge
- (d) **Router**

Why: A **router** works at the **Network layer (3)** using a routing table, and it separates broadcast domains. A **gateway** goes further, translating between different protocols across all layers.

Q. [EXPECTED] At which OSI layer does a switch operate?

- (a) Layer 1
- (b) **Layer 2**
- (c) Layer 3
- (d) Layer 4

Why: Switch and bridge = **Data Link (2)**, using MAC addresses. Hub and repeater = **Physical (1)**. Router = **Network (3)**. NIC spans 1 and 2.

Q. [PYQ] The refractive index of the core of an optical fibre is greater than that of cladding because:

- (a) the light gets reflected at the entrance end of the optical fibre
- (b) **the light gets totally internally reflected into the core**
- (c) the light gets restricted at the entrance end
- (d) the light gets refracted into cladding

Why: **Total internal reflection** requires light to travel from a denser medium (higher refractive index) into a rarer one at greater than the critical angle. If the cladding had the higher index, light would simply refract out and be lost.

Q. [EXPECTED] Which transmission medium is immune to electromagnetic interference?

- (a) Twisted pair
- (b) Coaxial cable
- (c) **Optical fibre**
- (d) Radio waves

Why: Fibre carries **light**, not electric current, so EMI cannot affect it. It also offers the highest bandwidth, longest range and best security (very hard to tap) - at higher cost and greater fragility.

Q. [PYQ] [TRAP] Which of the following statements is true?

- (a) Circuit setup is not required in both a datagram network and virtual-circuit network.
- (b) Circuit setup is required in both.
- (c) **Circuit setup is required in a virtual-circuit network.**
- (d) Circuit setup is required in a datagram network.

Why: A **virtual circuit** establishes a logical path (assigning a VC identifier) before data flows. A **datagram** network routes each packet independently with **no setup at all**, which is how IP works.

Q. [EXPECTED] Half duplex transmission means:

- (a) data flows in one direction only
- (b) **data flows in both directions but only one at a time**
- (c) data flows both ways simultaneously
- (d) no data flows

Why: **Simplex** = one direction only (TV broadcast). **Half duplex** = both directions alternately (walkie-talkie). **Full duplex** = both simultaneously (telephone).

7.2 OSI and TCP/IP Models

WEIGHTAGE: 3-4 marks | Data units and layer functions appear every year | Priority CRITICAL

7.2.1 Questions

Q. [PYQ] The internet layer in the TCP/IP model is associated with ___ data.

- (a) segments
- (b) frames
- (c) bytes
- (d) **packets**

Why: Lock the chain in memory: **Transport** = segments, **Internet/Network** = packets, **Data Link** = frames, **Physical** = bits. Some form of this is asked almost every year.

Q. [PYQ] Which of the following are networking transport layer protocols?

- (a) FTP and DNS
- (b) Telnet and ICMP
- (c) **TCP, UDP and SCTP**
- (d) ARP and RARP

Why: FTP, DNS and Telnet are **Application** layer; ICMP, ARP and RARP are **Network** layer. Only TCP, UDP and SCTP are Transport.

Q. [PYQ] [TRAP] Which of the following is NOT a network layer function?

- (a) **Error control and flow control**
- (b) Intra-routing
- (c) Routing
- (d) Congestion control

Why: Routing, intra-domain routing and congestion control are all Network-layer duties. **Error and flow control** belong to the **Data Link** layer (hop by hop) and the **Transport** layer (end to end). IP itself is best-effort with only a header checksum.

Q. [PYQ] What feature of the transport layer prevents data loss?

- (a) Encryption
- (b) Packet switching
- (c) Addressing
- (d) **Flow control**

Why: **Flow control** stops a fast sender from overrunning a slow receiver's buffer - buffer overflow is exactly how data would be lost. TCP implements it with a **sliding window** and the receiver's advertised window size.

Q. [PYQ] [TRAP] Which layer is responsible for sending data as a bit stream?

- (a) Presentation layer
- (b) Network layer
- (c) Physical layer
- (d) **Data link layer**

Why: KEY NOTE. This paper's key said Data Link. Standard theory assigns the **raw bit stream over the medium** to the **PHYSICAL** layer; the Data Link layer deals in **frames**. If the question says "raw bit stream" or "over the transmission medium", answer **Physical**.

Q. [EXPECTED] Encryption and compression are performed at which OSI layer?

- (a) Application
- (b) **Presentation**
- (c) Session
- (d) Transport

Why: The **Presentation** layer (6) handles translation (ASCII/EBCDIC), **encryption/decryption** and **compression** - it is also called the syntax layer. The **Session** layer (5) does dialog control, synchronisation and checkpointing.

Q. [EXPECTED] Port addressing is a function of which layer?

- (a) Network
- (b) Data Link
- (c) **Transport**
- (d) Application

Why: **Transport** uses **port** numbers to reach a specific process. Network uses **IP** (logical) addresses; Data Link uses **MAC** (physical) addresses.

Q. [EXPECTED] How many layers does the TCP/IP model have?

- (a) 7
- (b) 5
- (c) **4**
- (d) 3

Why: TCP/IP has **4**: Application (= OSI 5,6,7), Transport, Internet, Network Access (= OSI 1,2). OSI has 7. Mnemonic for OSI top-down: **All People Seem To Need Data Processing**.

7.3 Data Link Layer, Error and Flow Control

WEIGHTAGE: 1-3 marks | Sliding window behaviour is the recurring question | Priority HIGH

7.3.1 Questions

Q. [PYQ] How does the receiver handle out-of-order frames in Selective Repeat?

- (a) Requests retransmission of all frames
- (b) Sends a cumulative acknowledgment
- (c) **Buffers them until all missing frames arrive**
- (d) Discards them immediately

Why: This is the defining difference: **Go-Back-N discards** out-of-order frames and uses cumulative ACKs; **Selective Repeat buffers** them and acknowledges each individually, retransmitting only what was actually lost.

Q. [EXPECTED] In Go-Back-N with a window size of 4, if frame 2 is lost, which frames are retransmitted?

- (a) Only frame 2
- (b) **Frame 2 and all subsequent frames in the window**
- (c) No frames
- (d) All frames from 0

Why: The name says it - the sender "goes back N" and resends the lost frame **plus everything after it**, because the receiver discarded them all. This wastes bandwidth but keeps the receiver simple.

Q. [EXPECTED] [TRAP] Which error detection method is most powerful and used in Ethernet?

- (a) Simple parity check
- (b) Checksum
- (c) **CRC**
- (d) Hamming code

Why: **CRC** (Cyclic Redundancy Check) treats data as a polynomial and detects all burst errors shorter than the CRC length. **Checksum** is weaker and used in TCP/UDP/IP headers. **Hamming code** actually *corrects* single-bit errors (forward error correction), which is a different goal.

Q. [EXPECTED] A simple even parity check fails to detect:

- (a) single-bit errors
- (b) **errors affecting an even number of bits**
- (c) any errors
- (d) burst errors only

Why: If two bits flip, parity is restored and the error passes unnoticed. This is why CRC replaced parity for serious links.

Q. [EXPECTED] [TRAP] Which access method is used in wireless LANs rather than wired Ethernet?

- (a) CSMA/CD
- (b) **CSMA/CA**
- (c) Token passing
- (d) Polling

Why: **CSMA/CD (Collision Detection)** is wired Ethernet. Wireless stations cannot reliably detect collisions while transmitting, so Wi-Fi uses **CSMA/CA (Collision Avoidance)** with RTS/CTS handshaking and random backoff.

Q. [EXPECTED] [NUMERICAL] The maximum channel efficiency of pure ALOHA is:

- (a) 36.8%
- (b) **18.4%**
- (c) 50%
- (d) 100%

Why: Pure ALOHA = **18.4%**; slotted ALOHA doubles it to **36.8%** by forcing transmissions to begin at slot boundaries, which halves the vulnerable period.

7.4 Network Layer, IP Addressing and Routing

WEIGHTAGE: 2-4 marks | A subnetting numerical is very likely | Priority CRITICAL

7.4.1 Questions

Q. [PYQ] Which of the following 4-byte IP addresses are used for internet protocol layer?

- (a) Network address and port address
- (b) Network address and MAC address
- (c) **Network address and host address**
- (d) Host address and port address

Why: An IPv4 address is 32 bits = 4 bytes, split logically into **network** and **host** parts. Port belongs to Transport; MAC to Data Link.

Q. [PYQ] [NUMERICAL] A network has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle?

- (a) 1024
- (b) 3072
- (c) 2048
- (d) **4094**

Why: Write the mask in binary: 240 = 11110000, so network bits = 8+8+4 = **20** (a /20) and host bits = 32-20 = **12**. Usable hosts = $2^{12} - 2 = 4094$. Subtract 2 for the network address (all host bits 0) and the broadcast address (all host bits 1). Note the distractor 4096 is the *total* addresses.

Q. [EXPECTED] [NUMERICAL] How many usable host addresses are available in a /26 network?

- (a) 64
- (b) **62**
- (c) 32
- (d) 30

Why: Host bits = 32 - 26 = 6, so $2^6 - 2 = 62$. Handy ladder: /24 gives 254, /25 gives 126, /26 gives 62, /27 gives 30, /28 gives 14, /30 gives 2.

Q. [EXPECTED] Which IP address class is reserved for multicasting?

- (a) Class A
- (b) Class B
- (c) Class C
- (d) **Class D**

Why: **Class D (224-239)** is multicast; **Class E (240-255)** is reserved/experimental. Class A is 1-126, B is 128-191, C is 192-223, and **127.x.x.x** is **loopback**.

Q. [EXPECTED] [TRAP] What is the size of an IPv6 address?

- (a) 32 bits
- (b) 64 bits
- (c) **128 bits**
- (d) 256 bits

Why: IPv6 is **128 bits**, written as eight groups of four hex digits. It has a fixed 40-byte header, **no checksum**, no router fragmentation, and **no broadcast** - it uses multicast and anycast instead.

Q. [PYQ] What is the ARP reply's scope in terms of message type?

- (a) Anycast
- (b) **Unicast**
- (c) Multicast
- (d) Broadcast

Why: **ARP request = BROADCAST** (the sender does not know who owns the IP, so it must ask everyone). **ARP reply = UNICAST** (the responder learned the asker's MAC from the request, so it answers directly). Memorise the pair.

Q. [EXPECTED] ARP resolves:

- (a) IP address to domain name
- (b) **IP address to MAC address**
- (c) MAC address to IP address
- (d) domain name to IP address

Why: **ARP**: IP to MAC. **RARP**: MAC to IP (obsolete, replaced by DHCP). **DNS**: domain name to IP. **ICMP**: error reporting and diagnostics (ping, traceroute).

Q. [EXPECTED] [TRAP] Which routing protocol uses Dijkstra's algorithm and converges quickly?

- (a) RIP
- (b) **OSPF**
- (c) BGP
- (d) EIGRP

Why: **OSPF** is a **link-state** protocol: every router builds a full topology map and runs Dijkstra. **RIP** is distance-vector using **hop count with a maximum of 15**, and converges slowly. **BGP** is the Internet's **exterior** path-vector protocol used between autonomous systems.

Q. [EXPECTED] The TTL field in an IP packet prevents:

- (a) congestion
- (b) **packets circulating forever in routing loops**
- (c) fragmentation
- (d) duplicate delivery

Why: Each router decrements TTL; at zero the packet is discarded and an ICMP time-exceeded message may be returned - which is exactly the mechanism **traceroute** exploits.

7.5 Transport Layer - TCP and UDP

WEIGHTAGE: 2-4 marks | TCP vs UDP is asked in some form nearly every year | Priority CRITICAL

7.5.1 Questions

Q. [PYQ] [TRAP] Which of the following statements is FALSE about TCP?

- (a) It uses a three-way handshake to establish a connection.
- (b) It receives data from application as a single stream.
- (c) It is a connection-oriented protocol.
- (d) **It is a connection-less protocol.**

Why: TCP is definitively **connection-oriented**; **UDP** is the connectionless one. The option list contains both claims, which is your signal for which is being tested.

Q. [PYQ] What does the TCP service model guarantee about the order of data delivery?

- (a) **Data is delivered in the order it is transmitted.**
- (b) Data is not guaranteed to arrive in any specific order.
- (c) Data may arrive out of order and is handled by the application layer.
- (d) Data is delivered as fast as possible.

Why: TCP uses **sequence numbers** to reorder segments before delivery. The "handled by the application layer" option describes **UDP**, where the application must sort things out itself.

Q. [EXPECTED] [TRAP] Which protocol would you choose for live video streaming, and why?

- (a) TCP, because it is reliable
- (b) **UDP, because low latency matters more than retransmitting lost frames**
- (c) TCP, because it is faster
- (d) SCTP, because it is connectionless

Why: Retransmitting a video frame that is already too late to display is useless and adds delay. **UDP** is used for streaming, VoIP, gaming, **DNS**, DHCP, TFTP and SNMP. Its header is a fixed **8 bytes** versus TCP's 20-60.

Q. [PYQ] What happens if a RST (reset) packet is received during the connection management process?

- (a) The receiver sends another SYN packet.
- (b) **The connection is immediately terminated.**
- (c) The sender continues retransmissions.
- (d) The connection remains active.

Why: **RST aborts immediately** and discards queued data - no acknowledgement expected. Contrast **FIN**, which begins a graceful four-way shutdown. RST is sent when a segment arrives for a port with no listener.

Q. [EXPECTED] The TCP three-way handshake consists of:

- (a) SYN, ACK, FIN
- (b) **SYN, SYN+ACK, ACK**
- (c) SYN, ACK, RST
- (d) FIN, ACK, FIN

Why: Connection setup is three steps; **termination takes four** (FIN, ACK, FIN, ACK).

Q. [PYQ] [TRAP] Which statement is true about the slow start phase of TCP congestion control?

- (a) Congestion window increases by 1 MSS every round trip time.
- (b) **Congestion window approximately doubles every round trip time.**
- (c) Congestion window increases by 2 MSS on every successful acknowledgement.
- (d) Congestion window approximately doubles on every successful acknowledgement.

Why: The window grows by **1 MSS per ACK received**; since a window of N segments produces N ACKs per RTT, the window goes from N to 2N each RTT - i.e. it **doubles per RTT**. "Increases by 1 MSS per RTT" describes the later **congestion avoidance** phase (additive increase).

Q. [EXPECTED] [NUMERICAL] Which port number does HTTPS use?

- (a) 80
- (b) **443**

(c) 22

(d) 25

Why: Learn the core set: 20/21 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 67/68 DHCP, 69 TFTP, 80 HTTP, 110 POP3, 143 IMAP, 161 SNMP, 443 HTTPS, 3306 MySQL.

Q. [EXPECTED] A socket is uniquely identified by:

(a) port number alone

(b) IP address alone

(c) IP address + port number (+ protocol)

(d) MAC address

Why: This combination is what lets one machine hold thousands of simultaneous connections.

7.6 Application Layer Protocols

WEIGHTAGE: 1-3 marks | Protocol-to-purpose matching is easy marks | Priority HIGH

7.6.1 Questions

Q. [EXPECTED] [TRAP] Which protocol is used to SEND email, and which to RETRIEVE it?

- (a) POP3 to send, SMTP to retrieve
- (b) **SMTP to send, POP3/IMAP to retrieve**
- (c) IMAP to send, SMTP to retrieve
- (d) HTTP for both

Why: **SMTP pushes mail out** (port 25). **POP3** (110) downloads and typically deletes from the server; **IMAP** (143) keeps mail on the server and supports folders and multi-device sync.

Q. [EXPECTED] Which protocol automatically assigns IP address, subnet mask, gateway and DNS to a host?

- (a) DNS
- (b) ARP
- (c) **DHCP**
- (d) ICMP

Why: **DHCP** uses UDP ports 67 and 68. If it fails, Windows self-assigns an **APIPA** address in 169.254.x.x.

Q. [EXPECTED] [TRAP] Which of the following is an insecure protocol that transmits credentials in plain text?

- (a) SSH
- (b) HTTPS
- (c) **Telnet**
- (d) SFTP

Why: **Telnet** (port 23) sends everything unencrypted; **SSH** (22) is its encrypted replacement. Similarly FTP is insecure and SFTP/FTPS are the secure alternatives.

Q. [EXPECTED] HTTP is described as a stateless protocol because:

- (a) it cannot transfer files
- (b) **each request is independent and the server retains no memory of previous requests**
- (c) it uses UDP
- (d) it has no error codes

Why: Statelessness is why **cookies** and server-side **sessions** exist - they supply the memory HTTP lacks. Status codes: 200 OK, 301 moved, **404 not found**, 500 server error.

7.7 Network Security and Cryptography

WEIGHTAGE: 2-4 marks | Symmetric vs asymmetric is near-certain | Priority CRITICAL

7.7.1 Questions

Q. [PYQ] The process of verifying a user's identity is called:

- (a) integrity
- (b) **authentication**
- (c) validation
- (d) confidentiality

Why: **Authentication** = "who are you?" (password, OTP, biometric). **Authorisation** = "what may you do?". **Validation** = checking *data* format. **Integrity** = data has not been altered.

Q. [PYQ] The information transformed during encryption is ____.

- (a) encrypted message
- (b) coded message
- (c) cipher text
- (d) **plain text**

Why: The question asks what goes **IN**. **Plain text** is transformed *into* cipher text. "Cipher text" is the tempting wrong answer because it is the *result*.

Q. [PYQ] [TRAP] Which of the following is NOT a symmetric-key cryptographic algorithm?

- (a) AES
- (b) RC4
- (c) **RSA**
- (d) DES

Why: **RSA** (Rivest-Shamir-Adleman) is the classic **asymmetric/public-key** algorithm. Asymmetric family: **RSA**, **Diffie-Hellman**, **ECC**, **DSA**, **ElGamal**. Everything else you commonly meet (AES, DES, 3DES, RC4, RC5, Blowfish, IDEA) is symmetric.

Q. [PYQ] In public key cryptography, which key is used for decrypting a message encrypted with the public key?

- (a) Another public key
- (b) A symmetric key
- (c) The sender's private key
- (d) **The recipient's private key**

Why: Public and private keys are mathematically matched pairs - only the **corresponding private key** can undo what its public key locked. The *sender's* private key belongs to a different pair entirely.

Q. [EXPECTED] [TRAP] To create a digital signature, the sender encrypts the message digest with:

- (a) the recipient's public key
- (b) the recipient's private key
- (c) **the sender's private key**
- (d) a session key

Why: Two separate use cases: for **confidentiality** encrypt with the **recipient's PUBLIC** key; for a **digital signature** encrypt with the **sender's PRIVATE** key so anyone can verify it using the sender's public key. Signatures give authentication, integrity and non-repudiation - but **not** confidentiality.

Q. [PYQ] Communication between end systems is encrypted using a key that is commonly referred to as a:

- (a) single key
- (b) **session key**
- (c) private key
- (d) public key

Why: A **session key** is a temporary symmetric key generated for one session and then discarded. HTTPS does exactly this: asymmetric cryptography to agree a key, then fast symmetric encryption for the data.

Q. [EXPECTED] Why do real systems combine asymmetric and symmetric cryptography?

- (a) To use more keys
- (b) **Asymmetric solves key distribution; symmetric is far faster for bulk data**
- (c) Symmetric is more secure
- (d) Asymmetric cannot encrypt

Why: Asymmetric encryption is roughly a thousand times slower. Using it only to exchange the session key gives both secure key distribution and high throughput.

Q. [PYQ] Which of the following describes the use of digital certificates in authentication?

- (a) They store users' passwords.
- (b) They compress large files for secure transfer.
- (c) **They verify the identity of entities using Public Key Infrastructure (PKI).**
- (d) They encrypt data transmission.

Why: A certificate (format **X.509**) binds a **public key to an identity** and is signed by a trusted **Certificate Authority**. It authenticates identity; the traffic itself is encrypted afterwards with session keys.

Q. [EXPECTED] A hash function is characterised by being:

- (a) reversible
- (b) **one-way and producing a fixed-length digest**
- (c) variable in output length
- (d) encryption with a key

Why: Hashes are **irreversible**. Used for password storage, integrity checks and digital signatures. **MD5** (128-bit) and **SHA-1** (160-bit) are broken; **SHA-256/512** is current.

Q. [EXPECTED] [TRAP] Which malware self-replicates and spreads across networks without any user action?

- (a) Virus
- (b) **Worm**
- (c) Trojan horse
- (d) Adware

Why: A **virus** needs a host file and a user action; a **worm** spreads autonomously. A **Trojan** masquerades as useful software; **ransomware** encrypts files for payment; a **rootkit** hides an intruder's presence.

Q. [EXPECTED] SQL injection is best prevented by:

- (a) using a firewall
- (b) **using parameterised queries / prepared statements**
- (c) encrypting the database
- (d) renaming tables

Why: Parameterised queries separate code from data so user input can never be interpreted as SQL. Related web attack: **XSS**, prevented by output encoding.

Q. [PYQ] Which of the following is NOT a risk management methodology?

- (a) Acceptance of threats
- (b) **Risk generation**
- (c) Avoidance of threats
- (d) Transfer of risks

Why: The four genuine strategies are **avoid, accept, transfer (e.g. insurance) and mitigate**. Nobody **generates** risk as a strategy.

Q. [EXPECTED] [TRAP] A passive attack is characterised by:

- (a) modifying messages
- (b) **eavesdropping without altering data**
- (c) denial of service
- (d) masquerading

Why: **Passive** = eavesdropping and traffic analysis; hard to detect but easy to prevent with encryption. **Active** = modification, masquerade, replay, DoS; hard to prevent but easier to detect.

7.8 Wireless and Mobile Computing

WEIGHTAGE: 1-3 marks in DSSSB | GSM registers and handover types recur | Priority MEDIUM

7.8.1 Questions

Q. [PYQ] BTS stands for ____.

- (a) Base Transfer Station
- (b) **Base Transceiver Station**
- (c) Basic Transfer System
- (d) Basic Transceiver Station

Why: **Base** (not Basic) **Transceiver** (transmitter + receiver, which is what tower equipment does) **Station**. Watch the deliberate near-miss spellings.

Q. [PYQ] [TRAP] Which statement is true about Home Location Register (HLR) and Visitor Location Register (VLR)?

- (a) VLR is not responsible for the MS to inform the MLR about location changes.
- (b) VLR is not responsible for the MS to inform the HLR about location changes.
- (c) **VLR is responsible for the MS to inform the HLR about location changes.**
- (d) HLR is responsible for the MS to inform the VLR about location changes.

Why: Reporting always flows **VLR to HLR**. As you move you register with a new VLR, which updates your permanent **HLR** so the home network always knows which VLR is serving you. Incoming calls query the HLR first.

Q. [PYQ] ____ is the handover from a satellite network to a terrestrial cellular network.

- (a) Gateway handover
- (b) Inter satellite handover
- (c) **Inter system handover**
- (d) Intra satellite handover

Why: The key word is **system** - two different systems (satellite and terrestrial) are involved. Gateway, intra-satellite and inter-satellite handovers all occur *within* the satellite system.

Q. [EXPECTED] [TRAP] In a soft handover, the new connection is established:

- (a) after the old one is released
- (b) **before the old one is released, so there is no interruption**
- (c) only if the old one fails
- (d) by the mobile station alone

Why: **Soft handover** (CDMA) makes before break - no interruption. **Hard handover** (GSM) breaks before make - a brief gap.

Q. [PYQ] ____ is the address of the current tunnel endpoint for MN that can be chosen via DHCP.

- (a) **FA**
- (b) COA
- (c) HA
- (d) CN

Why: The tunnel runs from the **Home Agent** (start) to the current point of attachment. The **Foreign Agent** sits in the visited network and terminates it, so it is the **tunnel endpoint**. **COA** is the address *value*, not the entity; **CN** is the far-end correspondent.

Q. [PYQ] Which technology is commonly used in Wi-Fi to enhance signal range and reliability?

- (a) **MIMO (Multiple Input Multiple Output)**
- (b) WEP (Wired Equivalent Privacy)
- (c) TDMA
- (d) CDMA

Why: **MIMO** uses multiple antennas at both ends to exploit multipath reflections instead of suffering from them, improving range, reliability and throughput. **WEP** is (broken) security; TDMA and CDMA are cellular channel-access methods.

Q. [EXPECTED] Which IEEE standard defines Wireless LAN (Wi-Fi)?

- (a) 802.3
- (b) **802.11**
- (c) 802.15
- (d) 802.16

Why: 802.3 = Ethernet, 802.11 = Wi-Fi, 802.15 = Bluetooth/WPAN, 802.16 = WiMAX.

Q. [EXPECTED] [TRAP] Which Wi-Fi security standard is considered broken and should not be used?

- (a) WPA2
- (b) WPA3
- (c) **WEP**
- (d) 802.1X

Why: **WEP** (Wired Equivalent Privacy) has fatal cryptographic flaws and can be cracked in minutes. Progression of strength: WEP, WPA, WPA2, WPA3.

PART 8

WEB TECHNOLOGIES, XML, PHP AND E-COMMERCE

Weightage: 4 to 8 marks in DSSSB, 3 to 8 expected. Almost entirely syntax recall - the cheapest marks in the Computer Science half.

8.1 Web Architecture and HTTP

WEIGHTAGE: 1-2 marks | Priority HIGH

8.1.1 Questions

Q. [PYQ] What is the primary function of a web browser?

- (a) To store large amounts of data
- (b) To send and receive emails
- (c) To create websites
- (d) **To display and navigate web pages**

Why: A browser **renders and navigates**. Creating sites needs an editor; email needs a mail client.

Q. [PYQ] _____ are a database applications' universal front end that connects to the back end via the internet.

- (a) **Web browsers**
- (b) Web server
- (c) Software programs
- (d) Web application

Why: Whatever the database or platform, the **same browser** can act as the interface - that universality is exactly the point of the question.

Q. [PYQ] [TRAP] _____ is the standard that defines how web servers and application programs communicate.

- (a) JDBC
- (b) ODBC
- (c) CGI
- (d) **HTTP**

Why: KEY NOTE. This paper's key said HTTP. Textbook theory: **CGI (Common Gateway Interface)** is the standard defining how a web *server* passes a request to an external *application program*; **HTTP** defines *browser-to-server* communication. Judge by the pairing named in the question.

Q. [PYQ] Communication between an application program and a database server occurs via:

- (a) **ODBC OR JDBC**
- (b) API
- (c) web server
- (d) HTTP

Why: **ODBC** is the language-independent driver standard; **JDBC** is its Java equivalent. "API" is too general; a web server serves pages, not SQL.

Q. [EXPECTED] [TRAP] Which HTTP method sends data in the request body rather than the URL?

- (a) GET
- (b) **POST**
- (c) HEAD
- (d) OPTIONS

Why: **GET** puts parameters in the URL - visible, bookmarkable and length-limited. **POST** puts them in the body - hidden from the URL and unlimited in size, so it is used for forms, passwords and file uploads.

Q. [EXPECTED] Which HTTP status code range indicates a client error?

- (a) 2xx
- (b) 3xx
- (c) **4xx**
- (d) 5xx

Why: **1xx** informational, **2xx** success (200 OK), **3xx** redirection (301, 304), **4xx** client error (**400, 401, 403, 404**), **5xx** server error (500, 502, 503).

8.2 HTML and CSS

WEIGHTAGE: 1-3 marks | Exact syntax is tested, so memorise the strings | Priority HIGH

8.2.1 Questions

Q. [PYQ] Which of the following parameters is used to create a hyperlink in HTML?

- (a) Link
- (b) url
- (c) Path
- (d) **Href**

Why: href = **H**ypertext **R**EFerence. The <a> tag does nothing without it. There is no link, url or path attribute on an anchor.

Q. [PYQ] [TRAP] Which of the following opens the linked document in the window's full body among the target values?

- (a) _self
- (b) **_top**
- (c) _blank
- (d) _parent

Why: "Full body of the window" defines **_top** - it escapes *all* enclosing framesets. **_parent** climbs only **one** level; **_blank** opens a brand-new tab; **_self** (the default) reuses the current frame.

Q. [PYQ] Which of the following is the syntax to write an external CSS?

- (a) <style rel="stylesheet" type="text/css" href="style.css">
- (b) <style href="style.css" </style>
- (c) **<link rel="stylesheet" type="text/css" href="style.css">**
- (d)

Why: The decisive element is the **<link>** tag placed inside <head>. Anything using <style> is **internal** CSS; <a> creates a hyperlink. Memorise the full line exactly.

Q. [EXPECTED] [TRAP] Which CSS declaration has the highest priority?

- (a) An element selector
- (b) A class selector
- (c) An ID selector
- (d) **An inline style attribute**

Why: Specificity order: **inline style > ID > class/attribute/pseudo-class > element**, with !important overriding everything. When specificity ties, the **later** rule wins - that is the "cascading" in CSS.

Q. [PYQ] How to change all images to black and white (100% grey)?

- (a) img { filter: graycolor(100%);}
- (b) **img { filter: grayscale(100%);}**
- (c) img { graycolor(100%);}
- (d) img { filter: "grayscale(100%)";}

Why: The property is **filter**, the function is **grayscale** (American spelling, not "graycolor"), and the value is **not quoted**. Other useful filters: blur, brightness, contrast, invert, opacity, sepia.

Q. [EXPECTED] In the CSS box model, the order from innermost outward is:

- (a) margin, border, padding, content
- (b) **content, padding, border, margin**
- (c) content, border, padding, margin
- (d) padding, content, border, margin

Why: **Padding is inside the border; margin is outside it.** With the default content - box model, declared width excludes padding and border - which is why **box-sizing: border-box** is so widely used.

Q. [EXPECTED] Which HTML tag is an empty (void) element?

- (a) <p>
- (b) <div>
- (c) **
**
- (d)

Why: Void elements have no closing tag: `<br>`, `<hr>`, `<img>`, `<input>`, `<meta>`, `<link>`.

Q. [EXPECTED] [TRAP] Which input type allows selecting only ONE option from a group?

- (a) checkbox
- (b) **radio**
- (c) select multiple
- (d) text

Why: Radio buttons sharing the same `name` permit exactly one selection; checkboxes allow many. A `<select>` without `multiple` also allows one, but radio is the canonical answer.

8.3 JavaScript, DHTML, AJAX and jQuery

WEIGHTAGE: 2-4 marks | DOM methods and AJAX benefits recur | Priority HIGH

8.3.1 Questions

- Q. [PYQ] What is the output of:** `let x = 10; if (x > 5) console.log("Greater"); else console.log("Smaller");`
- (a) "Smaller"
 - (b) "undefined"
 - (c) **"Greater"**
 - (d) Error

Why: `10 > 5` is true.

- Q. [PYQ] [TRAP] What will the output be?** `document.write("Hello")` then `document.write("\nIndia")`
- (a) Hello
 - (b) HelloIndia
 - (c) India
 - (d) **Hello India**

Why: HTML collapses all whitespace - newlines, tabs and multiple spaces - into a **single space**. So `\n` does not create a line break but does not vanish either. For a real break you must write `<br>`. Note `\n` **does** work in a console, an alert box, or a plain text file.

- Q. [PYQ] Which DHTML method is used to change the style of an HTML element using CSS dynamically?**
- (a) `document.createElement("style")`
 - (b) `document.getElementById("element").setCSS()`
 - (c) **`element.style.backgroundColor = "red"`**
 - (d) `window.addEventListener("resize")`

Why: Every DOM element exposes a `style` object. Note the naming rule: CSS `background-color` (hyphenated) becomes JavaScript `backgroundColor` (camelCase), because hyphens are illegal in identifiers. There is no `setCSS()` method.

- Q. [PYQ] Which DHTML method is used to control the browser window size?**
- (a) **`window.resizeTo()`**
 - (b) `window.setSize()`
 - (c) `window.changeSize()`
 - (d) `document.resizeWindow()`

Why: Only `resizeTo()` and `resizeBy()` exist; the other three names are invented. Related real methods: `window.open()`, `window.moveTo()`, `window.scrollTo()`.

- Q. [EXPECTED] Which method selects a single element by its id?**
- (a) `document.getElementsByClassName()`
 - (b) **`document.getElementById()`**
 - (c) `document.createElement()`
 - (d) `document.write()`

Why: Note the singular/plural distinction: `getElementById` returns **one** element; `getElementsByClassName` and `getElementsByTagName` return **collections**. Modern equivalents: `querySelector()` and `querySelectorAll()`.

- Q. [EXPECTED] DHTML is best described as:**
- (a) a programming language
 - (b) a markup language
 - (c) **the combination of HTML, CSS, JavaScript and the DOM used to change a page after it loads**
 - (d) a server-side technology

Why: DHTML is **not a language** - it is the technique of combining those four to make a page dynamic without fetching a new page.

Q. [PYQ] Which advantage of AJAX helps in reducing bandwidth consumption?

- (a) AJAX uses less JavaScript.
- (b) AJAX only loads images on the page.
- (c) AJAX increases data transfer.
- (d) **AJAX sends and retrieves only small amounts of data from the server.**

Why: Only the needed fragment is exchanged instead of a full page with all its HTML, CSS and images. AJAX = **Asynchronous JavaScript And XML**, though today the payload is usually **JSON**.

Q. [EXPECTED] [TRAP] What is the main drawback of AJAX?

- (a) It is slower than page reloads
- (b) **The browser back button and bookmarking break unless specially handled**
- (c) It cannot access servers
- (d) It requires Java

Why: Because the URL does not change, history and bookmarks break by default. AJAX content may also be invisible to search engines, and it fails entirely if JavaScript is disabled.

Q. [PYQ] Which of the following methods is used to slide down an element?

- (a) **slideDown()**
- (b) moveBelow()
- (c) moveDown()
- (d) slideBelow()

Why: jQuery effects come in triplets: **slideDown/slideUp/slideToggle** and **fadeIn/fadeOut/fadeToggle**. There is no **moveBelow** or **slideBelow**.

Q. [PYQ] Which directive is used for binding the model data to the view?

- (a) ng-model-view
- (b) ng-model-app
- (c) **ng-model**
- (d) ng-app-model

Why: ``ng-model`` provides AngularJS **two-way data binding** - type in the input and the model updates, change the model and the input updates. The other three names do not exist.

8.4 XML

WEIGHTAGE: 1-3 marks | Parsers and DTD syntax recur | Priority HIGH

8.4.1 Questions

Q. [PYQ] ___ reads XML documents and provides access to their content and structure.

- (a) XML schema
- (b) XML pre-processor
- (c) **XML processor**
- (d) XML codes

Why: An **XML schema** *describes* what a valid document looks like; it reads nothing. The **processor (parser)** does the reading and also checks well-formedness.

Q. [PYQ] Which of the following is NOT an XML Parser?

- (a) **Shell**
- (b) JDOM
- (c) DOM
- (d) SAX

Why: A **shell** is an OS command interpreter (bash, csh, ksh). DOM, SAX and JDOM are genuine XML parsers.

Q. [EXPECTED] [TRAP] Which XML parser loads the entire document into memory as a tree?

- (a) SAX
- (b) StAX
- (c) **DOM**
- (d) JDOM

Why: **DOM** builds a full in-memory tree - allows random access and modification, but is memory-hungry. **SAX** is **event-based**, streaming through the document firing events; it is very memory-efficient but **read-only and forward-only**.

Q. [PYQ] How to write the external DTD?

- (a) <!DOCTYPE SYSTEM "file-name">
- (b) <!DOCTYPE element ROOT "file-name">
- (c) <!DOCTYPE SYSTEM "file-name" root-element>
- (d) **<!DOCTYPE root-element SYSTEM "file-name">**

Why: **Order is what is being tested:** <!DOCTYPE, then the **root element name**, then SYSTEM, then the quoted filename. Use PUBLIC instead of SYSTEM for a widely published standard DTD.

Q. [EXPECTED] [TRAP] Which of the following is TRUE of XML but not of HTML?

- (a) Tags are predefined
- (b) Closing tags are optional
- (c) **It is case sensitive and every tag must be closed**
- (d) Whitespace is collapsed

Why: XML is **case sensitive**, requires **every** tag to be closed, requires **quoted** attribute values, **preserves** whitespace, and needs exactly **one root element**. HTML is forgiving on all these counts.

Q. [EXPECTED] XSD is preferred over DTD because XSD:

- (a) is shorter
- (b) **is written in XML itself and supports data types and namespaces**
- (c) is older
- (d) needs no parser

Why: DTD uses its own non-XML syntax and has **no data types**. **XSD** (XML Schema Definition) is written in XML, supports integer/date/decimal types, namespaces and far richer constraints.

8.5 PHP

WEIGHTAGE: 0-2 marks | Array sorting functions are the classic question | Priority MEDIUM

8.5.1 Questions

Q. [PYQ] Which PHP function is used to sort an array in ascending order without affecting the keys?

- (a) ksort()
- (b) array_sort()
- (c) **asort()**
- (d) sort()

Why: Decode the naming scheme and you never memorise the table: plain **sort** sorts values and discards keys; prefix **a** (associative) **preserves the key-value pairing**; prefix **k** sorts by **key**; prefix **r** means reverse (descending); prefix **u** means a user-defined comparison. So "ascending, by value, keys preserved" = **asort()**. Note `array_sort()` does not exist in PHP.

Q. [EXPECTED] In PHP, which operator concatenates two strings?

- (a) +
- (b) **.**
- (c) &
- (d) ++

Why: PHP uses the **dot** for concatenation (`$a . $b`); + is strictly arithmetic. (JavaScript and Java use +, which is a common source of confusion.)

Q. [EXPECTED] [TRAP] What is the difference between include and require in PHP?

- (a) They are identical
- (b) **include gives a warning and continues if the file is missing; require gives a fatal error and stops**
- (c) require is faster
- (d) include cannot be used twice

Why: Use **require** for files the script genuinely cannot run without. The `_once` variants prevent accidental double inclusion.

Q. [EXPECTED] In PHP, variables enclosed in which quotes are interpolated (substituted)?

- (a) Single quotes
- (b) **Double quotes**
- (c) Backticks
- (d) Either

Why: `"Hello $name"` substitutes the value; `'Hello $name'` prints the literal text. PHP variables always begin with \$.

8.6 E-Commerce

WEIGHTAGE: 1-3 marks in DSSSB | Business-model classification is the standard question | Priority MEDIUM

8.6.1 Questions

Q. [PYQ] ___ primarily deals with buying and selling, particularly on a large scale.

- (a) **Commerce**
- (b) Finance
- (c) Supply chain
- (d) Distribution

Why: Commerce is *defined* by buying and selling at scale. Finance concerns money and capital; supply chain and distribution are logistics functions.

Q. [PYQ] The ___ E-commerce segment(s) is/are represented by OLX.

- (a) C2C, B2C and B2B
- (b) B2B
- (c) **C2C**
- (d) B2C

Why: On OLX an **individual sells to another individual** and the platform merely connects them - textbook **Consumer-to-Consumer**. Compare Amazon (mainly **B2C**), IndiaMART (**B2B**) and GeM (**B2G**).

Q. [EXPECTED] [TRAP] A freelancer offering services to a company on Upwork is an example of:

- (a) B2C
- (b) C2C
- (c) **C2B**
- (d) B2B

Why: C2B (Consumer-to-Business) - an individual supplies value to a business. Also covers stock-photo contributors and influencer marketing. It is the least-known model and therefore a favourite distractor.

Q. [PYQ] A fixed sum paid by customers for a service, typically on a monthly, quarterly or annual basis, is called ___ in electronic commerce.

- (a) licensing
- (b) affiliate
- (c) marketing
- (d) **subscription**

Why: "Fixed sum" plus "monthly/quarterly/annual" defines a recurring **subscription** (Netflix, Spotify). *Licensing* is a right-to-use fee; *affiliate* is referral commission.

Q. [PYQ] ___ is the process of visiting different web sites on the internet hosted by various companies, organisations, educational institutions, magazines, individuals, etc.

- (a) Webbing
- (b) Searching
- (c) Chatting
- (d) **Web surfing**

Why: **Surfing** = browsing around generally; **searching** implies a specific query. "Webbing" is not a term.

Q. [PYQ] [TRAP] Who is the father of e-commerce in India?

- (a) Bryan Eisenberg
- (b) Michael Aldrich
- (c) **K Vaitheeswaran**
- (d) Tim Ferris

Why: K Vaitheeswaran founded Fabmart/Indiaplaza in 1999, India's first e-commerce venture. **Michael Aldrich** invented online shopping **globally** in 1979 - examiners deliberately place both in the same question, so read whether it asks about India or the world.

Q. [EXPECTED] EDI stands for:

- (a) Electronic Data Integration
- (b) **Electronic Data Interchange**
- (c) Electronic Document Indexing
- (d) Encrypted Data Interface

Why: EDI is the structured computer-to-computer exchange of standard business documents (purchase orders, invoices) between organisations - a pre-internet forerunner of B2B e-commerce.

Q. [EXPECTED] Which of the following is NOT a security requirement of an e-commerce transaction?

- (a) Confidentiality
- (b) Integrity
- (c) Non-repudiation
- (d) **Portability**

Why: The security goals are confidentiality, integrity, authentication, **non-repudiation** and availability. **Portability** is a software quality attribute. Note **non-repudiation** - the sender cannot later deny sending - is achieved by digital signatures.

PART 9

**SOFTWARE ENGINEERING, GRAPHICS,
MIS AND EMERGING TECHNOLOGY**

Weightage: 8 to 18 marks combined. Chapter 9.4 on emerging technology is the one place where I am preparing you for a paper that may not look like the last one.

9.1 Software Engineering

WEIGHTAGE: 3-5 marks in DSSSB, 3-6 expected | SDLC models and coupling/cohesion are near-certain | Priority MEDIUM-HIGH

9.1.1 Questions

Q. [PYQ] ___ activities are at the very core of project completion work tasks, whereas ___ activities are not mandatory.

- (a) Requirements, arbitrary
- (b) Framework, umbrella
- (c) **Functional, non-functional**
- (d) Non-functional, functional

Why: **Functional** requirements define what the software must DO - without them the project is incomplete.

Non-functional requirements are quality attributes (performance, security, usability). Do not confuse with the other pair: **framework** activities are the main phases, **umbrella** activities run across the whole project (risk management, quality assurance, configuration management).

Q. [PYQ] The ___ design is a high-abstraction version of a system, followed by ___ design that breaks the concept into less-abstracted sub-systems, and later the ___ design showcases the implementation at module level.

- (a) detailed, system, architectural
- (b) detailed, architectural, system
- (c) architectural, detailed, system
- (d) **system, architectural, detailed**

Why: Always most abstract to most concrete: **System, Architectural, Detailed**. You picture the whole *system*, draw its *architecture*, then fill in the *details*.

Q. [PYQ] In building software, ___ should preferably be lower and ___ is preferred to be on the higher side.

- (a) inheritance from parent class, dependency with peer classes
- (b) aggregation with peer components, composition among classes
- (c) none of the given answer
- (d) **coupling among components, cohesion in bonding components**

Why: **LOW coupling, HIGH cohesion** - the single most important design maxim. Low coupling means a change in one module does not ripple outward; high cohesion means each module has one clear responsibility.

Q. [EXPECTED] [TRAP] Cohesion refers to:

- (a) interdependence between modules
- (b) **how strongly the elements within a single module belong together**
- (c) the number of modules
- (d) the depth of inheritance

Why: **Coupling is BETWEEN modules; cohesion is WITHIN one module**. Best cohesion is **functional**; worst is **coincidental**. Best coupling is **data/message**; worst is **content**.

Q. [PYQ] [TRAP] Within software engineering, which prescriptive model is INCOMPATIBLE with circumstances in which the requirements alter throughout development?

- (a) **Linear models**
- (b) Evolutionary models
- (c) Agile models
- (d) Incremental models

Why: "Linear model" = **Waterfall**. Each phase is frozen once complete, so a change discovered during coding cannot flow back into design. Evolutionary, agile and incremental models are all built to absorb change.

Q. [PYQ] [TRAP] Which of the following is NOT true about the Waterfall model?

- (a) Long duration
- (b) **High cost**
- (c) High-risk involvement

- (d) Low cost

Why: Waterfall is described as a **low-cost** model - no prototyping, no repeated risk analysis, minimal customer interaction. Its genuine drawbacks are long duration, high risk and total rigidity. Note the option list contains both "high cost" and "low cost", signalling which is tested.

Q. [PYQ] Which of the following is NOT a phase of the RAD model?

- (a) **Development modelling**
- (b) Business modelling
- (c) Process modelling
- (d) Data modelling

Why: RAD phases: **Business modelling, Data modelling, Process modelling, Application Generation, Testing and Turnover.** "Development modelling" is invented.

Q. [EXPECTED] [TRAP] Which SDLC model explicitly includes risk analysis in every iteration?

- (a) Waterfall
- (b) RAD
- (c) **Spiral**
- (d) V-Model

Why: The **Spiral model** (Barry Boehm) has four quadrants per loop: planning, **risk analysis**, engineering and evaluation. It suits large, expensive, high-risk projects. The **V-Model** pairs each development phase with a matching test phase.

Q. [EXPECTED] Which model is most appropriate when the customer cannot clearly state the requirements?

- (a) Waterfall
- (b) V-Model
- (c) **Prototype model**
- (d) Big Bang

Why: A quick working mock-up is built, shown to the user, and refined from their reaction - the fastest way to surface requirements the user could not articulate in the abstract.

Q. [EXPECTED] In Agile, a short fixed-length iteration producing a shippable increment is called a:

- (a) phase
- (b) milestone
- (c) **sprint**
- (d) baseline

Why: A sprint is typically 2-4 weeks. Scrum roles: **Product Owner** (owns the backlog), **Scrum Master** (facilitator), **Development Team**. Agile values working software over documentation and welcomes changing requirements.

Q. [PYQ] CMM stands for:

- (a) Cognitive Modularity Model
- (b) Capability Modularity Model
- (c) Cognitive Measurable Model
- (d) **Capability Maturity Model**

Why: Both words matter: **Capability** (what the organisation can do) and **Maturity** (how disciplined its process is). Levels: **Initial, Repeatable, Defined, Managed, Optimising** - "I Really Do My Own work."

Q. [EXPECTED] [NUMERICAL] At which CMM level is the process quantitatively measured and statistically controlled?

- (a) Level 2
- (b) Level 3
- (c) **Level 4**
- (d) Level 5

Why: **Level 4 = Managed (quantitatively managed)** - measurement and statistical control. **Level 5 = Optimising** adds continuous improvement and defect prevention.

Q. [EXPECTED] [TRAP] Which type of testing requires knowledge of the internal code structure?

- (a) Black box testing
- (b) **White box testing**

- (c) Acceptance testing
- (d) Beta testing

Why: **White box** (structural/glass box) examines internal logic - statement, branch and path coverage, cyclomatic complexity. **Black box** (functional) tests only inputs and outputs using equivalence partitioning and boundary value analysis.

Q. [EXPECTED] [TRAP] Beta testing is performed:

- (a) by developers at the development site
- (b) **by real users at the customer's site**
- (c) only by the QA team
- (d) before unit testing

Why: **Alpha testing** = at the developer's site by internal staff. **Beta testing** = at the customer's site by real users. Both are forms of **acceptance testing**.

Q. [EXPECTED] Re-running previously passed tests after a code change is called:

- (a) Unit testing
- (b) Smoke testing
- (c) **Regression testing**
- (d) Stress testing

Why: Regression testing confirms that a change has not broken something that used to work.

Q. [EXPECTED] [NUMERICAL] If a control flow graph has 12 edges and 9 nodes, the cyclomatic complexity is:

- (a) 3
- (b) 4
- (c) **5**
- (d) 21

Why: $V(G) = E - N + 2 = 12 - 9 + 2 = 5$. It equals the number of independent paths (and also decision points + 1), giving the minimum test cases for branch coverage.

Q. [EXPECTED] [TRAP] Verification answers which question?

- (a) Are we building the right product?
- (b) **Are we building the product right?**
- (c) Will the customer accept it?
- (d) Is it fast enough?

Why: **Verification** = conformance to specification, usually **without executing** the code (reviews, walkthroughs, inspections). **Validation** = does it meet the user's actual need, usually **by executing** it.

Q. [PYQ] Which of the following is the easiest to be transformed into a computer program?

- (a) **Pseudo-code**
- (b) Algorithm
- (c) Flowchart
- (d) None of the above

Why: Pseudo-code already uses programming constructs (IF, WHILE, assignment) in a code-like layout, so conversion is nearly line-by-line. A plain-English algorithm is more abstract; a flowchart is graphical.

Q. [PYQ] Program links with other parts of the program in a flowchart are represented by ____.

- (a) rectangles
- (b) **circles**
- (c) trapezoids
- (d) rhombuses

Why: The **connector** symbol is a circle, used to join parts of a flowchart across pages. **Rectangle** = processing, **diamond/rhombus** = decision, **parallelogram** = input/output, **oval** = start/stop.

Q. [EXPECTED] COCOMO is used for:

- (a) testing
- (b) **effort and cost estimation**
- (c) configuration management
- (d) requirement gathering

Why: **CONstructive COst MOdel** (Boehm) estimates effort in person-months from size in KLOC. Its three modes are **Organic**, **Semi-detached** and **Embedded**. The alternative, language-independent technique is **Function Point analysis**.

Q. [EXPECTED] The longest path through a PERT network, which determines the minimum project duration, is called the:

- (a) slack path
- (b) **critical path**
- (c) Gantt path
- (d) baseline

Why: Any delay on the **critical path** delays the whole project. A **Gantt chart** shows tasks against a calendar; **PERT** shows dependencies as a network.

9.2 Computer Graphics and Multimedia

WEIGHTAGE: 2-4 marks in DSSSB | Algorithms and projections recur | Priority MEDIUM

9.2.1 Questions

Q. [PYQ] In order to move objects in computer graphics that generate animation effects, which of the following is necessary?

- (a) **Various transformation operations**
- (b) Line rasterisation sequencing
- (c) Phong model of lighting
- (d) Eigen value computations

Why: Motion is repeated **translation, rotation and scaling** between frames. Rasterisation draws a line onto pixels; the Phong model shades surfaces.

Q. [EXPECTED] [TRAP] Why are homogeneous coordinates used in 2D transformations?

- (a) To reduce memory
- (b) **So that translation can also be expressed as matrix multiplication**
- (c) To support colour
- (d) To speed up rasterisation

Why: Scaling and rotation are naturally multiplicative, but translation is additive. Writing a point as (x, y, 1) makes translation multiplicative too, so several transformations can be **concatenated into a single composite matrix**.

Q. [PYQ] Which of the following is a clipping algorithm?

- (a) Simple DDA
- (b) **Liang Barsky's algorithm**
- (c) Bresenham's algorithm
- (d) Mid-point Algorithm

Why: DDA and Bresenham draw lines; mid-point draws circles. Liang-Barsky and Cohen-Sutherland clip lines; Sutherland-Hodgman clips polygons.

Q. [EXPECTED] [TRAP] Bresenham's line algorithm is preferred over DDA because it:

- (a) draws thicker lines
- (b) **uses only integer arithmetic, making it faster and free of round-off error**
- (c) supports colour
- (d) works in 3D

Why: DDA uses **floating-point** addition and rounding, which is slower and accumulates error. Bresenham uses an integer decision parameter with only addition, subtraction and bit shifts.

Q. [PYQ] [TRAP] ___ and ___ views cannot be framed together for rendering a computer visual appropriately.

- (a) **Orthogonal projection, horizontal view**
- (b) Vertical view, rear projection
- (c) Front-side, rear-side
- (d) Parallel projection, perspective projection

Why: Orthographic projection uses projectors **perpendicular** to the view plane, which conflicts with simultaneously presenting a horizontal viewing direction in the same frame. Note that "parallel projection, perspective projection" is also a genuinely incompatible pair in principle - reason from **which two views demand contradictory projector geometry**.

Q. [EXPECTED] In which projection do distant objects appear smaller?

- (a) Orthographic
- (b) Oblique
- (c) **Perspective**
- (d) Isometric

Why: **Perspective** projectors converge at a centre of projection, producing vanishing points and realism - like the eye or a camera. **Parallel/orthographic** projection preserves true dimensions and is used for engineering drawings.

Q. [PYQ] [TRAP] Which of these is NOT a parameter upon which the influence of lighting is based?

- (a) Light Source
- (b) Position of Observer
- (c) **Rendering Algorithm**
- (d) Surface

Why: The physical inputs to an illumination calculation are the **light source**, the **surface** properties and the **observer's position** (needed for specular highlights). The **rendering algorithm** is the *method of computing* the image, not a parameter of the lighting - the recipe is not an ingredient.

Q. [EXPECTED] Which shading method interpolates the surface normal and computes lighting per pixel?

- (a) Flat shading
- (b) Gouraud shading
- (c) **Phong shading**
- (d) Constant shading

Why: **Gouraud** interpolates *intensity* computed at the vertices - fast but can miss highlights inside a polygon. **Phong** interpolates the *normal* and lights each pixel - slower but far better highlights.

Q. [EXPECTED] Which hidden surface removal technique stores a depth value for every pixel?

- (a) Back-face detection
- (b) Painter's algorithm
- (c) **Z-buffer algorithm**
- (d) Ray tracing

Why: The **Z-buffer (depth buffer)** keeps the nearest surface per pixel - simple and universally used in hardware. **Ray tracing** is far more realistic (reflection, refraction, shadows) but computationally expensive.

Q. [PYQ] The construction of computer-generated simulations that simulate real-world activities using interactive graphics software and hardware is called:

- (a) **virtual reality systems**
- (b) hybrid systems
- (c) genetic algorithms
- (d) fuzzy neural networks

Why: The other three are **Artificial Intelligence** techniques. VR's "three I's" are **Immersion, Interaction, Imagination**; **AR** overlays content on the real world rather than replacing it.

Q. [PYQ] Which tool is used to adjust the brightness and contrast of an image in Adobe Photoshop?

- (a) Healing brush tool
- (b) Paint bucket tool
- (c) **Adjustments panel**
- (d) Selection tool

Why: The **Adjustments panel** holds brightness/contrast, levels, curves, hue/saturation. The **healing brush** repairs blemishes; the **paint bucket** fills areas.

Q. [PYQ] Which of the following formats is commonly used for exporting images for the web?

- (a) **PNG**
- (b) PSD
- (c) BMP
- (d) TIFF

Why: **PNG** = lossless with transparency, ideal for logos and sharp-edged graphics. **PSD** is Photoshop's editable working file; **BMP** is uncompressed; **TIFF** is for print/archival - all far too large for the web. For photographs, **JPEG** is preferred.

Q. [EXPECTED] [TRAP] Which image format scales to any size without loss of quality?

- (a) JPEG
- (b) PNG
- (c) GIF
- (d) **SVG**

Why: **SVG** is **vector** - stored as mathematical descriptions of shapes. JPEG, PNG, GIF and BMP are **raster** (a fixed pixel grid) and blur when enlarged.

Q. [PYQ] What is the main function of the 'cutting' tool in image editing?

- (a) To remove parts of the image**
- (b) To change the image resolution
- (c) To apply color effects
- (d) To resize the image

Why: Cutting/cropping removes content; it does not resample or recolour.

9.3 Management Information Systems and Knowledge Management

WEIGHTAGE: 3-6 marks in DSSSB | Appeared in all three papers supplied | Priority MEDIUM-HIGH

9.3.1 Questions

Q. [PYQ] ___ is a set of interrelated components that collect (or retrieve), process, store and distribute information to support decision-making and control in an organisation.

- (a) An information technology
- (b) Mini model of a processed system
- (c) Max model of a processed system
- (d) **An information system**

Why: IT is the technology; an information **SYSTEM** is technology + people + processes working toward an organisational purpose. The distractor "information technology" is deliberately close.

Q. [PYQ] In a management information system, what is/are the formal social unit that is/are devoted to attaining specified goals?

- (a) Organisation, management and marketing
- (b) **Organisation**
- (c) Management
- (d) Marketing

Why: An **organisation** is the formal social structure that takes in resources and produces outputs toward stated goals. **Management** is a *function*; **marketing** is an *activity*. Only organisation is the social *unit*.

Q. [PYQ] DSS is a computer program application used to improve a company's ___ capabilities.

- (a) **decision-making capabilities**
- (b) marketing capabilities
- (c) delegation capabilities
- (d) resources utilisation capabilities

Why: The name states it: **D**ecision **S**upport **S**ystem. It analyses semi-structured problems with models and what-if analysis.

Q. [PYQ] Which of the following systems comprises word processing, electronic filing, electronic mail, message switching, data storage and data and voice communication?

- (a) Executive Support Systems
- (b) **Office Automation Systems**
- (c) Customer Relationship Management
- (d) Decision Support Systems

Why: Every item listed is a routine **office** function. An ESS gives executives strategic dashboards; a DSS runs analytical models; CRM manages customers.

Q. [EXPECTED] [TRAP] Which system answers "what happened?" with routine, fixed-format reports?

- (a) TPS
- (b) **MIS**
- (c) DSS
- (d) ESS

Why: **TPS** records raw transactions. **MIS** summarises them into routine reports - "what happened?". **DSS** answers "what if?" and "what is best?". **ESS** answers strategic questions with highly summarised internal and external data.

Q. [PYQ] The set of processes developed in an organisation to create, gather, store, transfer and apply knowledge best describes:

- (a) **knowledge management**
- (b) organisational learnings
- (c) knowledge asset
- (d) organisational memory

Why: A **knowledge asset** is the thing being managed; **organisational memory** is the store; **organisational learning**

is the outcome. Only **KM** is the set of processes.

Q. [PYQ] Which of the following is NOT the purpose of knowledge management system?

- (a) Driving strategy
- (b) Integration
- (c) Sharing of knowledge
- (d) **Portability**

Why: **Portability** is a software quality attribute (running on different platforms), unrelated to KM goals.

Q. [PYQ] An individual's shared experience results in ____.

- (a) systematic knowledge
- (b) shared knowledge
- (c) **tacit knowledge**
- (d) tacit, systematic and shared knowledge

Why: **Tacit** knowledge is personal, experience-based and hard to write down - knowing how to ride a bicycle, or an experienced teacher's classroom instinct. **Explicit** knowledge is codified in manuals and databases.

Q. [EXPECTED] [TRAP] In the SECI model, converting tacit knowledge into explicit knowledge is called:

- (a) Socialisation
- (b) **Externalisation**
- (c) Combination
- (d) Internalisation

Why: Nonaka and Takeuchi's four modes: **Socialisation** (tacit to tacit), **Externalisation** (tacit to explicit), **Combination** (explicit to explicit), **Internalisation** (explicit to tacit).

Q. [PYQ] The lowest cost solution to developing a KM system is:

- (a) **in-house development**
- (b) outsourcing
- (c) development by end users
- (d) off-the-shelf solution

Why: Building internally uses existing staff and infrastructure, avoiding vendor fees, licences and consultancy charges.

Q. [PYQ] Which qualitative statistic is difficult to quantify precisely?

- (a) Model
- (b) Infrastructure
- (c) **Value**
- (d) Decision making

Why: **Value** is inherently subjective and context-dependent, so it resists precise measurement. Infrastructure can be counted; decision-making can be measured through outcomes and cycle times.

9.4 Emerging Technology - AI, Cloud, IoT and Cyber Security

WEIGHTAGE: 0-2 marks observed in DSSSB | 2-6 and RISING across KVS / NVS / EMRS | Priority MEDIUM-HIGH

Why this chapter exists

The three DSSSB papers you shared barely touched these topics. But newer papers across all these bodies are adding them, and **EMRS already has an explicit ICT and cyber-security component** in its published syllabus. Recent school curricula (CBSE Class XI-XII Computer Science and Informatics Practices) also now include Python, data handling and cyber safety - and TGT papers tend to follow the school syllabus they are recruiting teachers for.

This is the one chapter that prepares you for the paper that may **not** look like the last one. Treat it as insurance: a single pass is enough, but do not skip it.

9.4.1 Questions

Q. [EXPECTED] Machine learning in which the model is trained on labelled input-output pairs is called:

- (a) Unsupervised learning
- (b) **Supervised learning**
- (c) Reinforcement learning
- (d) Deep learning

Why: **Supervised** = labelled data (classification, regression). **Unsupervised** = unlabelled, finds structure (clustering, association). **Reinforcement** = learning from rewards and penalties through interaction.

Q. [EXPECTED] [TRAP] Which of the following is an unsupervised learning technique?

- (a) Linear regression
- (b) Decision tree classification
- (c) **K-means clustering**
- (d) Logistic regression

Why: **K-means clustering** groups unlabelled data. The other three all need labelled training data.

Q. [EXPECTED] The Turing Test is used to evaluate:

- (a) processing speed
- (b) **whether a machine's behaviour is indistinguishable from a human's**
- (c) memory capacity
- (d) network latency

Why: Proposed by **Alan Turing** (1950), who is regarded as the father of theoretical computer science and AI.

Q. [EXPECTED] Which of the following is NOT a characteristic of cloud computing?

- (a) On-demand self-service
- (b) Broad network access
- (c) Resource pooling
- (d) **Permanent local storage on the user's device**

Why: The NIST characteristics are on-demand self-service, broad network access, resource pooling, **rapid elasticity** and **measured service**. Cloud deliberately moves storage *off* the local device.

Q. [EXPECTED] [TRAP] In which cloud service model does the provider supply ready-to-use application software over the internet?

- (a) IaaS
- (b) PaaS
- (c) **SaaS**
- (d) DaaS

Why: **IaaS** gives raw infrastructure (virtual machines, storage - e.g. AWS EC2). **PaaS** gives a development platform (e.g. Google App Engine). **SaaS** gives finished applications (Gmail, Google Docs, Salesforce). Memorise by what you receive: Infrastructure, Platform, Software.

Q. [EXPECTED] IoT stands for:

- (a) Internet of Technology
- (b) **Internet of Things**
- (c) Integration of Things
- (d) Internet over Telephony

Why: IoT = everyday physical objects fitted with sensors and network connectivity - smart meters, wearables, connected appliances. Key concerns are **security** and **data privacy**, because such devices are often weakly protected.

Q. [EXPECTED] A blockchain is best described as:

- (a) a centralised database
- (b) **a distributed, append-only ledger where blocks are linked by cryptographic hashes**
- (c) a type of firewall
- (d) a cloud storage service

Why: Each block stores the **hash of the previous block**, so altering any past block invalidates every block after it - that is the source of tamper resistance.

Q. [EXPECTED] [TRAP] Which of the following best defines phishing?

- (a) Flooding a server with traffic
- (b) **Fraudulent messages or websites that trick users into revealing credentials**
- (c) Encrypting a victim's files for ransom
- (d) Intercepting traffic between two parties

Why: The distractors define **DoS**, **ransomware** and a **man-in-the-middle** attack respectively. Phishing exploits the *human*, not the software - which is why user awareness is the main defence.

Q. [EXPECTED] Two-factor authentication improves security because it requires:

- (a) two passwords
- (b) **two different categories of evidence, such as something you know plus something you have**
- (c) two user accounts
- (d) encryption twice

Why: The three authentication factors are **something you know** (password), **something you have** (phone, token) and **something you are** (biometric). Combining categories is what makes 2FA strong.

Q. [EXPECTED] Which Indian law gives legal recognition to electronic records and digital signatures?

- (a) RTI Act 2005
- (b) **IT Act 2000**
- (c) Companies Act 2013
- (d) RPwD Act 2016

Why: The **Information Technology Act, 2000** also defines cybercrimes and penalties. Useful for both the Computer Science and General Awareness sections.

Q. [EXPECTED] Which of the following is a strong password practice?

- (a) Using your date of birth
- (b) Reusing one password everywhere
- (c) **Using a long passphrase with mixed character types, unique per site**
- (d) Sharing it with a trusted colleague

Why: Length matters more than complexity. Related concepts worth knowing for ICT sections: **password managers**, **OTP**, **biometric authentication**, **VPN**, **firewall**, **antivirus**, **regular patching**.

Q. [EXPECTED] [TRAP] What is the primary purpose of a firewall?

- (a) To remove viruses from files
- (b) **To filter incoming and outgoing network traffic according to security rules**
- (c) To encrypt stored data
- (d) To back up data

Why: A firewall controls **traffic**; an **antivirus** removes malware; **encryption** protects stored or transmitted data. An **IDS** detects intrusions and an **IPS** blocks them.

PART 10

MATHEMATICS, STATISTICS AND THE DSSSB-SPECIFIC TOPICS

Weightage: 4 to 8 marks (Maths/Stats) plus 6 to 10 marks (Physics / Accounting / Economics) in DSSSB. Largely absent from KVS and NVS Computer Science papers.

Read this before you study this Part

This Part exists because **DSSSB draws its Discipline sections from the wider graduate stream**, so Physics, Accountancy and Business Economics questions appear alongside the computing ones. In the three papers you shared these were worth **6 to 10 marks per paper**.

If you are sitting only KVS or NVS Computer Science, you may safely de-prioritise Chapter 10.3 onwards. If you are sitting DSSSB, you cannot - it is too many marks to abandon.

The strategy is narrow and deliberate: learn the **specific formulas** below, not the subjects. Almost every question in those papers was a one-step formula application. If a question needs more than one unfamiliar step, skip it.

10.1 Numerical Methods

WEIGHTAGE: 1-3 marks in DSSSB | Interpolation and difference operators recur | Priority MEDIUM

10.1.1 Questions

Q. [PYQ] Interpolation is carried out using data from:

- (a) regression analysis
- (b) curve fitting and regression analysis
- (c) linear interpolation
- (d) **curve fitting and linear analysis**

Why: The distinguishing fact: **interpolation passes exactly through every known data point**, whereas **regression** finds a best-fit line that need not pass through any of them. So any option mentioning regression is wrong.

Q. [PYQ] Which method creates interpolants that are softer?

- (a) Polynomial and spline interpolation
- (b) **Piecewise constant interpolation**
- (c) Polynomial interpolation
- (d) Linear interpolation and spline interpolation

Why: Piecewise constant (nearest-neighbour) interpolation produces a plain **step function** with no curvature at all - the crudest and "softest" interpolant. **Spline** interpolation is the smoothest practical choice; high-degree **polynomial** interpolation oscillates badly near the edges (Runge's phenomenon).

Q. [PYQ] The ___ interpolation method is used in probability theory and statistics, and the Gaussian process is one example.

- (a) linear
- (b) polynomial
- (c) spline
- (d) **nonlinear**

Why: The **Gaussian process** (also called kriging) is a nonlinear, probabilistic interpolation technique used heavily in statistics and machine learning. Linear, polynomial and spline interpolation are deterministic curve fits.

Q. [PYQ] [NUMERICAL] The values of x are 1, 3, 4 and those of f(x) are 1, 27, 64. What is the polynomial using the Lagrange interpolation formula?

- (a) **$8x^2 - 19x + 12$**
- (b) $8x^2 + 9x + 12$
- (c) $8x^2 - 9x + 12$
- (d) $8x^2 + 9x + 12$

Why: Do not expand the Lagrange formula under exam pressure - verify the options instead. The correct polynomial must reproduce every data point. Test $8x^2 - 19x + 12$: at $x=1$, $8-19+12 = 1$; at $x=3$, $72-57+12 = 27$; at $x=4$, $128-76+12 = 64$. All match. Test a distractor $8x^2+9x+12$ at $x=1$: $8+9+12 = 29$, not 1 - fails immediately.

Substitute the smallest x into each option; usually three fail at once.

Q. [PYQ] [NUMERICAL] What is the value of Δ^n of $e^{(ax+b)}$?

- (a) $(e^{(ah)} - 1)^n e^{(ax-b)}$
- (b) **$(e^{(ah)} - 1)^n e^{(ax+b)}$**
- (c) $(e^{(ah)} + 1)^n e^{(ax-b)}$
- (d) $(e^{(ah)} + 1)^n e^{(ax+b)}$

Why: Compute one difference and spot the pattern: $\Delta e^{(ax+b)} = e^{(a(x+h)+b)} - e^{(ax+b)} = e^{(ax+b)}(e^{(ah)} - 1)$. So each application multiplies by the factor **$(e^{(ah)} - 1)$** , and n applications give **$(e^{(ah)} - 1)^n e^{(ax+b)}$** . Check two things in the options: the sign inside the bracket must be **minus**, and the exponent must remain **$ax + b$** (differencing never alters b).

Q. [PYQ] [NUMERICAL] If μ is an averaging operator and δ is the central difference operator, what is μ^2 ?

- (a) **$\mu^2 = (1/4)(\delta^2 + 4)$**
- (b) $\mu^2 = (1/2)(\delta^2 + 4)$

(c) $\mu^2 = (1/4)(\delta^2 + 6)$

(d) $\mu^2 = (1/4)(\delta^2 + 8)$

Why: From $\mu = (E^{1/2} + E^{-1/2})/2$, squaring gives $\mu^2 = (E + 2 + E^{-1})/4$. Since $\delta = E^{1/2} - E^{-1/2}$, $\delta^2 = E - 2 + E^{-1}$, so $E + E^{-1} = \delta^2 + 2$. Substituting: $\mu^2 = (\delta^2 + 2 + 2)/4 = (\delta^2 + 4)/4$.

Remember the pair of numbers: **one quarter, plus four**.

Q. [EXPECTED] The relation between the forward difference operator and the shift operator is:

(a) $\Delta = E + 1$

(b) $\Delta = E - 1$

(c) $\Delta = 1 - E$

(d) $\Delta = E$

Why: Since $\Delta f(x) = f(x+h) - f(x)$ and $E f(x) = f(x+h)$, we get $\Delta = E - 1$, equivalently $E = 1 + \Delta$. Also: $\nabla = 1 - E^{-1}$, and $\delta = E^{1/2} - E^{-1/2}$.

10.2 Probability and Statistics

WEIGHTAGE: 2-4 marks in DSSSB | Priority MEDIUM

10.2.1 Questions

Q. [PYQ] Who invented the probability definition?

- (a) **Simon Laplace**
- (b) Euclid
- (c) None of the given answer
- (d) Einstein

Why: **Pierre-Simon Laplace** gave the classical definition (favourable outcomes over total equally likely outcomes) in 1812. Euclid is geometry.

Q. [PYQ] Which probability calculus of views obeys particular rules?

- (a) Standard deviation
- (b) Variance
- (c) **Bayesian probability**
- (d) Frequency probability

Why: The phrase "calculus of **views**" is the clue: **Bayesian** probability treats probability as a **degree of belief** that is updated as evidence arrives, according to fixed rules. Standard deviation and variance are measures of *spread*, not interpretations of probability.

Q. [PYQ] [NUMERICAL] If two dice are thrown, find the probability that the sum is neither 5 nor 9.

- (a) 3/9
- (b) 5/9
- (c) 1/9
- (d) **7/9**

Why: Total outcomes = 36. Sum 5 occurs 4 ways (1+4, 2+3, 3+2, 4+1); sum 9 occurs 4 ways (3+6, 4+5, 5+4, 6+3). So $P(5 \text{ or } 9) = 8/36 = 2/9$, and $P(\text{neither}) = 1 - 2/9 = 7/9$. **Whenever a question says "neither/nor" or "not", count the unwanted cases and subtract from 1.**

Q. [PYQ] [NUMERICAL] A six-faced die is thrown. The probability that the number on top is even is:

- (a) 1/4
- (b) 1/6
- (c) **1/2**
- (d) 1/9

Why: Even outcomes {2, 4, 6} out of 6 gives $3/6 = 1/2$. The distractor 1/6 is the probability of one *specific* number.

Q. [PYQ] [NUMERICAL] Eighty-eight residents are classified as male/female and as tea drinkers/non-drinkers. For testing association between sex and tea drinking, the test statistic follows:

- (a) t distribution with 1 degree of freedom
- (b) Chi Square distribution with 2 degrees of freedom
- (c) Normal distribution
- (d) **Chi square distribution with 1 degree of freedom**

Why: Testing **association between two categorical variables** means a **chi-square test of independence**, which eliminates the t and normal options. Degrees of freedom = $(r-1)(c-1) = (2-1)(2-1) = 1$. Note the sample size 88 is a red herring - it never enters the df calculation.

Q. [PYQ] [TRAP] The algebraic sum of deviations is zero when measured from the ____.

- (a) second quartile
- (b) geometric mean
- (c) **arithmetic mean**
- (d) harmonic mean

Why: Proof in one line: $\sum(x - \text{mean}) = \sum x - n(\text{mean})$, and since $\text{mean} = (\sum x)/n$ we have $n(\text{mean}) = \sum x$, so the difference is **0**. Verify with 2, 4, 6: mean is 4 and deviations -2, 0, +2 sum to zero. Related property: the sum of *squared* deviations is **minimum** about the arithmetic mean.

Q. [PYQ] Identify the generic probability density function that corresponds with discrete random variables.

- (a) Cumulative distribution
- (b) Gaussian distribution
- (c) **Poisson distribution**
- (d) Rayleigh distribution

Why: **Poisson** is the discrete one - it counts whole numbers of rare events (0, 1, 2, ...) and has the special property **mean = variance = lambda**. Gaussian and Rayleigh are **continuous**; "cumulative distribution" is a general concept applying to both.

Q. [EXPECTED] For a binomial distribution, the mean is:

- (a) $np(1-p)$
- (b) **np**
- (c) p/n
- (d) n/p

Why: Mean = **np** , variance = **npq** where $q = 1-p$. Binomial counts successes in n independent trials.

Q. [EXPECTED] [TRAP] In a perfectly symmetric normal distribution:

- (a) mean > median > mode
- (b) mean < median < mode
- (c) **mean = median = mode**
- (d) mode is undefined

Why: This equality is the signature of the normal (Gaussian) bell curve. In a **positively skewed** distribution mean > median > mode; in a **negatively skewed** one the order reverses.

Q. [EXPECTED] Which measure of central tendency is least affected by extreme values?

- (a) Arithmetic mean
- (b) **Median**
- (c) Geometric mean
- (d) Range

Why: The **median** depends only on position, not magnitude, so a single outlier cannot drag it. The mean is highly sensitive. Also remember **AM $\geq$ GM $\geq$ HM** always.

10.3 Calculus, Matrices and Differential Equations

WEIGHTAGE: 2-4 marks in DSSSB | Priority MEDIUM (skip for KVS/NVS)

10.3.1 Questions

Q. [PYQ] [NUMERICAL] The series $\sum 1/(n^2)$ raised to the power $1/5$ is:

- (a) None of the given options
- (b) divergent
- (c) oscillatory convergent
- (d) **convergent**

Why: KEY NOTE. Simplify first: $1/(n^2)^{(1/5)} = 1/n^{(2/5)}$, a **p-series with $p = 0.4$** . The standard rule is that $\sum 1/n^p$ **converges if $p > 1$ and diverges if $p \leq 1$** , which makes this **divergent** - yet the official key marked it convergent. Learn the rule properly: $\sum 1/n$ diverges (harmonic), $\sum 1/n^2$ converges, $\sum 1/\sqrt{n}$ diverges.

Q. [PYQ] [NUMERICAL] If $u = \tan^{-1}(y/x)$, then $x \frac{du}{dx} + y \frac{du}{dy} = ?$

- (a) $\sin 2u$
- (b) **0**
- (c) None of the given options
- (d) $\cos 2u$

Why: Use **Euler's theorem**: if f is homogeneous of degree n then $x \frac{f}{dx} + y \frac{f}{dy} = n f$. Here $u(tx, ty) = \arctan(ty/tx) = \arctan(y/x) = u$, so u is homogeneous of **degree 0** and the answer is $0 \times u = 0$. Key insight: **any function of the pure ratio y/x is homogeneous of degree zero**, so this expression is always 0.

Q. [PYQ] [NUMERICAL] Find a unit vector normal to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$.

- (a) $(1/\sqrt{2})(i - 3j - 2k)$
- (b) $(1/\sqrt{14})(i - 3j + 2k)$
- (c) **$(1/\sqrt{14})(-i + 3j + 2k)$**
- (d) $(i + 3j + 2k)$

Why: The **gradient is normal to the surface**. Partial derivatives: $dF/dx = 3x^2 + 3yz = 3 - 6 = -3$; $dF/dy = 3y^2 + 3xz = 12 - 3 = 9$; $dF/dz = 3xy = 6$. So $\text{grad } F = -3i + 9j + 6k = 3(-i + 3j + 2k)$. Magnitude of the bracket = $\sqrt{1+9+4} = \sqrt{14}$. Unit normal = **$(-i + 3j + 2k)/\sqrt{14}$** . Always factor out the common term *before* computing the magnitude.

Q. [PYQ] [NUMERICAL] The volume of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$ and $x/a + y/b + z/c = 1$ is:

- (a) $abc/4$
- (b) $abc/2$
- (c) $abc/10$
- (d) **$abc/6$**

Why: A standard result worth memorising. The plane cuts the axes at $(a,0,0)$, $(0,b,0)$, $(0,0,c)$, forming a tetrahedron with three mutually perpendicular edges. Volume = $(1/3)(\text{base area})(\text{height}) = (1/3)(ab/2)(c) = \mathbf{abc/6}$. Hook: the corner tetrahedron is exactly **one sixth** of the cuboid with sides a, b, c .

Q. [PYQ] [NUMERICAL] The characteristic equation of $A = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$ is:

- (a) $\lambda^2 + 6\lambda - 7 = 0$
- (b) $\lambda^2 - 6\lambda + 7 = 0$
- (c) **$\lambda^2 - 6\lambda - 7 = 0$**
- (d) $\lambda^2 + 6\lambda + 7 = 0$

Why: For any 2×2 matrix use **$\lambda^2 - (\text{trace})\lambda + (\text{determinant}) = 0$** . Trace = $1+5 = 6$; determinant = $(1)(5) - (4)(3) = -7$. So $\lambda^2 - 6\lambda - 7 = 0$. Watch both signs: the trace term is always *minus* trace, and here the determinant is itself negative.

Q. [PYQ] [NUMERICAL] If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ then A^6 is:

- (a) $\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 6 & 0 & 6 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 6 & 0 & 1 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

(d) $[[1,0,0],[0,1,0],[6,0,1]]$

Why: Compute A^2 , spot the pattern, then jump. A^2 gives $[[1,0,0],[0,1,0],[2,0,1]]$ - the (2,2) entry becomes $(-1)^n$ and the (3,1) entry becomes n . For $n = 6$ (even), $(-1)^6 = +1$ and the (3,1) entry = **6**. Use **parity**: an even power always turns a -1 diagonal entry into +1.

Q. [PYQ] [NUMERICAL] Which differential equation corresponds to $y = a e^{(2x)} + b e^{(-x)}$?

- (a) $y'' - 2y' + y = 0$
- (b) $y'' + 2y' + y = 0$
- (c) $y'' + y' - 2y = 0$
- (d) $y'' - y' - 2y = 0$

Why: Read the exponents off the solution - they are the roots of the auxiliary equation. Roots 2 and -1 give $(m-2)(m+1) = m^2 - m - 2 = 0$, and replacing m^2 with y'' , m with y' and the constant with y gives $y'' - y' - 2y = 0$. Verify with $y = e^{(2x)}$: $4e - 2e - 2e = 0$. Correct.

Q. [PYQ] [NUMERICAL] The general solution of $dy/dx = e^{(x+y)}$ is:

- (a) $e^{(-x)} - e^{(-y)} = c$
- (b) $e^{(-x)} + e^y = c$
- (c) $e^x + e^{(-y)} = c$
- (d) $e^{(-x)} + e^y = c$

Why: Separate: $e^{(x+y)} = e^x \cdot e^y$, so $e^{(-y)} dy = e^x dx$. Integrating gives $-e^{(-y)} = e^x + k$, and rearranging gives $e^x + e^{(-y)} = c$. The sign flip comes from integrating $e^{(-y)}$, which yields *minus* $e^{(-y)}$.

Q. [PYQ] [NUMERICAL] [TRAP] The particular solution of $y'' - 6y' + 9y = 2e^{(3t)}$ is:

- (a) $e^{(3t)}/2$
- (b) $t^2 e^{(3t)}$
- (c) $e^{(3t)}$
- (d) $e^{(3t)}/9$

Why: The auxiliary equation $m^2 - 6m + 9 = (m-3)^2 = 0$ has $m = 3$ as a **DOUBLE root**, and the forcing term is $e^{(3t)}$ - the resonance case. **Rule: if r is not a root, PI is proportional to $e^{(rt)}$; if r is a simple root, multiply by t ; if r is a double root, multiply by t^2 .** Hence $t^2 e^{(3t)}$. The same idea appeared again as $(D^4 - 18D^2 + 81)y$, which factors to $(D-3)^2(D+3)^2$ and gives $x^2 e^{(3x)}$.

10.4 Physics Topics That Appeared

WEIGHTAGE: 3-5 marks in DSSSB only | Priority LOW for KVS/NVS

10.4.1 Questions

Q. [PYQ] [NUMERICAL] An LED emits green light of wavelength 5511.11 Angstrom. Find Eg.

- (a) 1.25 eV
- (b) **2.25 eV**
- (c) 3.606 eV
- (d) 36.06 eV

Why: Memorise the shortcut: $E \text{ (in eV)} = 12400 / \lambda \text{ (in Angstrom)} = 12400/5511.11 = 2.25 \text{ eV}$. The distractor 3.606 is the answer in units of 10^{-19} joules mistaken for eV - always finish the conversion.

Q. [PYQ] What is the lifetime of an atom's ground state in stimulated absorption?

- (a) 2 minutes
- (b) 11 hours
- (c) 5 seconds
- (d) **Infinity**

Why: The **ground state is the lowest energy level** - there is nowhere lower for the atom to fall to, so without external excitation it remains there indefinitely. Contrast an ordinary excited state (about 10^{-8} s) and a **metastable** state (about 10^{-3} s), whose long life is what makes population inversion possible.

Q. [PYQ] Which transition generates laser output in the CO₂ molecule?

- (a) **Vibrational states**
- (b) Pulsed output
- (c) Molecular states
- (d) Energy states

Why: CO₂ is a **molecule**, so it has vibrational and rotational levels in addition to electronic ones; lasing occurs between **vibrational** states, emitting at 10.6 micrometres. "Pulsed output" describes a mode of operation, not a transition.

Q. [PYQ] Which phenomenon causes a phase shift in the retardation plates?

- (a) Phasor plates
- (b) Retardation plates
- (c) Polaroid filters
- (d) **Double refraction**

Why: The causal chain: the plate is made of a **birefringent (doubly refracting)** crystal, so the two polarisation components see different refractive indices, travel at different speeds and emerge with a **phase difference**. Note "retardation plates" is offered as an option - but that is the **device**, not the **phenomenon**.

Q. [PYQ] Does polarisation vary with direction?

- (a) Polarising axis
- (b) **Transverse wave**
- (c) Propagating axis
- (d) Pass axis

Why: Polarisation exists **only because light is a transverse wave** - the oscillation is perpendicular to the direction of travel, so there is a choice of direction. **Longitudinal waves such as sound cannot be polarised**, which is the classic proof that light is transverse.

Q. [PYQ] [NUMERICAL] A body of mass 4.9 kg hangs from a spring and oscillates with period 0.6 s. How much will the spring shorten when the body is removed?

- (a) 0.809 metre
- (b) 1.089 metre
- (c) 1.009 metre
- (d) **0.089 metre**

Why: From $T = 2\pi \sqrt{m/k}$ we get $m/k = T^2/(4\pi^2)$, and the static extension $x = (m/k)g = gT^2/(4\pi^2) = 9.8 \times 0.36 / 39.478 = 0.089 \text{ m}$. Note the **mass cancels out entirely** - it is a red herring, because a heavier mass stretches

more but also oscillates more slowly.

Q. [PYQ] [NUMERICAL] Two identical cells of emf 1.5 V and internal resistance 0.5 ohm are in series. If the permitted current is 1 A, the effective terminal potential difference is:

- (a) 1 V
- (b) 0.75 V
- (c) 2 V
- (d) 0.5 V

Why: In **series**, emf adds and internal resistance adds: $EMF = 3 \text{ V}$, $r = 1 \text{ ohm}$. Then $V = EMF - Ir = 3 - (1)(1) = 2 \text{ V}$. (In parallel, emf stays the same and r becomes r/n .)

Q. [PYQ] The magnetic moment of a bar magnet is _ the magnetic moment of an equivalent solenoid if the solenoid's magnetic field is ____ that of the bar magnet.

- (a) equal to ; same as
- (b) more than ; same as
- (c) less than ; same as
- (d) equal to ; different from

Why: Definitional: an "**equivalent** solenoid" is one reproducing the magnet's behaviour. Since a dipole's external field is determined entirely by its magnetic moment, **same field implies equal magnetic moment**.

10.5 Accountancy and Business Economics

WEIGHTAGE: 3-5 marks in DSSSB only | Priority LOW for KVS/NVS

10.5.1 Questions

Q. [PYQ] According to the double entry system, every transaction will have ___ entries.

- (a) Right and Left
- (b) Up and Down
- (c) **Debit and Credit**
- (d) Up and Left

Why: Every transaction has a **debit** side and a **credit** side of equal value (the paper spelled it "Debt"). Total debits always equal total credits, which is what a **trial balance** checks.

Q. [PYQ] Which of the following accounts is/are prepared for net profit?

- (a) **Profit and loss account**
- (b) Trading account, profit and loss account and balance sheet
- (c) Trading account
- (d) Balance sheet

Why: Trading account gives **GROSS profit**; Profit and Loss account gives **NET profit**; Balance sheet shows **position, not profit**. Keep the pair straight: Trading = Gross, P&L = Net.

Q. [PYQ] What is a long-term asset in accounting?

- (a) Intangible assets
- (b) Current liabilities
- (c) Fixed liabilities
- (d) **Fixed assets**

Why: **Fixed assets** are held for use in the business for more than one year and not for resale - land, building, machinery, vehicles. **Current assets** convert to cash within a year.

Q. [PYQ] Any account or goods used by the owner of the business for his personal use is called:

- (a) sales
- (b) **drawings**
- (c) general expenses
- (d) expenses

Why: **Drawings** reduce the owner's capital. Remember the accounting equation: **Assets = Liabilities + Capital**.

Q. [PYQ] The part of the firm's capital required for conducting day-to-day expenses is called:

- (a) **working capital**
- (b) asset
- (c) expenses
- (d) fixed capital

Why: "Day to day" is the trigger. **Working Capital = Current Assets - Current Liabilities**; long-term needs are met by **fixed capital**.

Q. [PYQ] [NUMERICAL] [TRAP] ABC Ltd's purchases were Rs. 1,90,000. Gross profit was 20% ON SALES and closing stock was Rs. 30,000 more than opening stock. Find the gross profit.

- (a) Rs. 35,000
- (b) Rs. 52,000
- (c) Rs. 25,000
- (d) **Rs. 40,000**

Why: Step 1: $\text{COGS} = \text{Opening} + \text{Purchases} - \text{Closing} = 1,90,000 - 30,000 = 1,60,000$ (the opening stock cancels out, so its value was never needed). Step 2: gross profit is 20% **of sales**, so $\text{COGS} = 80\%$ of sales, giving $\text{sales} = 1,60,000 / 0.8 = 2,00,000$. Step 3: $\text{GP} = 20\%$ of $2,00,000 = \text{Rs. } 40,000$. **The trap: "20% on sales" is NOT "20% on cost"** - on cost it would have been Rs. 32,000. Always check the base.

Q. [PYQ] [NUMERICAL] Rent of Rs. 2,50,000 was received for the year ending 31 March 2024, of which Rs. 50,000 relates to next year. The amount credited to P&L is ___ and the advance is shown as ___.

- (a) Rs. 1,50,000; Liability
- (b) Rs. 2,00,000; Asset
- (c) Rs. 3,00,000; Asset
- (d) **Rs. 2,00,000; Liability**

Why: Two principles. **Accrual concept:** only income *earned this year* goes to the P&L, so $2,50,000 - 50,000 = 2,00,000$. **Income received in advance:** the business still owes the service, and an obligation is a **LIABILITY**. The trap is marking it an asset because cash came in - the *cash* is an asset, but the unearned portion creates a matching liability.

Q. [PYQ] A public company must have at least ___ members.

- (a) 14
- (b) **7**
- (c) 2
- (d) 3

Why: Learn the pair: **private company minimum 2, public company minimum 7**. A company also has **limited liability** because it is a separate legal person, unlike a partnership or sole proprietorship.

Q. [PYQ] The most common type of business organisation is:

- (a) partnership
- (b) **sole proprietorship**
- (c) NGO
- (d) corporation

Why: Easiest and cheapest to start with no registration formalities, so it covers the vast majority of small shops and traders - despite carrying **unlimited liability**.

Q. [PYQ] Which of the following is NOT a tertiary business activity?

- (a) Transportation
- (b) Warehousing
- (c) Insurance
- (d) **Manufacturing**

Why: **Manufacturing is SECONDARY** (it physically makes goods). Transportation, warehousing and insurance are **auxiliaries to trade**, which defines tertiary/service activity. **Primary** = extraction from nature.

Q. [PYQ] Which of the following statement is correct for a business organisation?

- (a) Responsibility leads to decentralisation
- (b) Authority leads to decentralisation
- (c) **Delegation leads to decentralisation**
- (d) Centralisation leads to autonomy

Why: Systematic **delegation** of authority throughout the organisation is decentralisation. Note the last option states the exact opposite of the truth - it is *decentralisation* that gives autonomy. Also: authority can be delegated, but **accountability cannot**.

Q. [PYQ] An organisation structure establishes relationships between:

- (a) **people, work and resources**
- (b) organisations and environment
- (c) suppliers and customers
- (d) organisations and society

Why: Structure defines who does what work with which resources, and who reports to whom.

Q. [PYQ] Elasticity of demand explains the ___ of the correlation between price and quantity.

- (a) level, degree and angle
- (b) angle
- (c) level
- (d) **degree**

Why: Elasticity measures **magnitude of responsiveness**, so "degree" is the right word. $E_d = (\% \text{ change in quantity}) / (\% \text{ change in price})$.

Q. [PYQ] Marginal revenue is calculated by:

- (a) dividing the change in total output quantity by the change in total revenue
- (b) dividing the change in total cost by the change in total output quantity
- (c) **dividing the change in total revenue by the change in total output quantity**
- (d) dividing the change in total revenue by the change in total profit

Why: $MR = \text{change in Total Revenue} / \text{change in Output}$. Revenue on top, quantity below. The total-*cost* version defines **marginal cost** instead.

Q. [PYQ] [TRAP] Which of the following inflations describes how demand for goods and services can drive up their prices?

- (a) Cost-push inflation
- (b) Built-in Inflation
- (c) **Demand-pull inflation**
- (d) Demand-push inflation

Why: Demand **PULLS** prices up; costs **PUSH** them up. "Demand-push" does not exist as a term - mixing the two words is the most common error.

Q. [PYQ] ___ is also referred to as the Sales-Force-Composite method or the Collective Opinion Method.

- (a) Consumer interview method
- (b) Substitute approach
- (c) Evolutionary approach
- (d) **Opinion survey**

Why: Both alternative names refer to collecting and combining the **opinions** of the firm's own sales staff in their territories.

Q. [PYQ] According to business economics, the growth of an existing product is used to estimate the demand for a new product using:

- (a) growth curve approach
- (b) vicarious approach
- (c) **substitute approach**
- (d) opinion polling approach

Why: The new product is assumed to **substitute** for the existing one, so the existing product's growth indicates its potential. For a brand-new product with no history, firms use the **evolutionary, substitute and vicarious** approaches together.

Q. [PYQ] Which of the following is NOT a type of business?

- (a) Experience driven
- (b) **Expense driven**
- (c) Efficiency driven
- (d) Expert driven

Why: The genuine categories are **experience-driven, efficiency-driven and expert-driven** - based on whether a firm competes through accumulated experience, operational efficiency or specialist expertise. No business is defined by its *expenses*.

Q. [PYQ] What does the term 'bootstrapping' refer to in the context of starting a business?

- (a) Raising funds through venture capitalists
- (b) **Self-funding the business using personal resources**
- (c) Obtaining bank loans for initial capital
- (d) Crowdfunding through online platforms

Why: The image is "pulling yourself up by your own bootstraps". All three distractors involve **external** money, which is exactly what bootstrapping avoids.

Q. [PYQ] Which of the following comes under applied mathematics?

- (a) Information processing
- (b) **Operations research**
- (c) Management accounting
- (d) Organisation

Why: **Operations Research** applies mathematical optimisation, queuing theory, simulation and game theory to real

decision problems - universally classified as applied mathematics.

PART 11

TEACHING METHODOLOGY AND CHILD PEDAGOGY

Weightage: 10 to 20 marks in DSSSB, 10 to 25 across KVS / NVS / EMRS. The highest marks-per-hour in the entire syllabus, and the section most candidates under-prepare.

11.1 Child Development

WEIGHTAGE: 3-6 marks | Newborn abilities and development principles recur | Priority CRITICAL

11.1.1 Questions

Q. [EXPECTED] [TRAP] Which of the following distinguishes growth from development?

- (a) They are identical
- (b) **Growth is quantitative and stops at maturity; development is qualitative and continues lifelong**
- (c) Growth is qualitative
- (d) Development stops at adolescence

Why: **Growth** = measurable increase in size, weight, vocabulary count - and it **stops**. **Development** = qualitative improvement in functioning and organisation - and it continues "womb to tomb".

Q. [EXPECTED] Development proceeding from head to toe is called:

- (a) Proximodistal
- (b) **Cephalocaudal**
- (c) Bilateral
- (d) Distal

Why: **Cephalocaudal** = head to toe (a baby controls its head before its legs). **Proximodistal** = centre outward (shoulder and arm control before finger control).

Q. [PYQ] Auditory and visual co-ordination system is present in child from ____.

- (a) **Birth**
- (b) 3 weeks of birth
- (c) 6 days of birth
- (d) 6 months of birth

Why: A newborn turns its head and eyes toward a sound - an **innate**, not learned, ability. Note that **vision is the weakest sense at birth** (acuity roughly 20/400, focal distance 8-12 inches) while **touch is the most mature**.

Q. [PYQ] [NUMERICAL] Regarding vision of new-born babies: I. Neonates can discriminate stationary black and white stripes 1/8 inch wide. II. By three months babies can see stripes as narrow as 1/64 inch.

- (a) **Both I and II**
- (b) Neither I nor II
- (c) Only I
- (d) Only II

Why: Both are standard infant-acuity findings. Learn the pair of fractions - **1/8 inch at birth, 1/64 inch by three months** - and note the direction: as vision sharpens, the denominator grows.

Q. [PYQ] Broca's area and Wernicke's area are concerned with language. These are the areas of: I. right hemisphere. II. left hemisphere.

- (a) **Only II**
- (b) Only I
- (c) Both I and II
- (d) Neither I nor II

Why: Language is **lateralised to the LEFT hemisphere** in about 95% of right-handed people. **Broca's area** (frontal lobe) handles speech **production**; **Wernicke's area** (temporal lobe) handles **comprehension**. Hook: Left for L anguage and Logic.

Q. [PYQ] Which of the following ailments is NOT caused by obesity?

- (a) High blood pressure
- (b) **AIDS**
- (c) Orthopedic problem
- (d) Diabetes

Why: **AIDS is an infectious disease caused by the HIV virus**; obesity is a metabolic condition. Obesity genuinely causes hypertension, type 2 diabetes, orthopaedic problems and sleep apnoea.

Q. [EXPECTED] [TRAP] Which deficiency causes night blindness?

- (a) Vitamin C
- (b) Vitamin D
- (c) **Vitamin A**
- (d) Iron

Why: Vitamin A = night blindness; Vitamin C = scurvy; Vitamin D = rickets; Iodine = goitre; Vitamin B1 = beri-beri; Iron = anaemia (very common among Indian schoolchildren).

11.2 Theories of Development

WEIGHTAGE: 3-6 marks | Piaget, Erikson, Marcia and Kohlberg are all fair game | Priority CRITICAL

11.2.1 Questions

Q. [PYQ] Piaget's formal operational stage begins at approximately ____.

- (a) **11 years**
- (b) 14 years
- (c) 8 years
- (d) 7 years

Why: Learn the boundary chain **0 - 2 - 7 - 11**: Sensorimotor 0-2, Pre-operational 2-7, Concrete operational 7-11, **Formal operational 11 onwards** (abstract and hypothetical reasoning).

Q. [EXPECTED] [TRAP] A child who cannot take another person's viewpoint is displaying:

- (a) conservation
- (b) **egocentrism**
- (c) reversibility
- (d) seriation

Why: Egocentrism is characteristic of the **pre-operational** stage (2-7), along with **centration** (focusing on one feature only), animism and irreversibility. **Conservation** and **reversibility** are achieved later, in the concrete operational stage.

Q. [EXPECTED] Object permanence develops during which Piagetian stage?

- (a) **Sensorimotor**
- (b) Pre-operational
- (c) Concrete operational
- (d) Formal operational

Why: Around **8 months**, the infant understands that a hidden object still exists. It marks the beginning of mental representation.

Q. [PYQ] Inconsistency in the development of different type of conservation is called ____.

- (a) **Horizontal Decalage**
- (b) Meta-cognition
- (c) Selective attention
- (d) Immersion

Why: *Decalage* is French for "gap"; **horizontal** means within the same stage. So it is a gap **across tasks at one stage** - a child may conserve number but not yet volume. Distractors: **metacognition** = thinking about one's own thinking; **immersion** = a language-teaching approach.

Q. [EXPECTED] [TRAP] The gap between what a learner can do alone and what they can do with guidance is called:

- (a) scaffolding
- (b) **the Zone of Proximal Development**
- (c) assimilation
- (d) equilibration

Why: ZPD is the *gap* (Vygotsky); **scaffolding** is the temporary *support* provided within it, gradually withdrawn. Teaching should target the ZPD.

Q. [EXPECTED] [TRAP] Which statement correctly contrasts Piaget and Vygotsky?

- (a) Both saw development as purely biological
- (b) **Piaget held that development leads learning; Vygotsky held that learning leads development**
- (c) Vygotsky ignored language
- (d) Piaget emphasised culture more

Why: Piaget: a solitary little scientist, universal biological stages, language follows thought. **Vygotsky:** a social apprentice, culture-dependent development, language drives thought.

Q. [EXPECTED] In Erikson's theory, the crisis of adolescence (12-19 years) is:

- (a) Trust vs Mistrust
- (b) Industry vs Inferiority
- (c) **Identity vs Role Confusion**
- (d) Intimacy vs Isolation

Why: Erikson gave **eight psychosocial stages** across the lifespan. Stage 4 (6-12) is Industry vs Inferiority; stage 5 is **Identity vs Role Confusion**; stage 6 (20-40) is Intimacy vs Isolation.

Q. [PYQ] Which identity status is described by 'Experimenting and actively searching for alternatives in their struggle to establish their identity'?

- (a) **Identity moratorium**
- (b) Identity diffusion
- (c) Identity foreclosure
- (d) Identity achievement

Why: Exploration is underway but **no commitment yet** - and *moratorium* literally means a pause. Contrast: **diffusion** = neither exploring nor committed; **foreclosure** = committed **without** exploring (identity handed down by parents); **achievement** = explored **and** committed.

Q. [EXPECTED] In Kohlberg's theory, obeying rules only to avoid punishment belongs to which level?

- (a) **Pre-conventional**
- (b) Conventional
- (c) Post-conventional
- (d) Universal

Why: **Pre-conventional** (up to about 9): punishment-obedience, then self-interest. **Conventional**: good-boy/good-girl approval, then law and order. **Post-conventional**: social contract, then universal ethical principles. What matters is the **reasoning**, not the choice.

Q. [PYQ] [TRAP] Which of the following is NOT a Neo-Freudian Theory of Development? I. Sullivan's Interpersonal Theory. II. Erikson's Psycho-social Development.

- (a) Only II
- (b) Both I and II
- (c) **Neither I nor II**
- (d) Only I

Why: A **double negative**. Both Sullivan and Erikson **ARE** Neo-Freudians, so neither qualifies as "not Neo-Freudian" - hence "Neither I nor II". Technique: first judge each statement's truth, *then* apply the question's negation. Neo-Freudians kept Freud's focus on the unconscious and early experience but rejected his emphasis on sexuality, stressing **social** factors instead.

Q. [PYQ] [TRAP] Which is correctly matched? I. A child calling all four-legged animals doggie - under-generalization. II. A child using 'duck' only for his toy duck - over-generalizing.

- (a) Both I and II
- (b) **Neither I nor II**
- (c) Only II
- (d) Only I

Why: **Both labels are swapped**. Calling all animals "doggie" applies the word **too broadly** = **OVER-generalisation**. Using "duck" for only one toy applies it **too narrowly** = **UNDER-generalisation**. Hook: **OVER** = too many things covered; **UNDER** = too few.

Q. [PYQ] In the context of Parenting, which of the following is correctly matched?

- (a) Clearly communicating parent - Child being less liked
- (b) Power assertive parents - Coercive tactics in peer used by child
- (c) Parents being harsh - Unattractive social behavior of child
- (d) **Give and take relationship of parents - Children show same relationship with peer**

Why: Children **model** the interaction style they experience at home, so reciprocal give-and-take parenting (the **authoritative** style) produces children who are reciprocal with peers. Baumrind's four styles: **Authoritative** (high control + high warmth, best outcomes), **Authoritarian** (high control, low warmth), **Permissive** (low control, high warmth), **Neglectful** (low both, worst outcomes).

11.3 Learning, Teaching Methods and Lesson Planning

WEIGHTAGE: 3-6 marks | Herbartian steps and method-to-founder matching recur | Priority CRITICAL

11.3.1 Questions

Q. [PYQ] David Kolb's model of learning style is based on which type of learning?

- (a) Inquiry-based learning
- (b) Integrative learning
- (c) Collaborative learning
- (d) **Experiential learning**

Why: Kolb defined learning as knowledge created through the **transformation of experience**. His four-stage cycle: **Concrete Experience, Reflective Observation, Abstract Conceptualisation, Active Experimentation**, giving four styles (Diverging, Assimilating, Converging, Accommodating).

Q. [EXPECTED] [TRAP] Learning by suddenly perceiving the whole pattern - the "aha!" moment - was demonstrated by:

- (a) Pavlov
- (b) Skinner
- (c) **Kohler**
- (d) Thorndike

Why: **Kohler** (Gestalt) with his chimpanzee experiments = **insight learning**. **Pavlov** = classical conditioning (association); **Skinner** = operant conditioning (consequences); **Thorndike** = trial and error with the Laws of Readiness, Exercise and **Effect**.

Q. [EXPECTED] Learning by observing and imitating a model was proposed by:

- (a) Piaget
- (b) **Bandura**
- (c) Ausubel
- (d) Maslow

Why: **Bandura's** social/observational learning, famously the Bobo doll experiment. Its four processes: attention, retention, reproduction, motivation.

Q. [PYQ] Heuristic method of teaching science was introduced by ____.

- (a) William James
- (b) **Henry Edward Armstrong**
- (c) Lev Vygotsky
- (d) Abraham Maslow

Why: **Armstrong**, an English chemist, argued students should learn science by **investigating like scientists** rather than being told. Remember the triple: **Armstrong - Heuristic - Science**. Compare **Kilpatrick - Project method** and **Dewey - learning by doing**.

Q. [EXPECTED] [TRAP] Which teaching method moves from specific examples to a general rule?

- (a) Deductive method
- (b) **Inductive method**
- (c) Lecture method
- (d) Demonstration method

Why: **Inductive** = specific to general (discovering a formula from examples). **Deductive** = general rule to specific applications. Good practice often pairs them: induce the rule, then deduce applications.

Q. [PYQ] When learning is meaning making, then students are the meaning makers. The learning process in this context is:

- (a) **student-centred**
- (b) teacher-centred
- (c) society-centred
- (d) subject-centred

Why: If the **students** make the meaning, the process revolves around the students - the **constructivist** view, whose instructional form is learner-centred teaching with the teacher as **facilitator**.

Q. [PYQ] [TRAP] _____ is a methodical approach. It suggests a logical and well-organised sequence of activities. It follows more definite steps.

- (a) teaching strategy
- (b) teaching method
- (c) teaching technique
- (d) **teaching approach**

Why: Learn the hierarchy from broadest to narrowest: **Approach** (philosophy and assumptions) to **Method** (overall plan) to **Strategy** (plan for a specific goal) to **Technique** (the actual classroom move). "Methodical approach" plus "well-organised sequence" points to **approach**.

Q. [PYQ] [TRAP] Which of the following is NOT one of the five steps of Herbartian lesson planning?

- (a) Association
- (b) Presentation
- (c) Application
- (d) **Evaluation**

Why: Herbart's five: **Preparation, Presentation, Association/Comparison, Generalisation, Application. Evaluation** was added by later lesson-plan models, not by Herbart. Hook: "Please **P**resent **A**nd **G**eneralise the **A**pplication."

Q. [PYQ] For preparing a unit plan, which procedure is followed? I. Formation of proper units and sub-units if not prescribed in the syllabus. II. Determination of methods of assessment.

- (a) **Both I and II**
- (b) Neither I nor II
- (c) Only I
- (d) Only II

Why: Both are genuine steps: a unit plan must **define its boundaries** and must **specify how learning will be assessed**, because objectives and assessment must align. In "Both I and II" questions, check each statement independently and do not reject one merely for sounding obvious.

Q. [PYQ] [TRAP] Which of the following statement is NOT correctly matched?

- (a) Child with special need - should be seated near the board
- (b) **Performing art - every child should have clear view of board**
- (c) Group work - sitting arrangement should be in clump or clustered
- (d) Group discussion - teacher should have eye contact with every child

Why: **Performing arts need open floor space** for movement and mutual observation - the blackboard is essentially irrelevant. The other three pairings are all correct.

Q. [EXPECTED] Micro-teaching involves:

- (a) teaching a full lesson to a large class
- (b) **practising one specific teaching skill with a small group for a short time, then getting feedback and re-teaching**
- (c) teaching without any plan
- (d) teaching only gifted children

Why: The cycle is **plan, teach, feedback, re-plan, re-teach, re-feedback**. It is a teacher-training technique for isolating and improving individual skills.

11.4 Assessment and Evaluation

WEIGHTAGE: 2-4 marks | The formative/summative/diagnostic distinction is near-certain | Priority CRITICAL

11.4.1 Questions

Q. [PYQ] Diagnostic assessment mainly helps in which of the following?

- (a) Grading performance
- (b) Standardising content
- (c) Promoting learners
- (d) **Identifying misconceptions**

Why: *Diagnostic* comes from medicine - a diagnosis finds **what exactly is wrong** so the right remediation can follow. **Grading** and **promoting** are functions of **summative** assessment.

Q. [EXPECTED] [TRAP] Assessment conducted during teaching to give feedback and adjust instruction is:

- (a) summative
- (b) **formative**
- (c) placement
- (d) prognostic

Why: **Formative** = **assessment FOR learning** (during, continuous, feedback-driven). **Summative** = **assessment OF learning** (at the end, produces marks and grades). **Placement** decides the starting point; **diagnostic** locates specific difficulties.

Q. [EXPECTED] CCE stands for:

- (a) Common Curriculum Evaluation
- (b) **Continuous and Comprehensive Evaluation**
- (c) Cumulative Class Examination
- (d) Central Certification Exam

Why: **Continuous** (throughout the year, not one final exam) and **Comprehensive** (both scholastic and co-scholastic areas - academics plus attitudes, values, health and arts).

Q. [EXPECTED] [TRAP] A test that measures what it claims to measure is said to have:

- (a) reliability
- (b) **validity**
- (c) objectivity
- (d) usability

Why: **Validity** = measures the right thing. **Reliability** = gives consistent results on repetition. **Objectivity** = different examiners award the same marks. Note the relationship: a test can be reliable **without** being valid, but it cannot be valid without being reliable.

Q. [PYQ] All of the given are the methods of Information and Communication Technology-based assessment except ____.

- (a) Concept map construction
- (b) **Writing reflection journal on paper**
- (c) Multimedia program development
- (d) Computer-based testing

Why: The qualifier "**on paper**" removes all technology. A reflection journal *would* be ICT-based if kept as a blog or digital document - examiners insert such phrases deliberately, so read every option to the end.

Q. [EXPECTED] In the revised Bloom's taxonomy, the highest cognitive level is:

- (a) Evaluating
- (b) Analysing
- (c) Synthesising
- (d) **Creating**

Why: Revised (Anderson) order: Remembering, Understanding, Applying, Analysing, Evaluating, **Creating**. In the **original** taxonomy, Evaluation was highest and Synthesis came below it - so read which version is being asked.

Q. [EXPECTED] A test that compares a student against a fixed learning standard rather than against classmates is:

- (a) norm-referenced
- (b) **criterion-referenced**
- (c) diagnostic
- (d) aptitude

Why: **Criterion-referenced** measures against a defined outcome. **Norm-referenced** ranks the student against peers and deliberately spreads scores.

11.5 Inclusive Education, Policies and the Community

WEIGHTAGE: 3-6 marks | Act years and scheme target groups are pure recall | Priority CRITICAL

11.5.1 Questions

Q. [PYQ] The philosophy of inclusive education was added in District Primary Education Programme (DPEP) in ____ year.

- (a) 1966
- (b) **1997**
- (c) 2012
- (d) 1950

Why: Sequence: **DPEP launched 1994**, the international **Salamanca Statement also 1994**, and India incorporated inclusive education into DPEP in **1997**.

Q. [PYQ] [TRAP] Which law legally mandates the inclusion of children with disabilities into mainstream schools in India?

- (a) **RPwD Act, 2016**
- (b) National Curriculum Framework, 2005
- (c) RTI Act, 2009
- (d) NEP, 2020

Why: Only the **RPwD Act 2016** is a **statute** mandating inclusion (it also recognises **21 disabilities**, with 4% job and 5% higher-education reservation). NCF 2005 is a **framework**; NEP 2020 is a **policy**. And note the trap: **RTI = Right to Information (2005)**, whereas the education law of 2009 is **RTE = Right to Education**.

Q. [EXPECTED] The RTE Act 2009 guarantees free and compulsory education for children aged:

- (a) 3 to 12 years
- (b) **6 to 14 years**
- (c) 6 to 18 years
- (d) 0 to 6 years

Why: The **86th Constitutional Amendment (2002)** inserted **Article 21A** making it a fundamental right; the RTE Act came into force on **1 April 2010**. It also mandates 25% reservation in private schools for disadvantaged groups and bans corporal punishment.

Q. [EXPECTED] [TRAP] The key difference between integrated and inclusive education is:

- (a) they are identical
- (b) **in integration the child adapts to the school; in inclusion the school adapts to the child**
- (c) inclusion uses separate special schools
- (d) integration is only for gifted children

Why: This one sentence is the whole distinction, and it is asked in many wordings. **Segregated** education places children with disabilities in entirely separate special schools.

Q. [PYQ] Integrated child development service provides nutritious food for children who are ____.

- (a) School going
- (b) School drop-out
- (c) **Below age of six years and too young to go to school**
- (d) Girls below 12 years of age

Why: **ICDS (launched 1975)** is an **early childhood (0-6)** programme delivered through **Anganwadi** centres, also covering pregnant and lactating mothers. School-going children are covered by the **Mid-Day Meal Scheme (PM POSHAN)** instead - keep the two separate.

Q. [EXPECTED] The NEP 2020 school structure is:

- (a) 10+2
- (b) **5+3+3+4**
- (c) 8+4
- (d) 5+5+2

Why: The **5+3+3+4** design: Foundational (5), Preparatory (3), Middle (3), Secondary (4). NEP 2020 also recommends

the **mother tongue as medium up to at least Grade 5** and a target of **6% of GDP** for education.

Q. [PYQ] Gender inequality refers to which of the following?

- (a) **Social and economic disparities between genders**
- (b) Biological differences only
- (c) Differences in hairstyles
- (d) Uniform distribution of resources

Why: Sex is biological; gender is socially constructed. Biological difference alone is not *inequality*, and "uniform distribution of resources" describes equality - the opposite of what is asked.

Q. [PYQ] What is the core principle of the multilingual approach in early education?

- (a) Prioritising grammar over communication
- (b) Replacing mother tongue with the target language quickly
- (c) **Welcoming the child's language and using it to build understanding**
- (d) Teaching only in the national language

Why: The home language is a **resource and a bridge**, not a problem to be removed. Rapid replacement is **submersion** ("sink or swim"), which harms both comprehension and self-esteem.

Q. [PYQ] Aesthetic education primarily develops ____.

- (a) speed in recall
- (b) logical reasoning
- (c) factual knowledge
- (d) **sensory responsiveness**

Why: Aesthetics is fundamentally about **perceiving and responding** to beauty, form, colour, sound and rhythm. Recall and factual knowledge are memory functions; logical reasoning is cognitive.

Q. [PYQ] You are designing a curriculum for vocational education. To ensure learners can transfer skills to actual work settings, which principle should you emphasise most?

- (a) **Principle of activity and practicality**
- (b) Principle of articulation
- (c) Principle of recency
- (d) Principle of economy

Why: Skills transfer to real workplaces only through **hands-on practice in realistic conditions**. *Articulation* concerns smooth progression between stages; *economy* concerns efficient use of time and resources.

Q. [PYQ] In school, a 'subject' differs from a 'discipline' because it is ____.

- (a) an external research field
- (b) more deeply organised
- (c) a philosophical field
- (d) **the branch of the discipline taught in the classroom**

Why: A **discipline** is the full scholarly domain produced by researchers; a **subject** is the curated, age-appropriate slice of it that appears on a timetable. School "Science" is a subject drawn from the disciplines of Physics, Chemistry and Biology.

Q. [PYQ] Participation in which activities can enable the community to accept a teacher as one of them? I. Problems of unhygienic conditions. II. Problems of agriculture.

- (a) Neither I nor II
- (b) Only I
- (c) **Both I and II**
- (d) Only II

Why: Acceptance comes from involvement in matters the community **actually cares about daily**. The wider principle: the teacher's role extends beyond the classroom into community service, so options restricting the teacher to purely academic work are usually wrong.

Q. [PYQ] [TRAP] Which of the following is the activity related to community services?

- (a) Collect picture about different musical instrument and dances
- (b) Collect information about handicrafts of different place
- (c) **Participating in decoration at the time of events**
- (d) Explore food habit and living style of the people of different state

Why: Apply one test: **is the student GIVING something to the community, or GETTING information from it?**

Collecting pictures, information and habits all benefit the student's own learning - those are *project* work. Only decorating for a community event contributes effort **for the community's benefit**, which is service.

Q. [EXPECTED] Action research is best described as:

- (a) large-scale research by universities
- (b) **small-scale research by the teacher in their own classroom to solve an immediate practical problem**
- (c) research on teaching methods by NCERT
- (d) a literature review

Why: Its cycle is **plan, act, observe, reflect**. Related concept: **reflective teaching** - systematically examining what happened in a lesson and why, then changing practice.

Q. [EXPECTED] [TRAP] Which of the following best describes the difference between guidance and counselling?

- (a) They are identical
- (b) **Guidance is broader and more informational; counselling is a focused, confidential one-to-one professional relationship**
- (c) Counselling is only for academic problems
- (d) Guidance is only for adults

Why: Guidance helps a person understand themselves and make informed choices (educational, vocational, personal). Counselling addresses a specific personal or psychological problem in confidence. The **cumulative record card** provides the basis for both.

Q. [EXPECTED] Who among the following is known as Lokhitwadi?

- (a) Ishwar Chandra Vidyasagar
- (b) **Gopal Hari Deshmukh**
- (c) Rabindranath Tagore
- (d) Ram Mohan Roy

Why: Gopal Hari Deshmukh, the Marathi social reformer, wrote under the pen name **Lokhitwadi** ("one who desires the people's welfare"). Social-reformer questions appear in both the pedagogy and General Awareness sections.

11.6 Final Revision Table

11.6.1 The 30 facts most likely to be asked

Fact	Answer
Piaget stage boundaries	0-2-7-11; formal operational begins at 11
Object permanence	About 8 months, sensorimotor stage
Auditory-visual coordination	Present from BIRTH
Newborn visual acuity	1/8 inch stripes at birth; 1/64 inch by 3 months
Broca's and Wernicke's areas	LEFT hemisphere
Horizontal decalage	Inconsistency across conservation tasks in the same stage
Over-generalisation	Word applied too BROADLY (all animals are "doggie")
Under-generalisation	Word applied too NARROWLY (only his toy duck)
Erikson stage 5 (12-19)	Identity vs Role Confusion
Identity moratorium	Actively exploring, not yet committed
Best parenting style	Authoritative (high control + high warmth)
Kolb	Experiential learning, 4-stage cycle
Heuristic method	Henry Edward Armstrong, for science
Project method	Kilpatrick (based on Dewey)
Insight learning	Kohler (Gestalt)
Observational learning	Bandura (Bobo doll)
Operant conditioning	Skinner
Classical conditioning	Pavlov
Laws of Readiness, Exercise, Effect	Thorndike
ZPD and scaffolding	Vygotsky
Herbartian five steps	Preparation, Presentation, Association, Generalisation, Application (NO Evaluation)
Hierarchy of terms	Approach > Method > Strategy > Technique
Diagnostic assessment	Identifies misconceptions
Formative vs summative	FOR learning vs OF learning
Revised Bloom's highest level	Creating
ICDS (1975)	Children below 6 years, via Anganwadi
Mid-Day Meal / PM POSHAN	School-going children
DPEP 1994; inclusion added	1997
RTE Act 2009	Ages 6-14, Article 21A, in force 1 April 2010
RPwD Act 2016	Legally mandates inclusion; 21 disabilities
NEP 2020 structure	5+3+3+4; mother tongue to Grade 5; 6% of GDP
Integration vs inclusion	Child adapts to school vs SCHOOL adapts to child
Aesthetic education develops	Sensory responsiveness
Vocational curriculum principle	Activity and practicality
Community service test	Student GIVES to community, not GETS information

The last thing to read before you close this book

Two habits decide most Computer Science teaching-exam results, and neither is about knowing more.

First, circle the negative word. A very large share of questions in these papers are framed as NOT, FALSE, EXCEPT or INCORRECT. Candidates who read fluently and answer the *positive* version of the question lose 5 to 10 marks per paper to nothing but haste. Physically mark that word on your rough sheet before you look at the options.

Second, eliminate before you guess. You will not know every answer. But if you can rule out two options, the expected value of attempting is **+0.375** - clearly worth taking. If you can rule out nothing, leave it blank. That single discipline is often the difference between a score near the cut-off and one comfortably above it.

Work through this bank twice, keep your leak list honest, and revise Part 1's weightage table so your hours go where the marks are.

All the best.